US012402563B2

(12) United States Patent
O'Connor et al.

(10) Patent No.: US 12,402,563 B2
(45) Date of Patent: Sep. 2, 2025

(54) CROSS MEMBER LOCATION AND AVOIDANCE DURING AN UNLOADING OPERATION

(71) Applicant: **Deere & Company**, Moline, IL (US)

(72) Inventors: **Sara C. O'Connor**, West Des Moines, IA (US); **Kellen E. O'Connor**, Clive, IA (US); **Jeremy J. Faust**, Grimes, IA (US); **Ryan R. White**, Polk City, IA (US); **Nicholas C. Baltz**, Port Byron, IL (US)

(73) Assignee: **Deere & Company**, Moline, IL (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 259 days.

(21) Appl. No.: **18/334,869**

(22) Filed: **Jun. 14, 2023**

(65) **Prior Publication Data**

US 2024/0138310 A1 May 2, 2024

Related U.S. Application Data

(60) Provisional application No. 63/381,187, filed on Oct. 27, 2022, provisional application No. 63/381,178, filed on Oct. 27, 2022.

(51) **Int. Cl.**

A01D 90/10	(2006.01)
G06T 7/70	(2017.01)
G06V 20/56	(2022.01)

(52) **U.S. Cl.**

CPC ***A01D 90/10*** (2013.01); ***G06T 7/70*** (2017.01); ***G06V 20/56*** (2022.01); *G06T 2207/30188* (2013.01); *G06T 2207/30252* (2013.01)

(58) **Field of Classification Search**

CPC A01D 90/10; A01D 43/07; A01D 43/08; A01D 41/1217; A01D 41/127; A01D 43/073; A01D 45/00; G06T 7/70; G06T 2207/30188; G06T 2207/30252; G06V 20/56; G01S 17/89

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,575,316 A * 11/1996 Pollklas A01D 43/087 141/96
7,277,784 B2 10/2007 Weiss
8,060,283 B2 11/2011 Mott et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN 108550141 A 9/2018
DE 102009027245 A1 12/2010

(Continued)

OTHER PUBLICATIONS

U.S. Appl. No. 18/465,580 Application and Drawings filed Sep. 12, 2023, 114 pages.

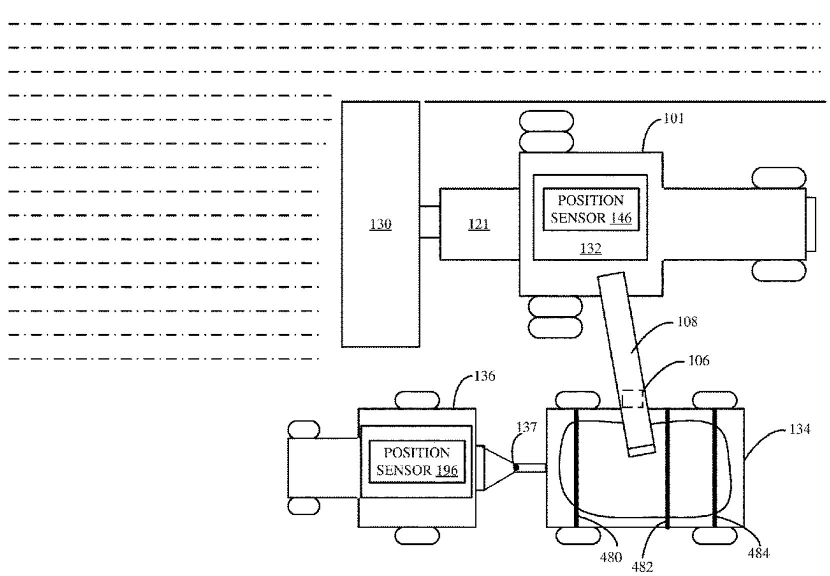
(Continued)

Primary Examiner — Jelani A Smith
Assistant Examiner — Alyssa Rorie
(74) *Attorney, Agent, or Firm* — Joseph R. Kelly; Kelly, Holt & Christenson, PLLC

(57) **ABSTRACT**

A cross member on a receiving vehicle which is coupled to a following vehicle, is located relative to a leading vehicle. The location of the cross member is tracked during an unloading operation so the leading vehicle avoids the cross member when unloading material into the receiving vehicle.

20 Claims, 29 Drawing Sheets



(56) **References Cited**

U.S. PATENT DOCUMENTS

8,868,304	B2	10/2014	Bonefas	
9,119,342	B2	9/2015	Bonefas	
9,326,444	B2	5/2016	Bonefas	
9,642,305	B2	5/2017	Nykamp et al.	
9,861,040	B2	1/2018	Bonefas	
9,992,931	B2	6/2018	Bonefas et al.	
2008/0147282	A1	6/2008	Kormann	
2012/0215381	A1*	8/2012	Wang	A01B 69/008 701/1
2012/0215394	A1	8/2012	Wang et al.	
2012/0215409	A1	8/2012	Wang et al.	
2013/0166157	A1	6/2013	Schleicher et al.	
2013/0213518	A1	8/2013	Bonefas	
2013/0231823	A1	9/2013	Wang et al.	
2014/0224377	A1	8/2014	Bonefas	
2014/0325422	A1*	10/2014	Madsen	G06F 3/0488 715/771
2015/0264866	A1	9/2015	Foster et al.	
2015/0366134	A1*	12/2015	Dollinger	A01D 41/127 701/50
2016/0057921	A1	3/2016	Pickett et al.	
2017/0042088	A1	2/2017	Nykamp et al.	
2017/0055446	A1	3/2017	Nykamp et al.	
2018/0024549	A1	1/2018	Hurd	
2018/0058861	A1	3/2018	Doria et al.	
2020/0039306	A1	2/2020	Rogan	
2020/0128738	A1*	4/2020	Suleman	A01D 41/1278
2020/0132822	A1	4/2020	Pimentel et al.	
2020/0133262	A1	4/2020	Suleman et al.	
2020/0164803	A1	5/2020	Jales Costa et al.	
2020/0196526	A1	6/2020	Koch et al.	
2020/0319655	A1	10/2020	Desai et al.	
2021/0195840	A1	7/2021	Puryk et al.	
2021/0294337	A1	9/2021	Van Mill et al.	
2021/0329840	A1	10/2021	Craig	
2021/0337729	A1	11/2021	O'Connor et al.	
2022/0015288	A1	1/2022	Christiansen et al.	
2022/0015290	A1	1/2022	Yao et al.	
2022/0122197	A1	4/2022	Hanrieder	
2022/0197302	A1*	6/2022	McClelland	G05D 1/667
2022/0204051	A1	6/2022	Rands et al.	
2022/0287239	A1	9/2022	Faust et al.	
2022/0410704	A1	12/2022	O'Connor et al.	
2023/0113645	A1*	4/2023	Christiansen	A01D 41/1275 701/50
2023/0180659	A1	6/2023	Grieshop et al.	
2023/0281896	A1*	9/2023	Christiansen	G01F 23/804 345/629
2023/0345856	A1	11/2023	Mahler et al.	
2023/0403969	A1	12/2023	Christiansen et al.	

FOREIGN PATENT DOCUMENTS

DE	102012211001	A1	1/2014
DE	102015109799	A1	12/2016
DE	102016214320	A1	2/2017
EP	1219153	A2	7/2002
EP	2044826	A2	4/2009
EP	2245917	A2	11/2010
EP	2995191	A1	3/2016
EP	3316218	A1	5/2018
EP	3815486	A1	5/2021
EP	3949714	A1	2/2022
EP	3955183	A1	2/2022
EP	4056019	A1	9/2022
EP	4062740	A1	9/2022
EP	4070636	A1	10/2022
EP	4101286	A1	12/2022
WO	WO 2022036114	A1	2/2022

OTHER PUBLICATIONS

U.S. Appl. No. 18/447,718 Application and Drawings filed Aug. 10, 2023, 53 pages.

U.S. Appl. No. 18/352,576 Application and Drawings filed Jul. 14, 2023, 39 pages.

U.S. Appl. No. 18/358,401 Application and Drawings filed Jul. 25, 2023, 47 pages.

U.S. Appl. No. 18/471,701 Application and Drawings filled Sep. 21, 2023, 50 pages.

Autodesk, "Determine Volume of Solid", Oct. 16, 2019, 3 pages. <https://knowledge.autodesk.com/support/autocad/learn-explore/caas/sfdcarticles/sfdcarticles/Determine-volume-of-solid.html#:~:text=To%20determine%20the%20volume%20of%20a%20solid%20object%2C%20use%20the,Press%20ENTER>.

Stanley, Nick, Solidworks Calculating Volume of a Solid. GoEngineer. 6 pages. Feb. 3. 2022. Retrieved from the Internet: <https://www.goengineer.com/blog/solidworks-calculating-volume-of-a-solid>.

U.S. Appl. No. 63/381,178 Application and Drawings filed Oct. 27, 2022. 101 pages.

U.S. Appl. No. 63/381,187 Application and Drawings filed Oct. 27, 2022, 104 pages.

U.S. Appl. No. 63/512,368 Application and Drawings filed Jul. 7, 2023, 101 pages.

Tesla Motors Club Community Thread entitled "Lane change notification" 4 pages, originally posted Dec. 29, 2021 by user "Horvin". Retrieved from internet: <https://teslamotorsclub.com/tmc/threads/lane-change-notification.252072/>.

U.S. Appl. No. 18/334,863 Application and Drawings filed Jun. 14, 2023. 73 pages.

U.S. Appl. No. 18/334,836 Application and Drawings filed Jun. 14, 2023, 47 pages.

U.S. Appl. No. 18/334,861 Application and Drawings filed Jun. 14, 2023, 96 pages.

U.S. Appl. No. 18/334,874 Application and Drawings filed Jun. 14, 2023, 59 pages.

Active Fill Control (https://www.deere.com/en/technology-products/precision-ag-technology/guidance/active-fill-control/) 2 pages. Retrieved on Jun. 27, 2023.

Machine Sync (https://www.deere.com/en/technology-products/precision-ag-technology/guidance/machine-sync/) 3 pages, Retrieved on Jun. 27, 2023.

Huang, Xiaoshul & Mei, Guofeng & Zhang, Jian & Abbas, Rana. (2021), A comprehensive survey on point cloud registration. Mar. 5, 2021, 17 pages https://www.researchgate.net/publication/349787695_A_comprehensive_survey_on_point_cloud_registration.

Extended European Search Report and Written Opinion issued in European Patent Application No. 23203537.8, dated Mar. 14, 2024, in 05 pages.

Extended European Search Report and Written Opinion issued in European Patent Application No. 23203536.0, dated Mar. 14, 2024, in 06 pages.

Extended European Search Report and Written Opinion issued in European Patent Application No. 23203519.6, dated Mar. 14, 2024, in 05 pages.

Extended European Search Report and Written Opinion issued in European Patent Application No. 23203069.2, dated Mar. 18, 2024, in 07 pages.

Extended European Search Report and Written Opinion issued in European Patent Application No. 23202822.5, dated Mar. 19, 2024, in 07 pages.

Extended European Search Report and Written Opinion issued in European Patent Application No. 23202466.1, dated Mar. 26, 2024, in 07 pages.

Extended European Search Report and Written Opinion issued in European Patent Application No. 23202485.1, dated Mar. 26, 2024, in 07 pages.

Extended European Search Report and Written Opinion issued in European Patent Application No. 23203067.6, dated Mar. 26, 2024, in 11 pages.

Extended European Search Report and Written Opinion issued in European Patent Application No. 23202510.6, dated Apr. 4, 2024, in 08 pages.

(56) **References Cited**

OTHER PUBLICATIONS

Extended European Search Report and Written Opinion issued in European Patent Application No. 23202817.5, dated Apr. 23, 2024, in 08 pages.

* cited by examiner

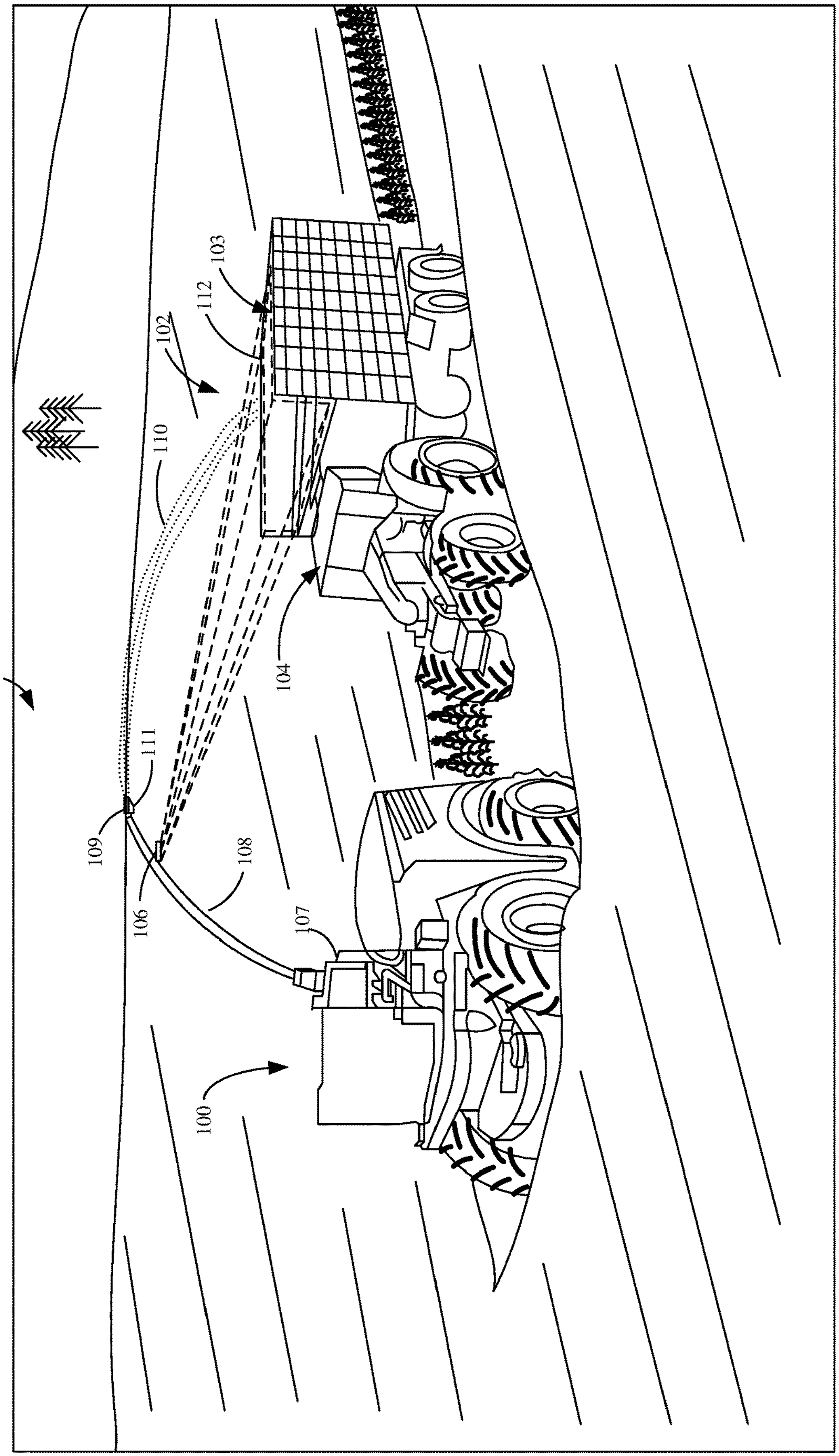


FIG. 1

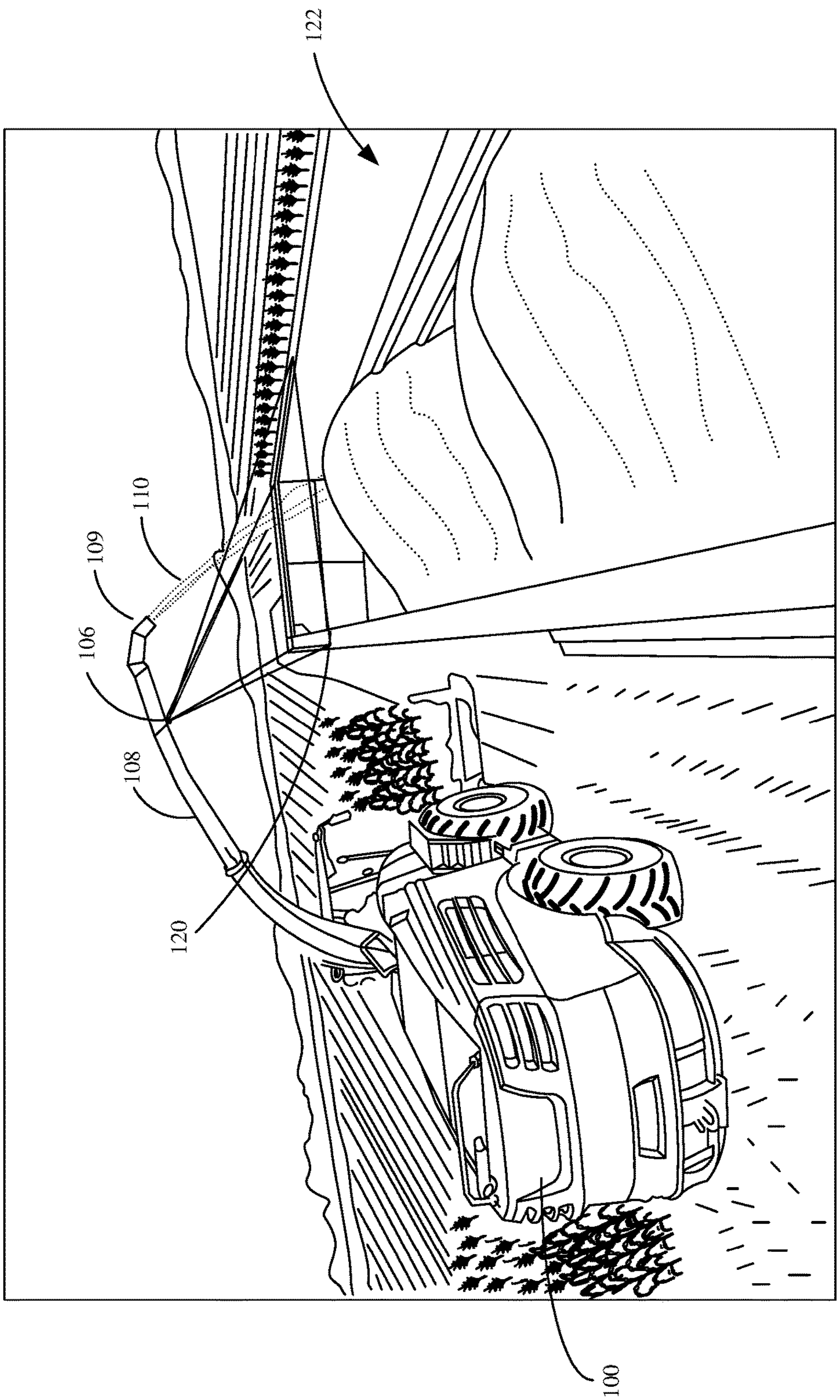


FIG. 2

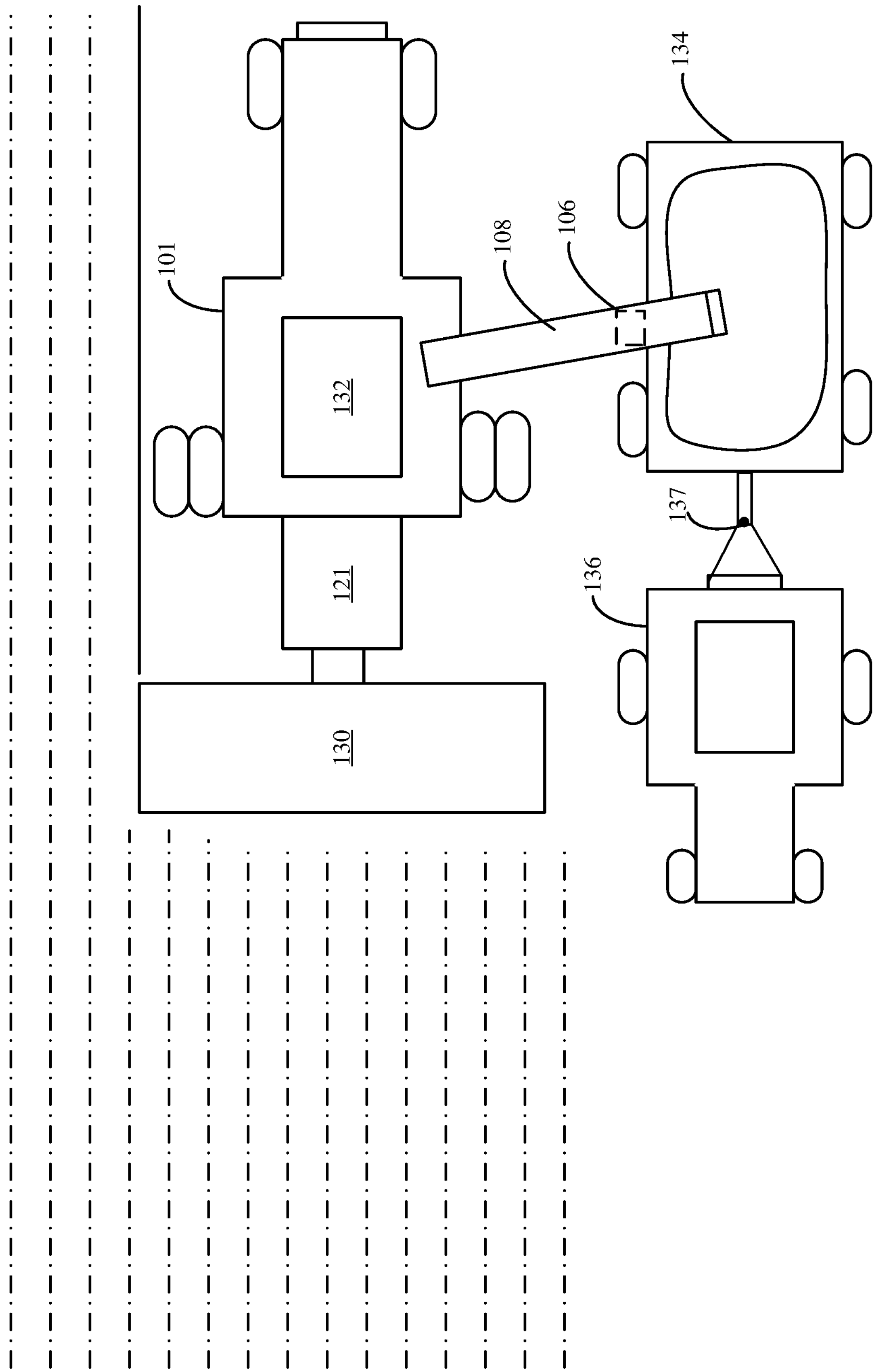


FIG. 3

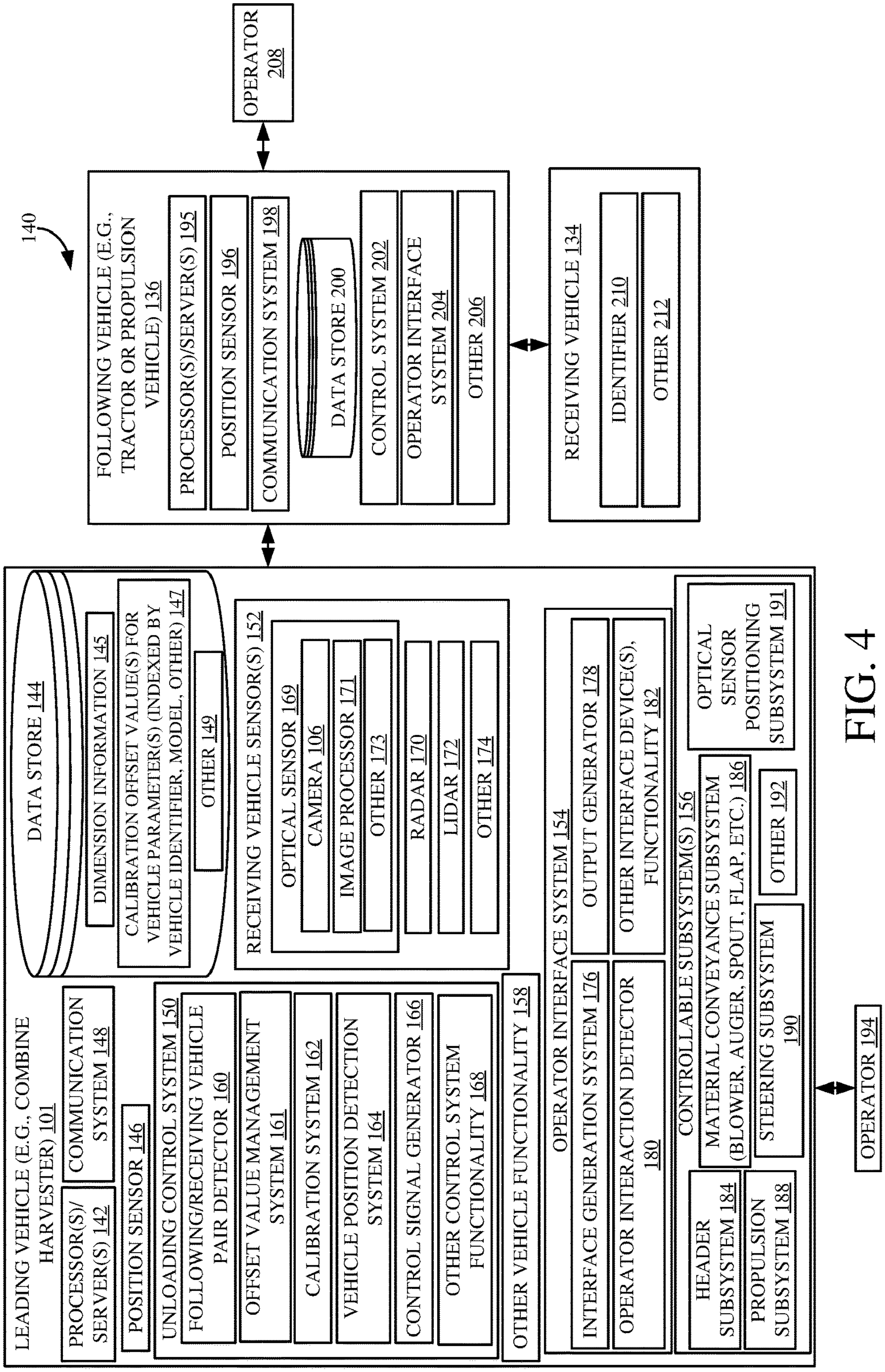

140
OPERATOR 208
FOLLOWING VEHICLE (E.G., TRACTOR OR PROPULSION VEHICLE) 136
PROCESSOR(S)/SERVER(S) 195
POSITION SENSOR 196
COMMUNICATION SYSTEM 198
DATA STORE 200
CONTROL SYSTEM 202
OPERATOR INTERFACE SYSTEM 204
OTHER 206
RECEIVING VEHICLE 134
IDENTIFIER 210
OTHER 212
LEADING VEHICLE (E.G., COMBINE HARVESTER) 101
PROCESSOR(S)/ SERVER(S) 142
COMMUNICATION SYSTEM 148
POSITION SENSOR 146
DATA STORE 144
DIMENSION INFORMATION 145
CALIBRATION OFFSET VALUE(S) FOR VEHICLE PARAMETER(S) (INDEXED BY VEHICLE IDENTIFIER, MODEL, OTHER) 147
OTHER 149
UNLOADING CONTROL SYSTEM 150
FOLLOWING/RECEIVING VEHICLE PAIR DETECTOR 160
OFFSET VALUE MANAGEMENT SYSTEM 161
CALIBRATION SYSTEM 162
VEHICLE POSITION DETECTION SYSTEM 164
CONTROL SIGNAL GENERATOR 166
OTHER CONTROL SYSTEM FUNCTIONALITY 168
RECEIVING VEHICLE SENSOR(S) 152
OPTICAL SENSOR 169
CAMERA 106
IMAGE PROCESSOR 171
OTHER 173
RADAR 170
LIDAR 172
OTHER 174
OTHER VEHICLE FUNCTIONALITY 158
OPERATOR INTERFACE SYSTEM 154
INTERFACE GENERATION SYSTEM 176
OUTPUT GENERATOR 178
OPERATOR INTERACTION DETECTOR 180
OTHER INTERFACE DEVICE(S), FUNCTIONALITY 182
CONTROLLABLE SUBSYSTEM(S) 156
HEADER SUBSYSTEM 184
MATERIAL CONVEYANCE SUBSYSTEM (BLOWER, AUGER, SPOUT, FLAP, ETC.) 186
PROPULSION SUBSYSTEM 188
STEERING SUBSYSTEM 190
OTHER 192
OPTICAL SENSOR POSITIONING SUBSYSTEM 191
OPERATOR 194

FIG. 4

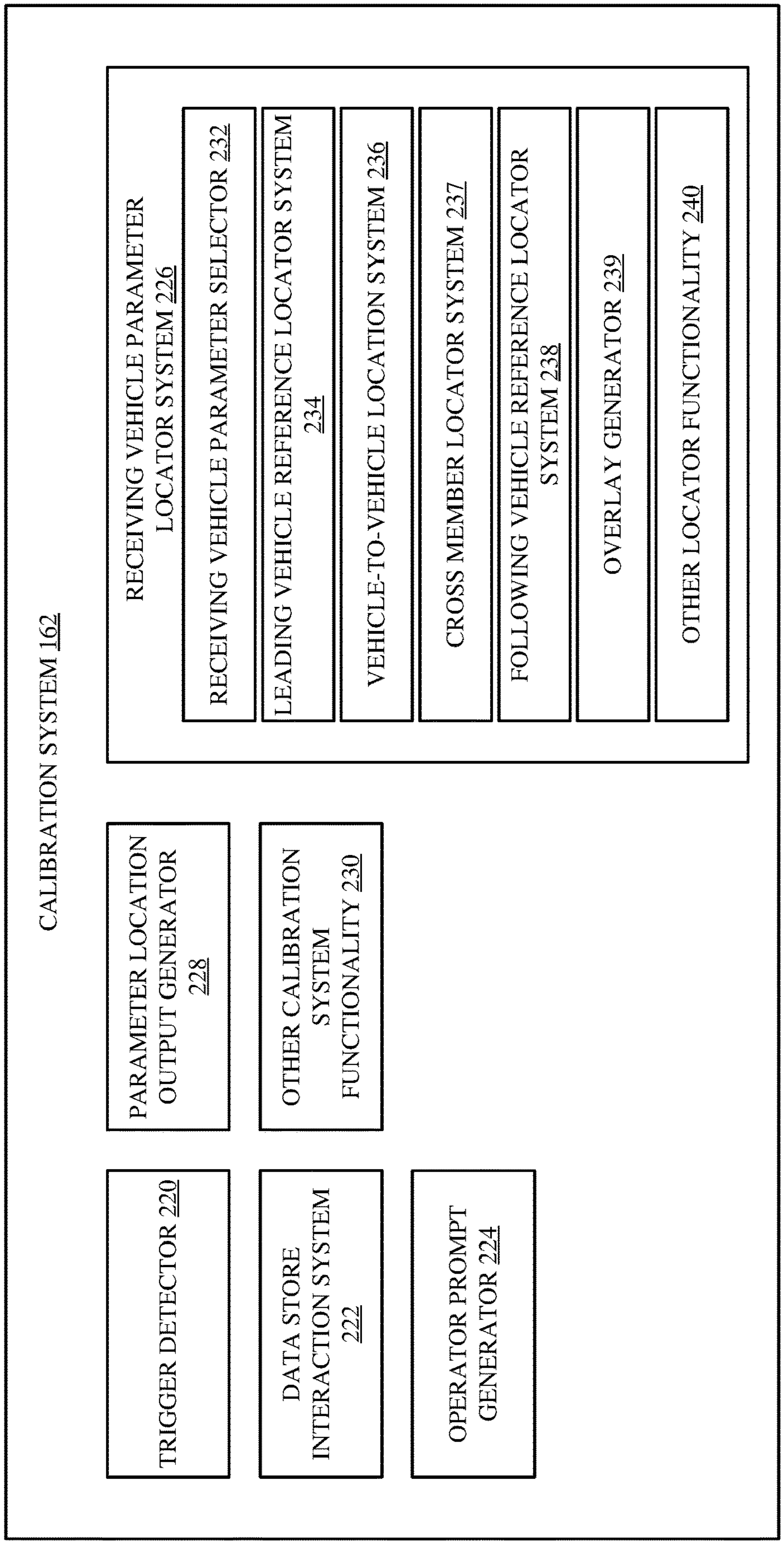

CALIBRATION SYSTEM 162
TRIGGER DETECTOR 220
DATA STORE INTERACTION SYSTEM 222
OPERATOR PROMPT GENERATOR 224
PARAMETER LOCATION OUTPUT GENERATOR 228
OTHER CALIBRATION SYSTEM FUNCTIONALITY 230
RECEIVING VEHICLE PARAMETER LOCATOR SYSTEM 226
RECEIVING VEHICLE PARAMETER SELECTOR 232
LEADING VEHICLE REFERENCE LOCATOR SYSTEM 234
VEHICLE-TO-VEHICLE LOCATION SYSTEM 236
CROSS MEMBER LOCATOR SYSTEM 237
FOLLOWING VEHICLE REFERENCE LOCATOR SYSTEM 238
OVERLAY GENERATOR 239
OTHER LOCATOR FUNCTIONALITY 240

FIG. 5

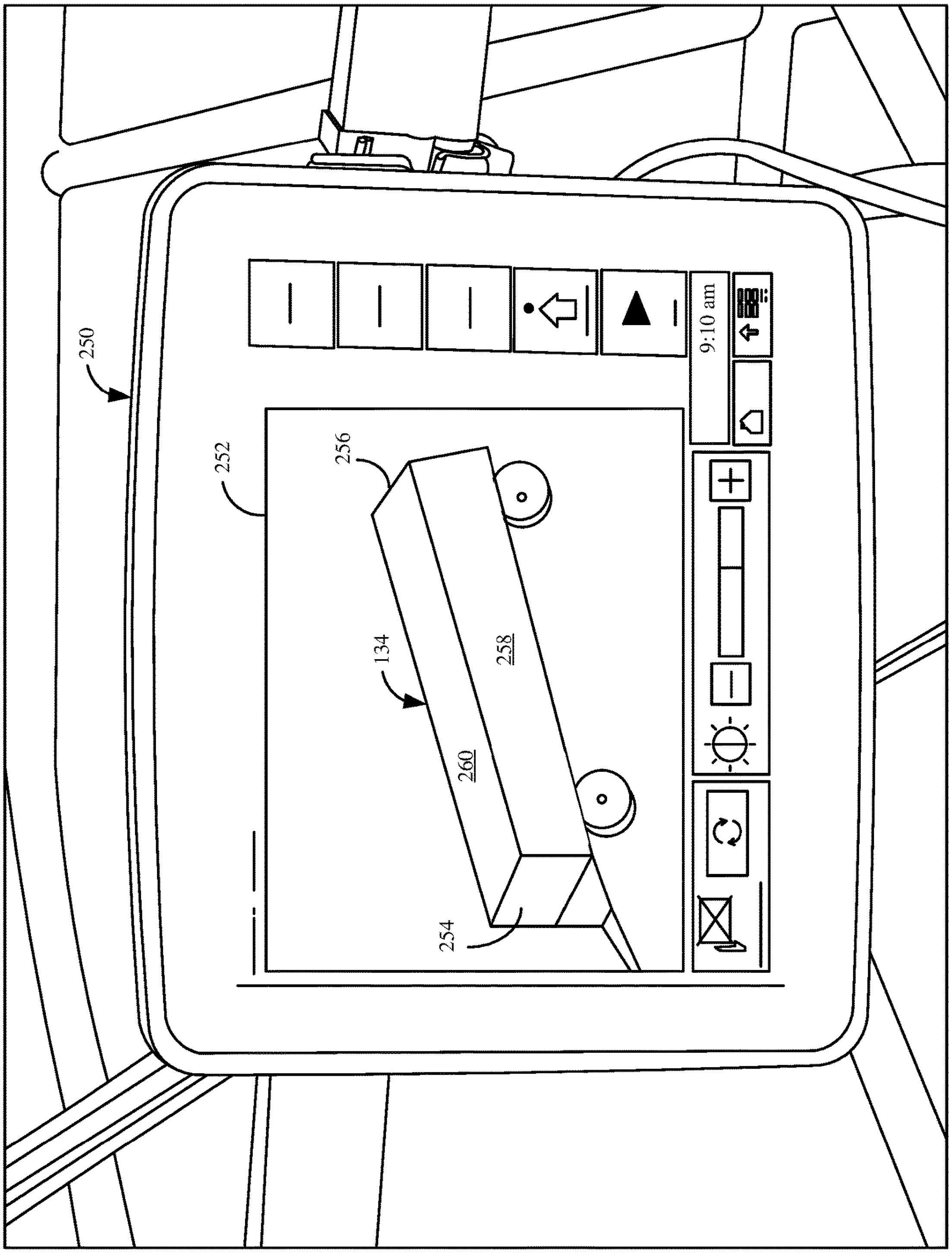


FIG. 6

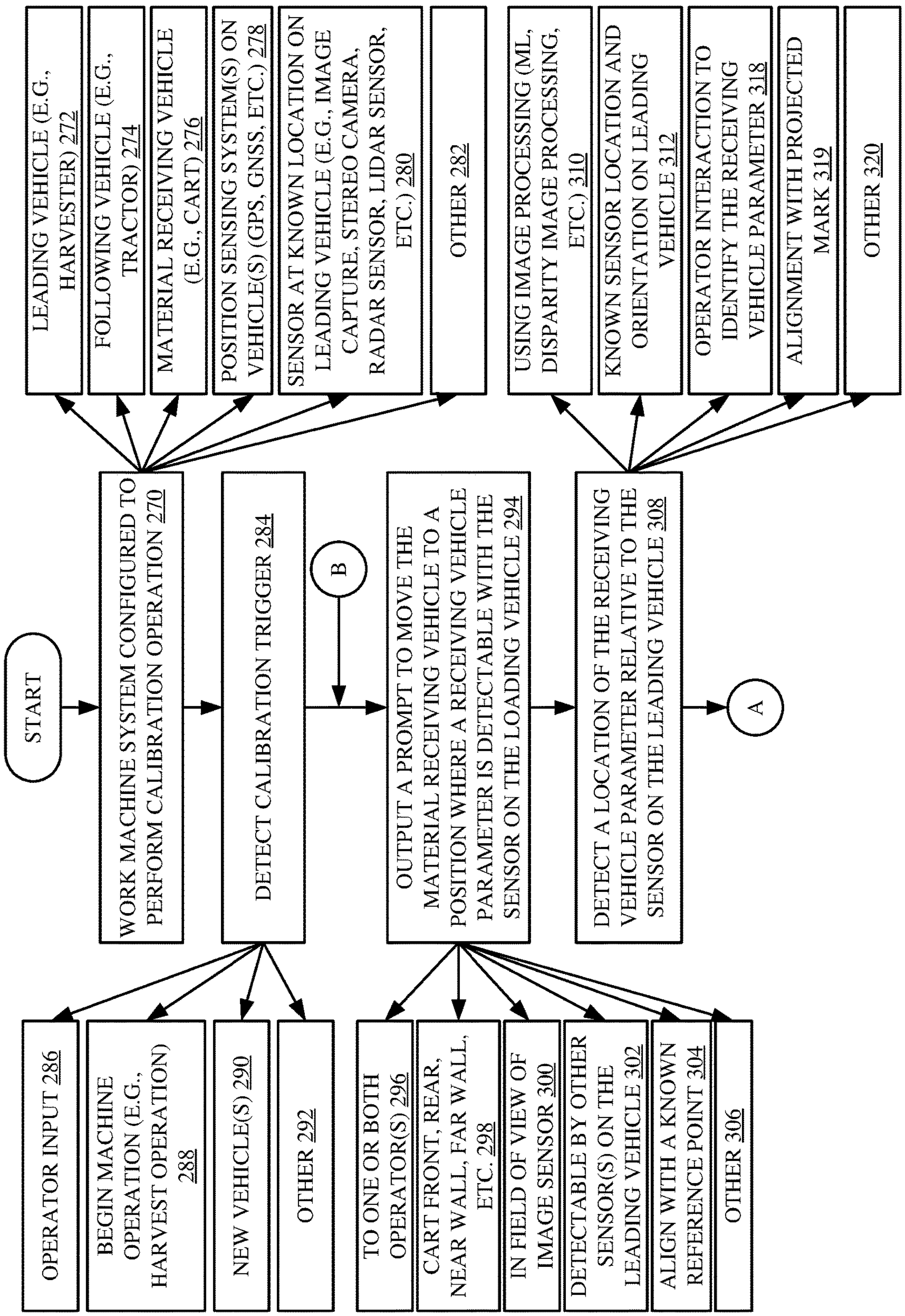

START
WORK MACHINE SYSTEM CONFIGURED TO PERFORM CALIBRATION OPERATION 270
LEADING VEHICLE (E.G., HARVESTER) 272
FOLLOWING VEHICLE (E.G., TRACTOR) 274
MATERIAL RECEIVING VEHICLE (E.G., CART) 276
POSITION SENSING SYSTEM(S) ON VEHICLE(S) (GPS, GNSS, ETC.) 278
SENSOR AT KNOWN LOCATION ON LEADING VEHICLE (E.G., IMAGE CAPTURE, STEREO CAMERA, RADAR SENSOR, LIDAR SENSOR, ETC.) 280
OTHER 282
DETECT CALIBRATION TRIGGER 284
OPERATOR INPUT 286
BEGIN MACHINE OPERATION (E.G., HARVEST OPERATION) 288
NEW VEHICLE(S) 290
OTHER 292
B
OUTPUT A PROMPT TO MOVE THE MATERIAL RECEIVING VEHICLE TO A POSITION WHERE A RECEIVING VEHICLE PARAMETER IS DETECTABLE WITH THE SENSOR ON THE LOADING VEHICLE 294
TO ONE OR BOTH OPERATOR(S) 296
CART FRONT, REAR, NEAR WALL, FAR WALL, ETC. 298
IN FIELD OF VIEW OF IMAGE SENSOR 300
DETECTABLE BY OTHER SENSOR(S) ON THE LEADING VEHICLE 302
ALIGN WITH A KNOWN REFERENCE POINT 304
OTHER 306
DETECT A LOCATION OF THE RECEIVING VEHICLE PARAMETER RELATIVE TO THE SENSOR ON THE LEADING VEHICLE 308
USING IMAGE PROCESSING (ML, DISPARITY IMAGE PROCESSING, ETC.) 310
KNOWN SENSOR LOCATION AND ORIENTATION ON LEADING VEHICLE 312
OPERATOR INTERACTION TO IDENTIFY THE RECEIVING VEHICLE PARAMETER 318
ALIGNMENT WITH PROJECTED MARK 319
OTHER 320
A

FIG. 7A

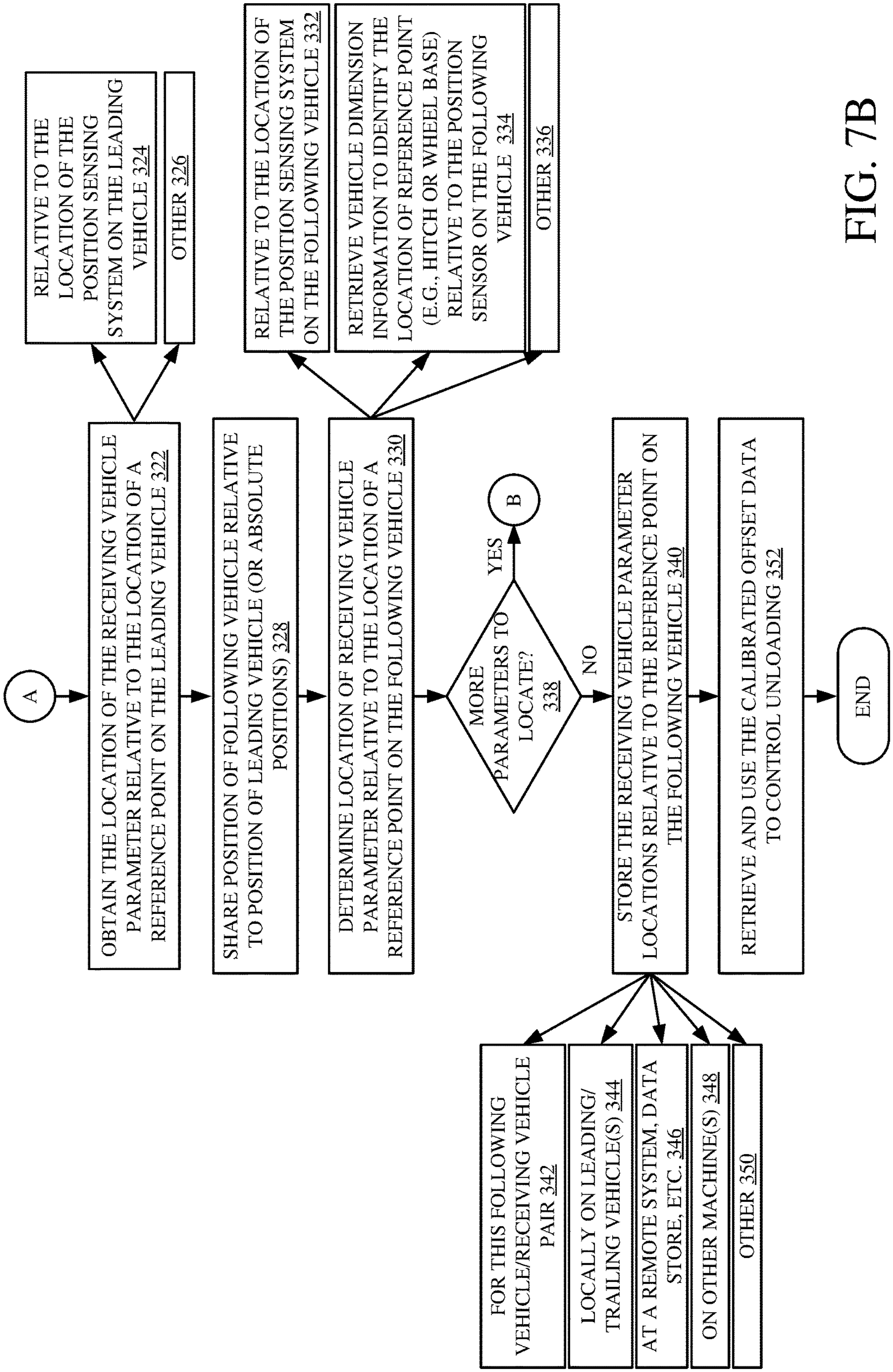

A
OBTAIN THE LOCATION OF THE RECEIVING VEHICLE PARAMETER RELATIVE TO THE LOCATION OF A REFERENCE POINT ON THE LEADING VEHICLE 322
RELATIVE TO THE LOCATION OF THE POSITION SENSING SYSTEM ON THE LEADING VEHICLE 324
OTHER 326
SHARE POSITION OF FOLLOWING VEHICLE RELATIVE TO POSITION OF LEADING VEHICLE (OR ABSOLUTE POSITIONS) 328
DETERMINE LOCATION OF RECEIVING VEHICLE PARAMETER RELATIVE TO THE LOCATION OF A REFERENCE POINT ON THE FOLLOWING VEHICLE 330
RELATIVE TO THE LOCATION OF THE POSITION SENSING SYSTEM ON THE FOLLOWING VEHICLE 332
RETRIEVE VEHICLE DIMENSION INFORMATION TO IDENTIFY THE LOCATION OF REFERENCE POINT (E.G., HITCH OR WHEEL BASE) RELATIVE TO THE POSITION SENSOR ON THE FOLLOWING VEHICLE 334
OTHER 336
MORE PARAMETERS TO LOCATE? 338
YES
B
NO
STORE THE RECEIVING VEHICLE PARAMETER LOCATIONS RELATIVE TO THE REFERENCE POINT ON THE FOLLOWING VEHICLE 340
FOR THIS FOLLOWING VEHICLE/RECEIVING VEHICLE PAIR 342
LOCALLY ON LEADING/ TRAILING VEHICLE(S) 344
AT A REMOTE SYSTEM, DATA STORE, ETC. 346
ON OTHER MACHINE(S) 348
OTHER 350
RETRIEVE AND USE THE CALIBRATED OFFSET DATA TO CONTROL UNLOADING 352
END

FIG. 7B

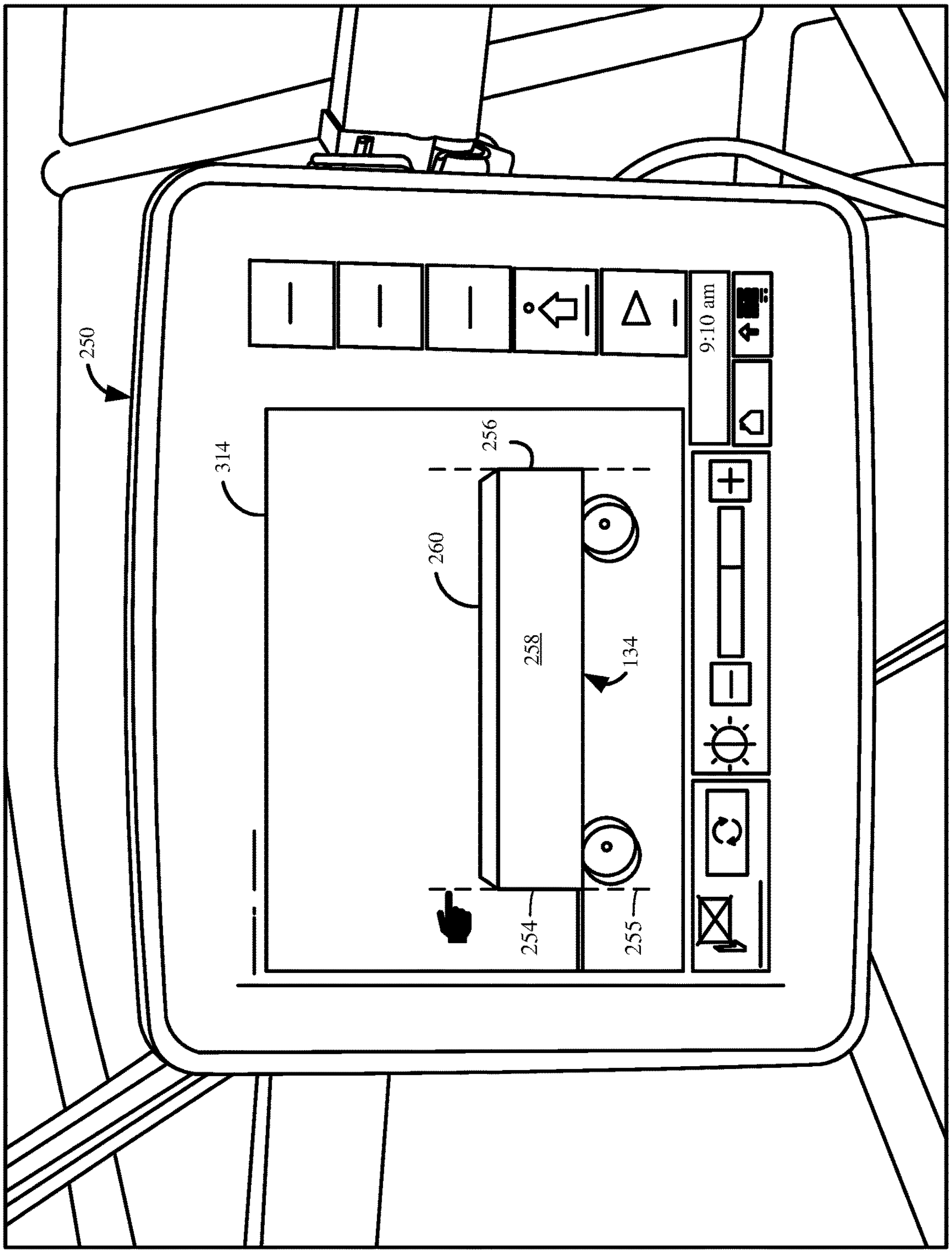


FIG. 8

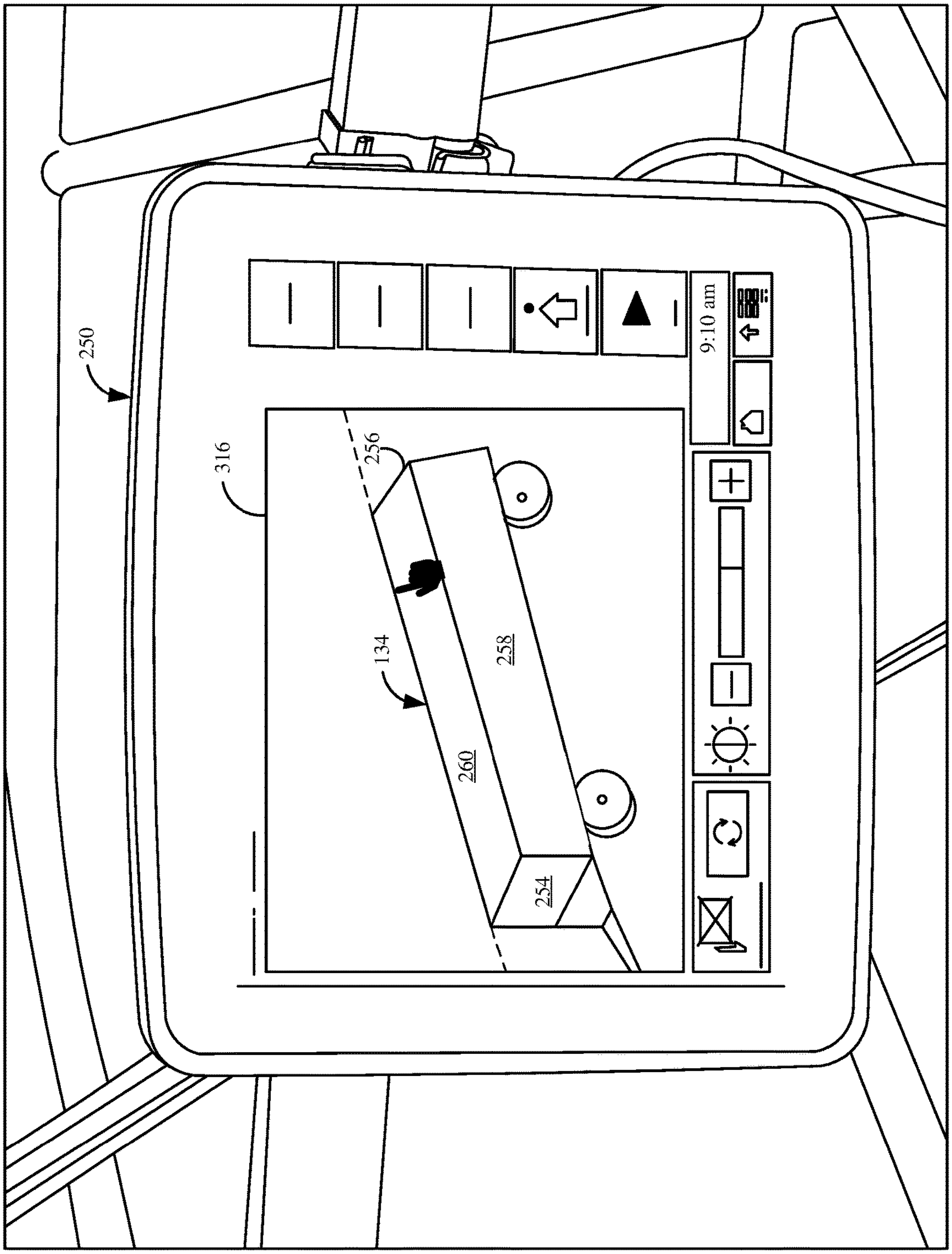
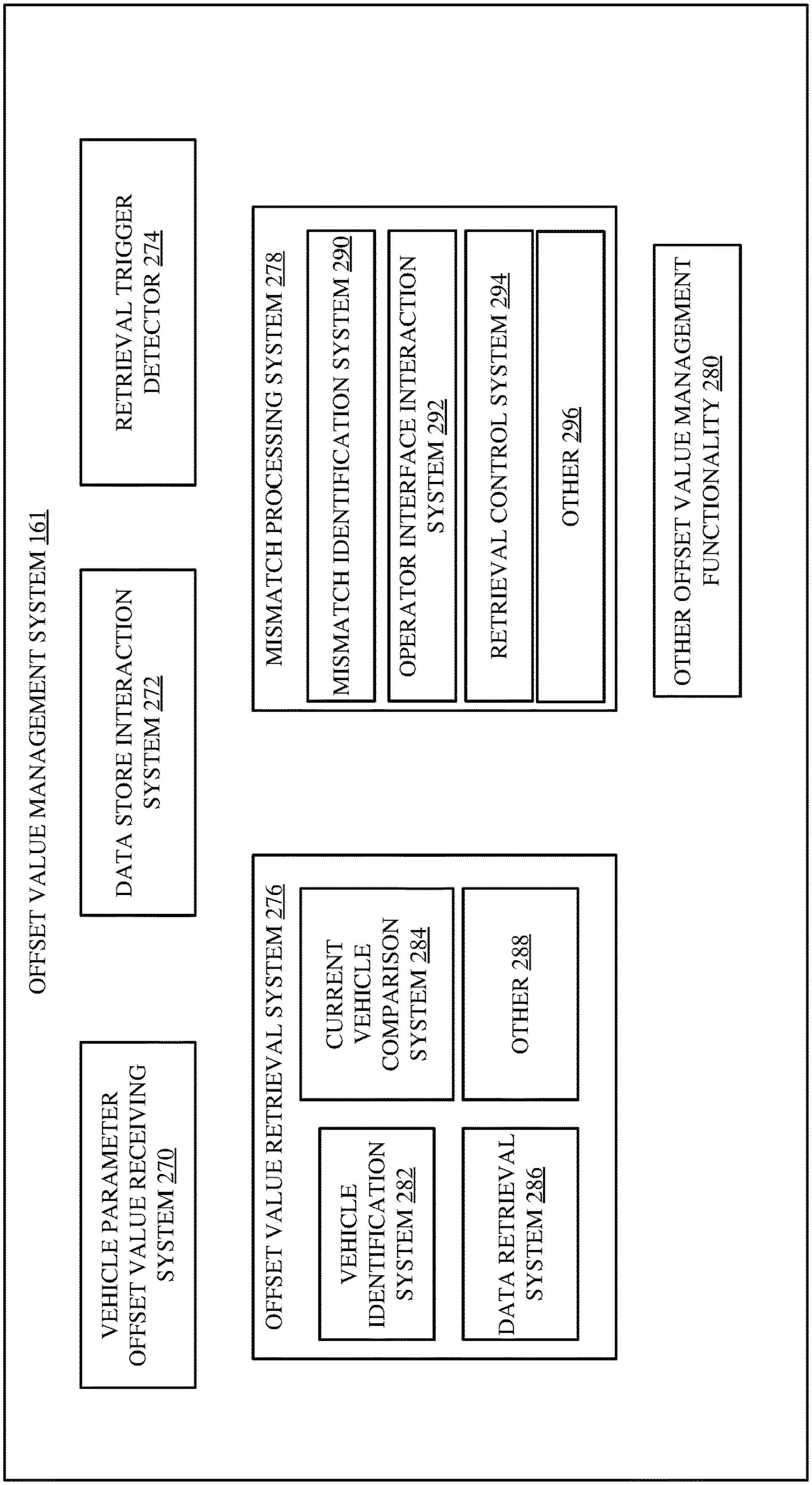


FIG. 9



OFFSET VALUE MANAGEMENT SYSTEM 161
VEHICLE PARAMETER OFFSET VALUE RECEIVING SYSTEM 270
DATA STORE INTERACTION SYSTEM 272
RETRIEVAL TRIGGER DETECTOR 274
OFFSET VALUE RETRIEVAL SYSTEM 276
VEHICLE IDENTIFICATION SYSTEM 282
CURRENT VEHICLE COMPARISON SYSTEM 284
DATA RETRIEVAL SYSTEM 286
OTHER 288
MISMATCH PROCESSING SYSTEM 278
MISMATCH IDENTIFICATION SYSTEM 290
OPERATOR INTERFACE INTERACTION SYSTEM 292
RETRIEVAL CONTROL SYSTEM 294
OTHER 296
OTHER OFFSET VALUE MANAGEMENT FUNCTIONALITY 280

FIG. 10

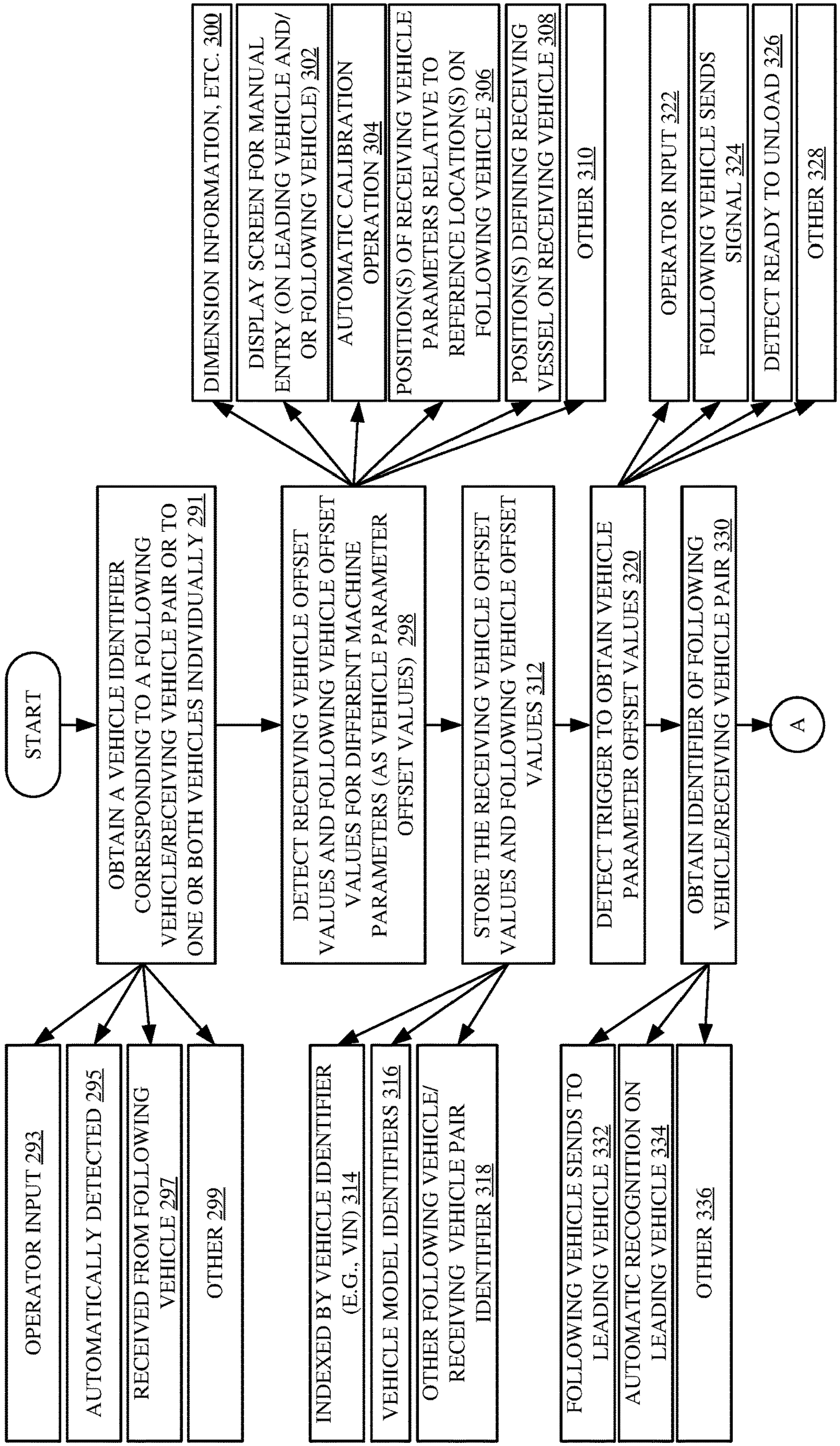

START
OBTAIN A VEHICLE IDENTIFIER CORRESPONDING TO A FOLLOWING VEHICLE/RECEIVING VEHICLE PAIR OR TO ONE OR BOTH VEHICLES INDIVIDUALLY 291
OPERATOR INPUT 293
AUTOMATICALLY DETECTED 295
RECEIVED FROM FOLLOWING VEHICLE 297
OTHER 299
DETECT RECEIVING VEHICLE OFFSET VALUES AND FOLLOWING VEHICLE OFFSET VALUES FOR DIFFERENT MACHINE PARAMETERS (AS VEHICLE PARAMETER OFFSET VALUES) 298
DIMENSION INFORMATION, ETC. 300
DISPLAY SCREEN FOR MANUAL ENTRY (ON LEADING VEHICLE AND/ OR FOLLOWING VEHICLE) 302
AUTOMATIC CALIBRATION OPERATION 304
POSITION(S) OF RECEIVING VEHICLE PARAMETERS RELATIVE TO REFERENCE LOCATION(S) ON FOLLOWING VEHICLE 306
POSITION(S) DEFINING RECEIVING VESSEL ON RECEIVING VEHICLE 308
OTHER 310
STORE THE RECEIVING VEHICLE OFFSET VALUES AND FOLLOWING VEHICLE OFFSET VALUES 312
INDEXED BY VEHICLE IDENTIFIER (E.G., VIN) 314
VEHICLE MODEL IDENTIFIERS 316
OTHER FOLLOWING VEHICLE/ RECEIVING VEHICLE PAIR IDENTIFIER 318
DETECT TRIGGER TO OBTAIN VEHICLE PARAMETER OFFSET VALUES 320
OPERATOR INPUT 322
FOLLOWING VEHICLE SENDS SIGNAL 324
DETECT READY TO UNLOAD 326
OTHER 328
OBTAIN IDENTIFIER OF FOLLOWING VEHICLE/RECEIVING VEHICLE PAIR 330
FOLLOWING VEHICLE SENDS TO LEADING VEHICLE 332
AUTOMATIC RECOGNITION ON LEADING VEHICLE 334
OTHER 336
A

FIG. 11A

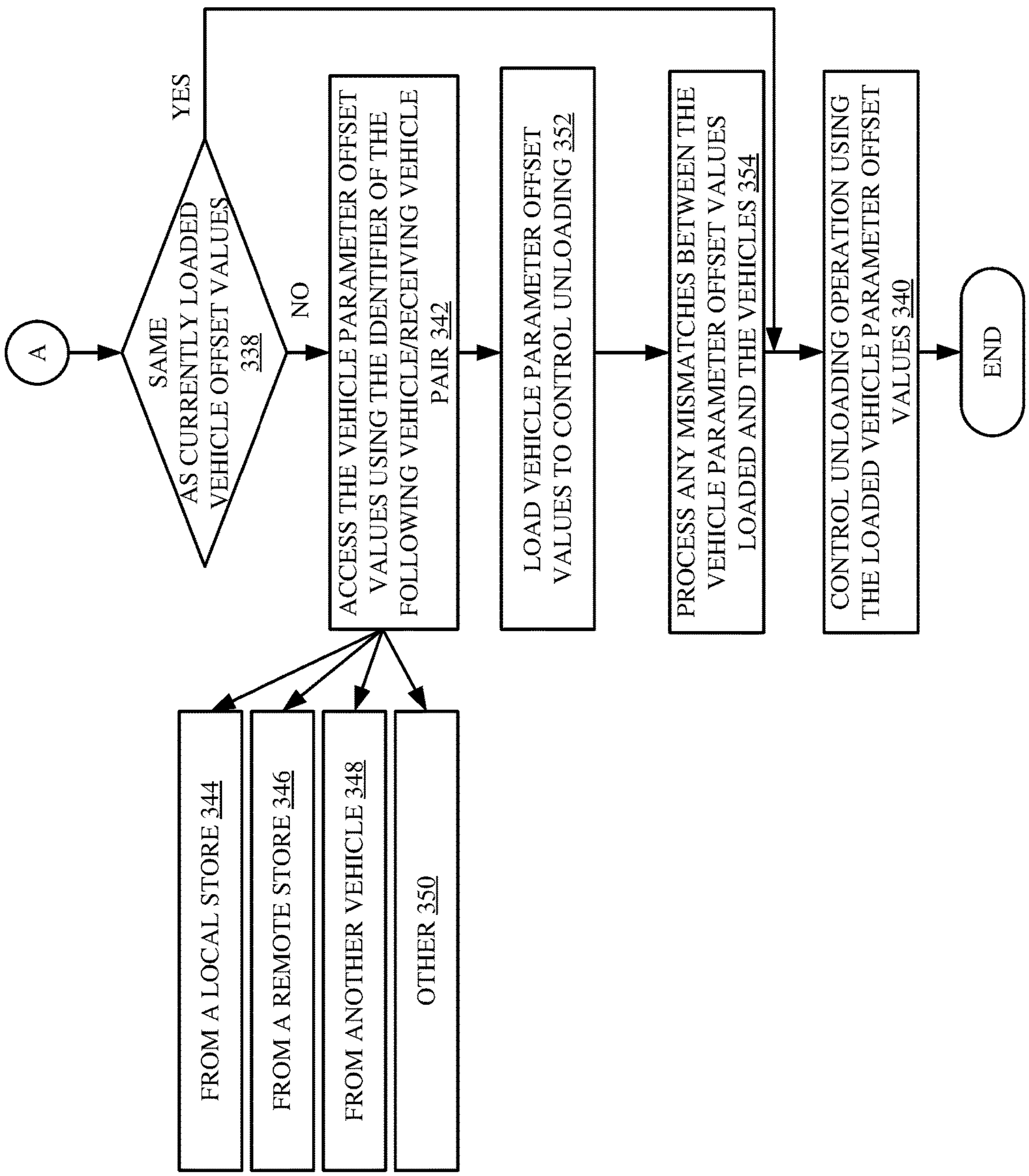

A
SAME AS CURRENTLY LOADED VEHICLE OFFSET VALUES 338
YES
NO
ACCESS THE VEHICLE PARAMETER OFFSET VALUES USING THE IDENTIFIER OF THE FOLLOWING VEHICLE/RECEIVING VEHICLE PAIR 342
FROM A LOCAL STORE 344
FROM A REMOTE STORE 346
FROM ANOTHER VEHICLE 348
OTHER 350
LOAD VEHICLE PARAMETER OFFSET VALUES TO CONTROL UNLOADING 352
PROCESS ANY MISMATCHES BETWEEN THE VEHICLE PARAMETER OFFSET VALUES LOADED AND THE VEHICLES 354
CONTROL UNLOADING OPERATION USING THE LOADED VEHICLE PARAMETER OFFSET VALUES 340
END

FIG. 11B

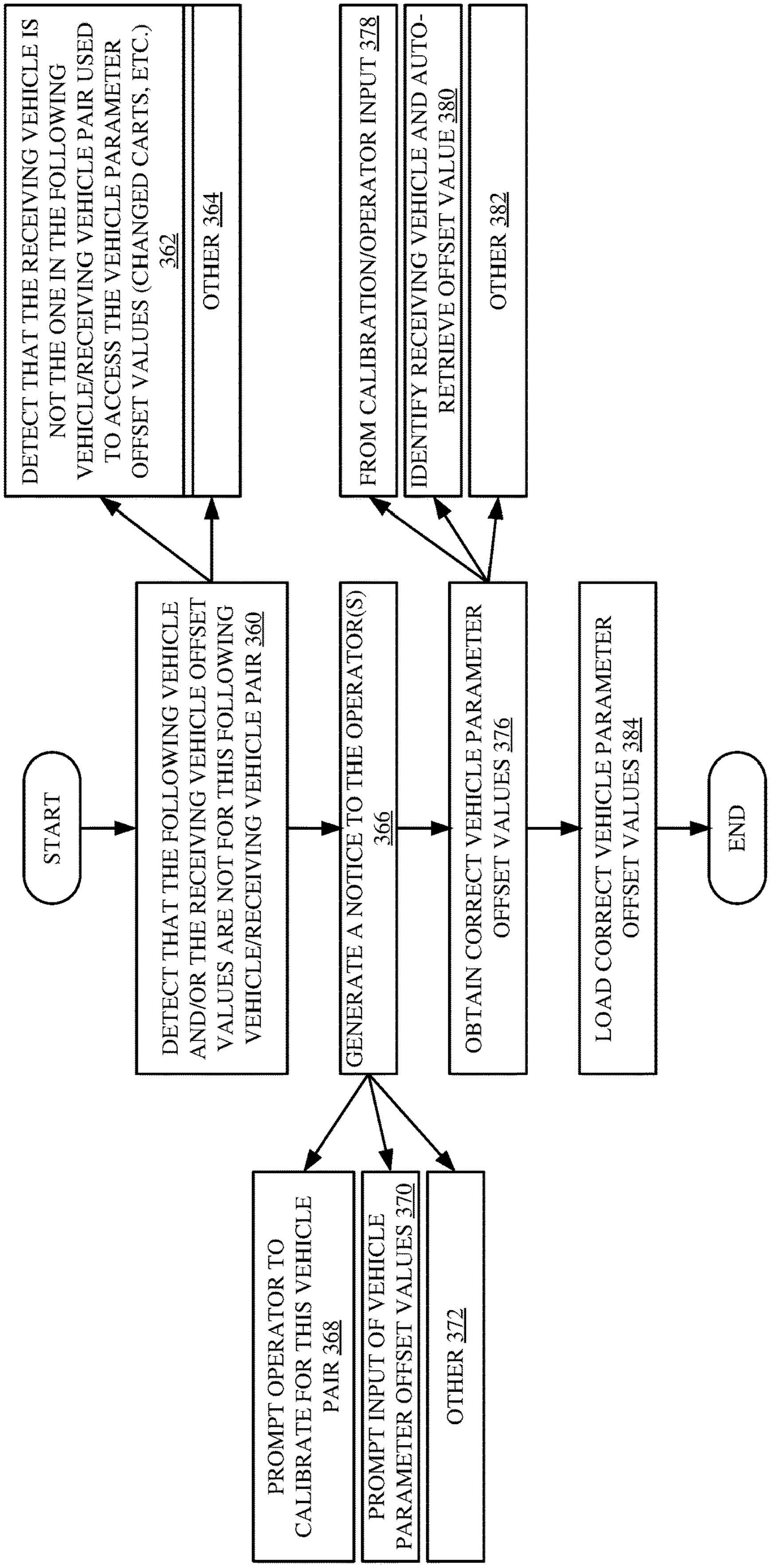

START
DETECT THAT THE FOLLOWING VEHICLE AND/OR THE RECEIVING VEHICLE OFFSET VALUES ARE NOT FOR THIS FOLLOWING VEHICLE/RECEIVING VEHICLE PAIR 360
DETECT THAT THE RECEIVING VEHICLE IS NOT THE ONE IN THE FOLLOWING VEHICLE/RECEIVING VEHICLE PAIR USED TO ACCESS THE VEHICLE PARAMETER OFFSET VALUES (CHANGED CARTS, ETC.) 362
OTHER 364
GENERATE A NOTICE TO THE OPERATOR(S) 366
PROMPT OPERATOR TO CALIBRATE FOR THIS VEHICLE PAIR 368
PROMPT INPUT OF VEHICLE PARAMETER OFFSET VALUES 370
OTHER 372
OBTAIN CORRECT VEHICLE PARAMETER OFFSET VALUES 376
FROM CALIBRATION/OPERATOR INPUT 378
IDENTIFY RECEIVING VEHICLE AND AUTO-RETRIEVE OFFSET VALUE 380
OTHER 382
LOAD CORRECT VEHICLE PARAMETER OFFSET VALUES 384
END

FIG. 12

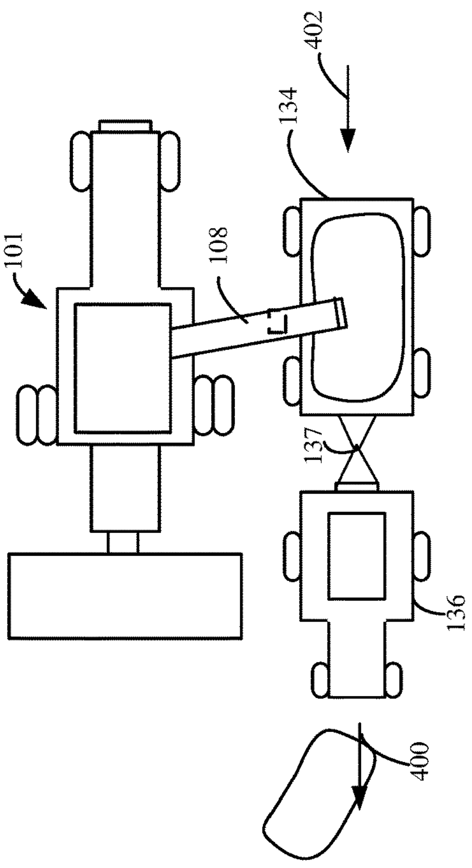


FIG. 13A

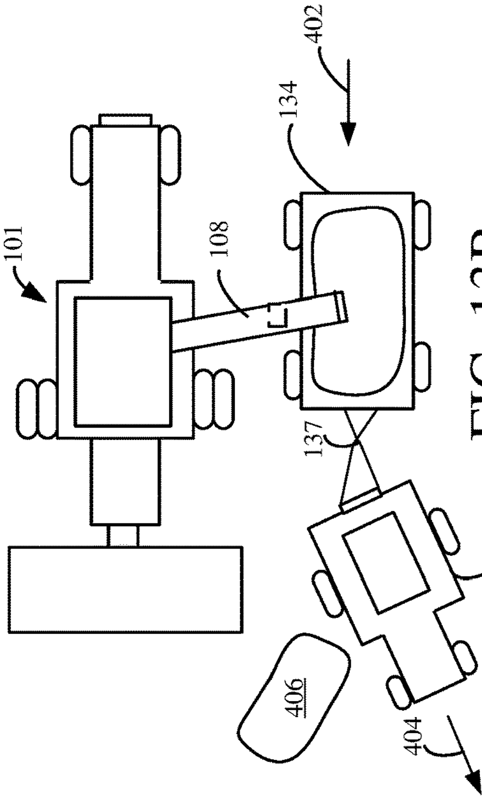


FIG. 13B

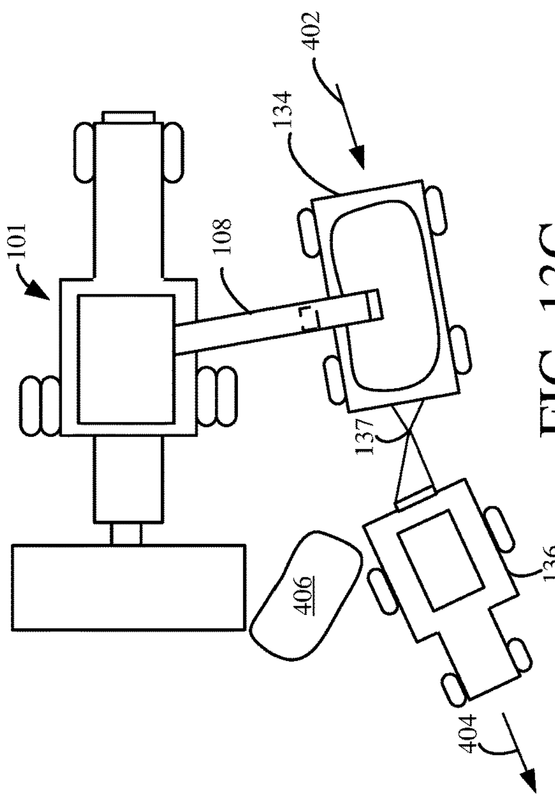


FIG. 13C

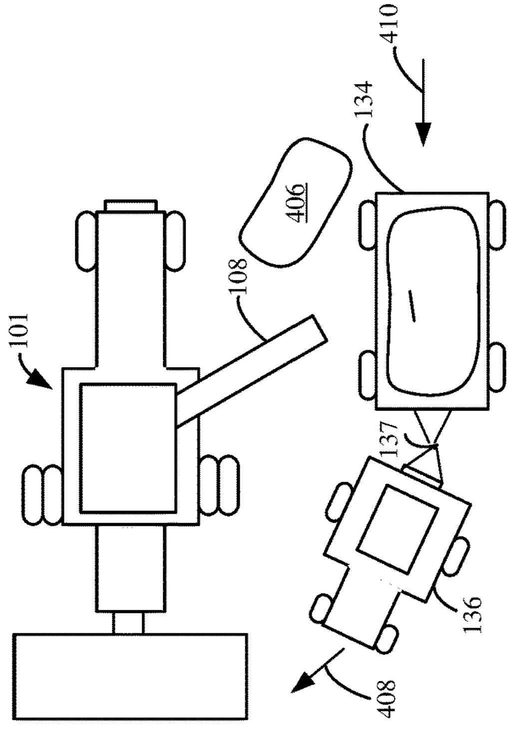


FIG. 13D

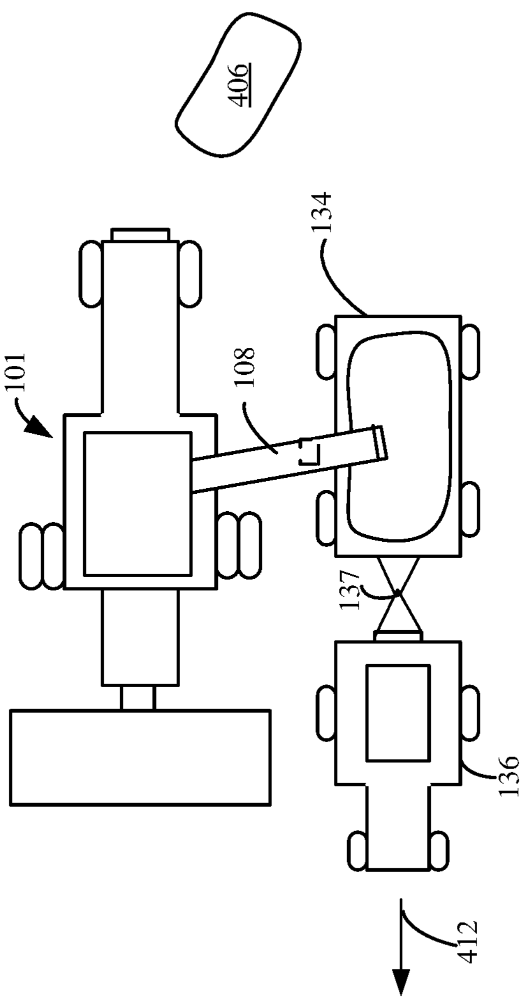


FIG. 13E

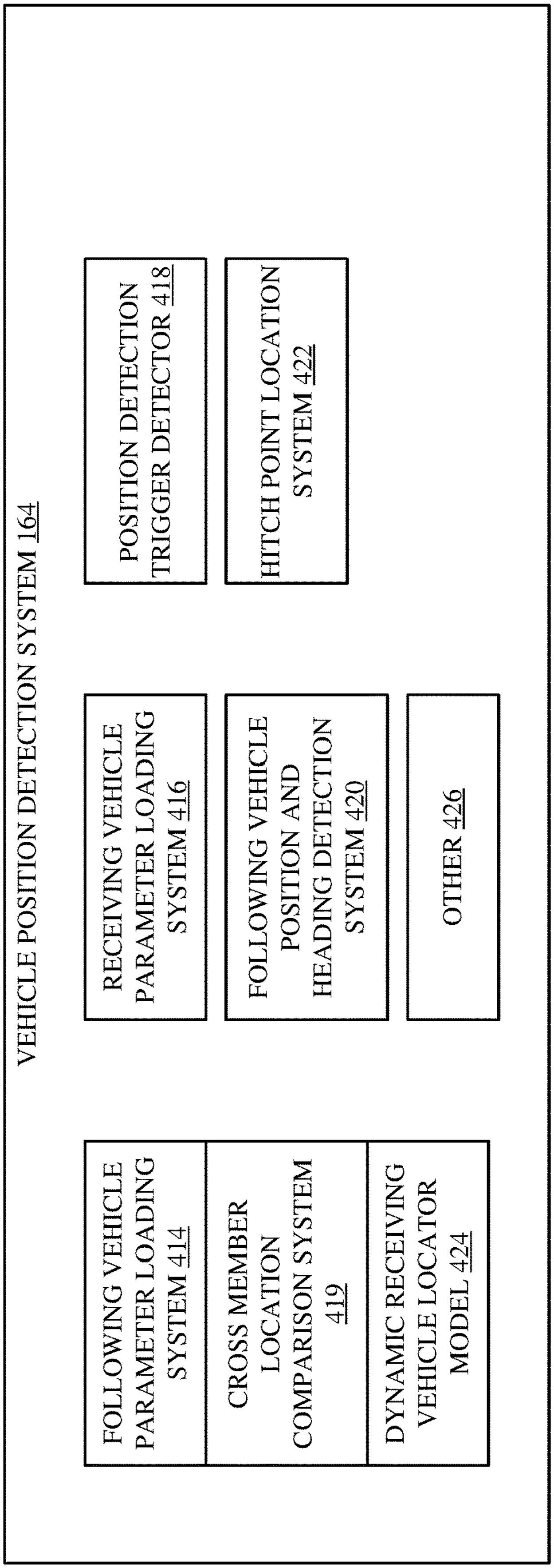

VEHICLE POSITION DETECTION SYSTEM 164
FOLLOWING VEHICLE PARAMETER LOADING SYSTEM 414
CROSS MEMBER LOCATION COMPARISON SYSTEM 419
DYNAMIC RECEIVING VEHICLE LOCATOR MODEL 424
RECEIVING VEHICLE PARAMETER LOADING SYSTEM 416
FOLLOWING VEHICLE POSITION AND HEADING DETECTION SYSTEM 420
OTHER 426
POSITION DETECTION TRIGGER DETECTOR 418
HITCH POINT LOCATION SYSTEM 422

FIG. 14

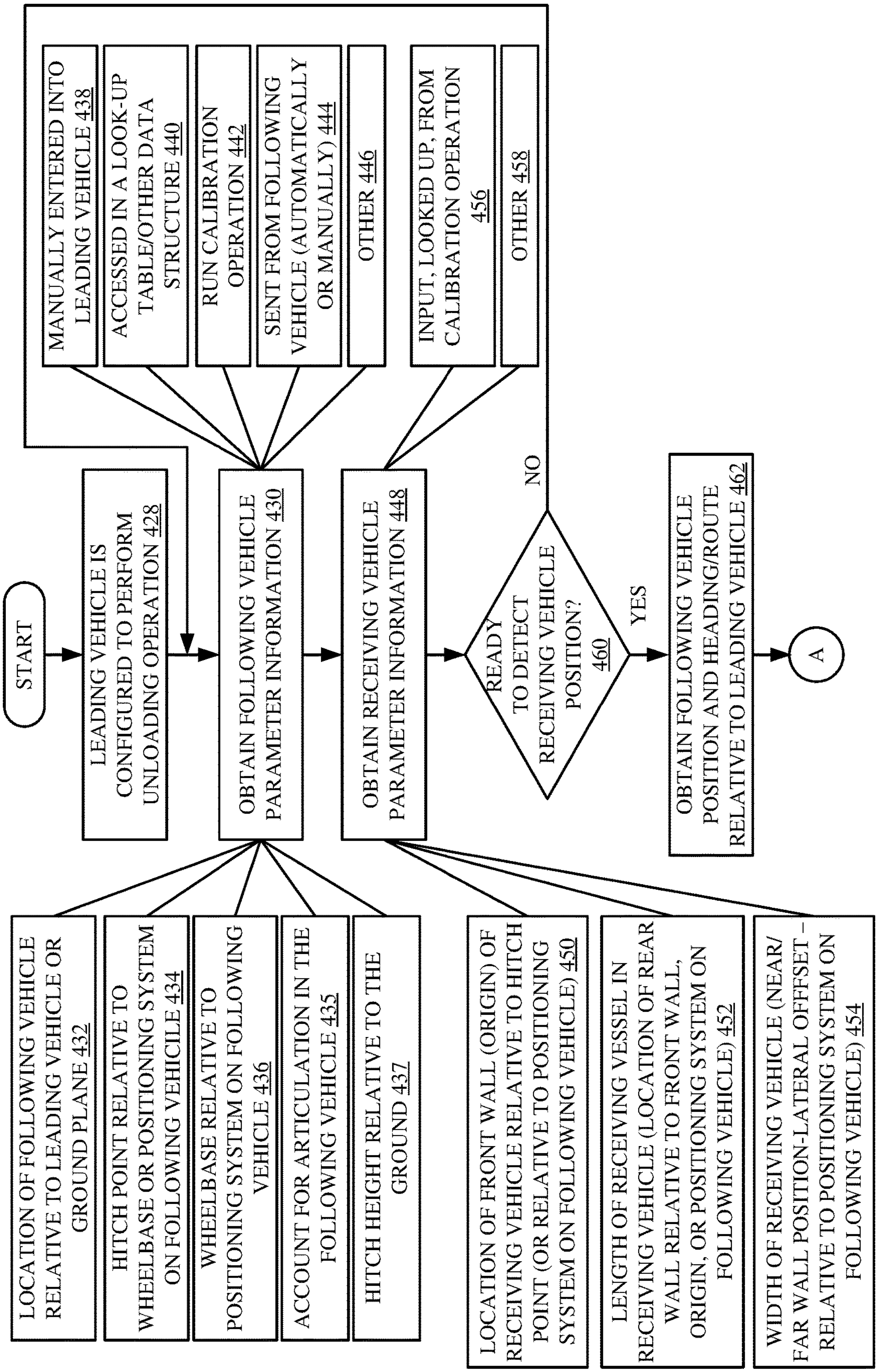

START
LEADING VEHICLE IS CONFIGURED TO PERFORM UNLOADING OPERATION 428
OBTAIN FOLLOWING VEHICLE PARAMETER INFORMATION 430
LOCATION OF FOLLOWING VEHICLE RELATIVE TO LEADING VEHICLE OR GROUND PLANE 432
HITCH POINT RELATIVE TO WHEELBASE OR POSITIONING SYSTEM ON FOLLOWING VEHICLE 434
WHEELBASE RELATIVE TO POSITIONING SYSTEM ON FOLLOWING VEHICLE 436
ACCOUNT FOR ARTICULATION IN THE FOLLOWING VEHICLE 435
HITCH HEIGHT RELATIVE TO THE GROUND 437
MANUALLY ENTERED INTO LEADING VEHICLE 438
ACCESSED IN A LOOK-UP TABLE/OTHER DATA STRUCTURE 440
RUN CALIBRATION OPERATION 442
SENT FROM FOLLOWING VEHICLE (AUTOMATICALLY OR MANUALLY) 444
OTHER 446
OBTAIN RECEIVING VEHICLE PARAMETER INFORMATION 448
LOCATION OF FRONT WALL (ORIGIN) OF RECEIVING VEHICLE RELATIVE TO HITCH POINT (OR RELATIVE TO POSITIONING SYSTEM ON FOLLOWING VEHICLE) 450
LENGTH OF RECEIVING VESSEL IN RECEIVING VEHICLE (LOCATION OF REAR WALL RELATIVE TO FRONT WALL, ORIGIN, OR POSITIONING SYSTEM ON FOLLOWING VEHICLE) 452
WIDTH OF RECEIVING VEHICLE (NEAR/ FAR WALL POSITION-LATERAL OFFSET – RELATIVE TO POSITIONING SYSTEM ON FOLLOWING VEHICLE) 454
INPUT, LOOKED UP, FROM CALIBRATION OPERATION 456
OTHER 458
READY TO DETECT RECEIVING VEHICLE POSITION? 460
NO
YES
OBTAIN FOLLOWING VEHICLE POSITION AND HEADING/ROUTE RELATIVE TO LEADING VEHICLE 462
A

FIG. 15A

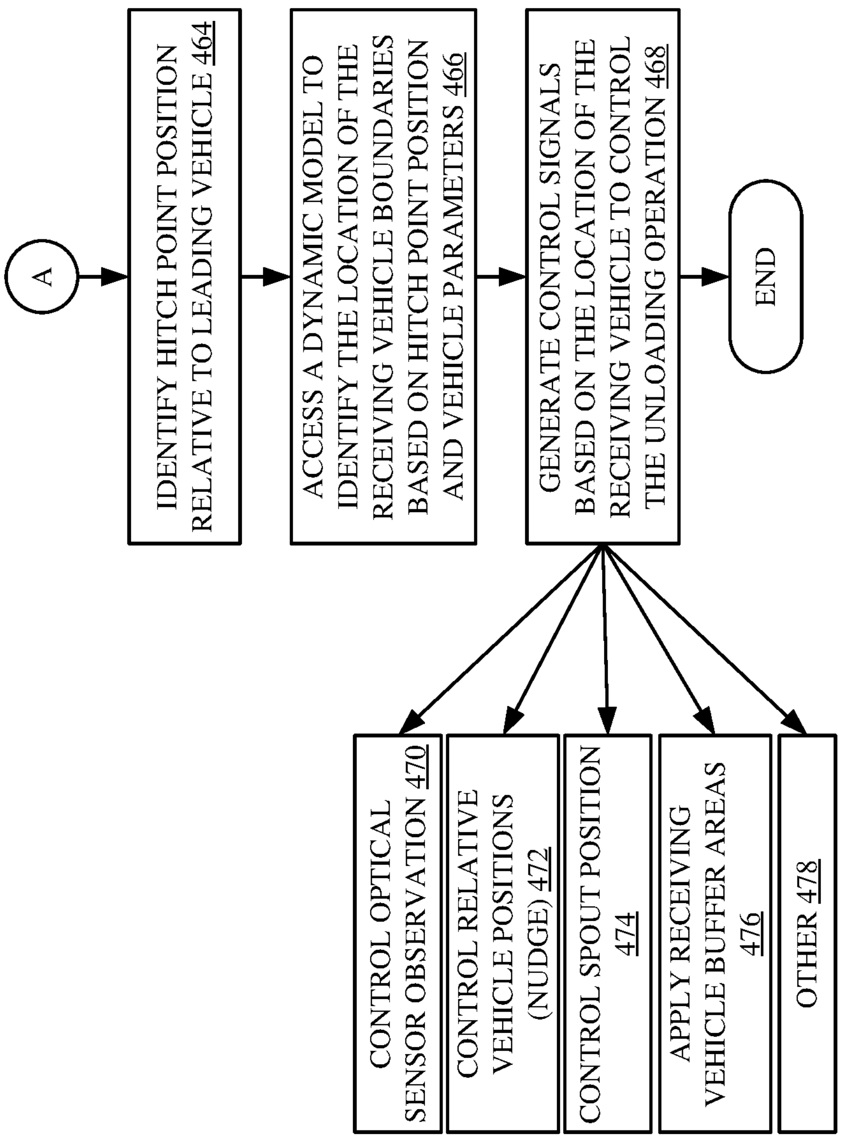

A
IDENTIFY HITCH POINT POSITION RELATIVE TO LEADING VEHICLE 464
ACCESS A DYNAMIC MODEL TO IDENTIFY THE LOCATION OF THE RECEIVING VEHICLE BOUNDARIES BASED ON HITCH POINT POSITION AND VEHICLE PARAMETERS 466
GENERATE CONTROL SIGNALS BASED ON THE LOCATION OF THE RECEIVING VEHICLE TO CONTROL THE UNLOADING OPERATION 468
END
CONTROL OPTICAL SENSOR OBSERVATION 470
CONTROL RELATIVE VEHICLE POSITIONS (NUDGE) 472
CONTROL SPOUT POSITION 474
APPLY RECEIVING VEHICLE BUFFER AREAS 476
OTHER 478

FIG. 15B

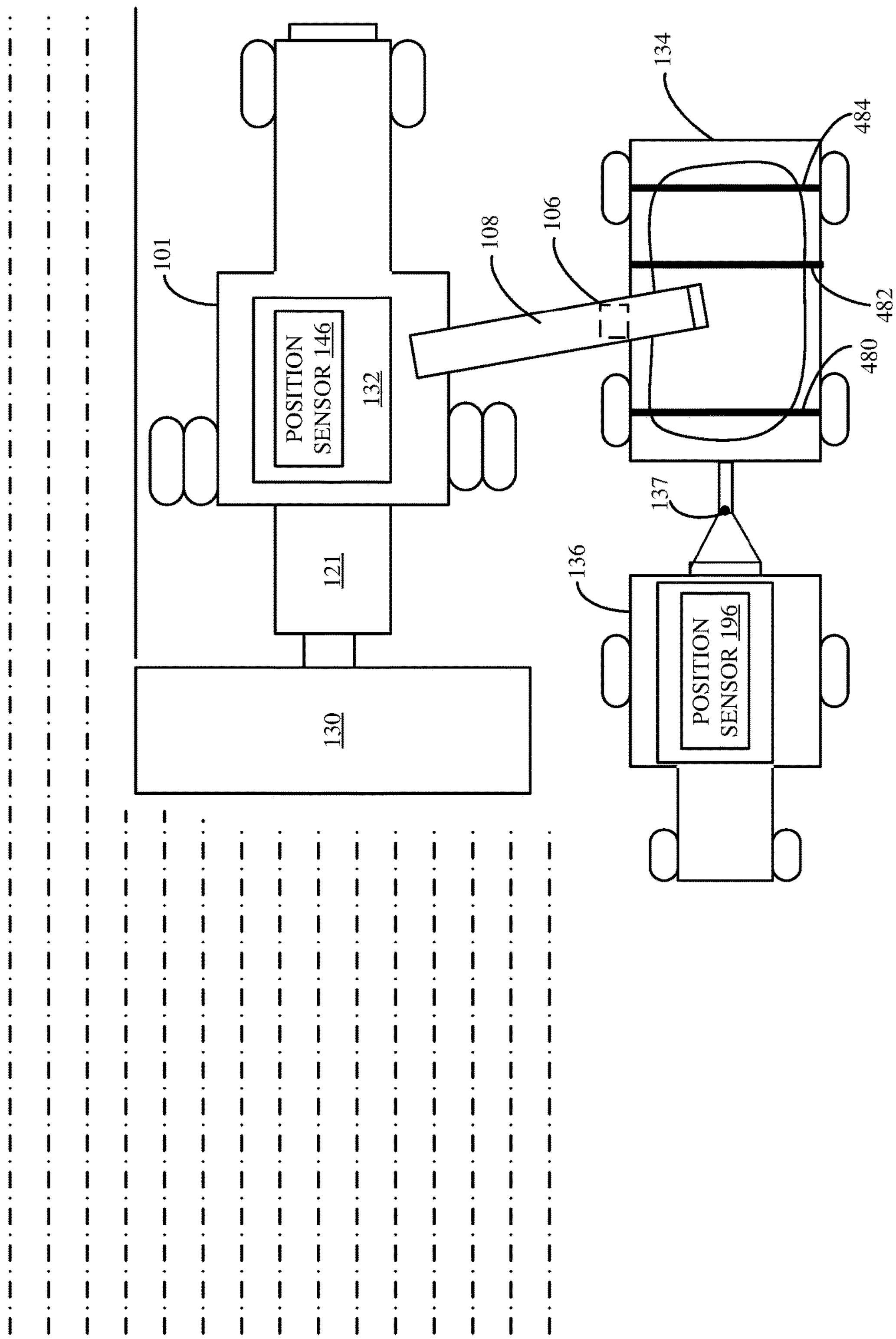

101
POSITION SENSOR 146
132
121
130
108
106
134
484
482
480
137
136
POSITION SENSOR 196

FIG. 16

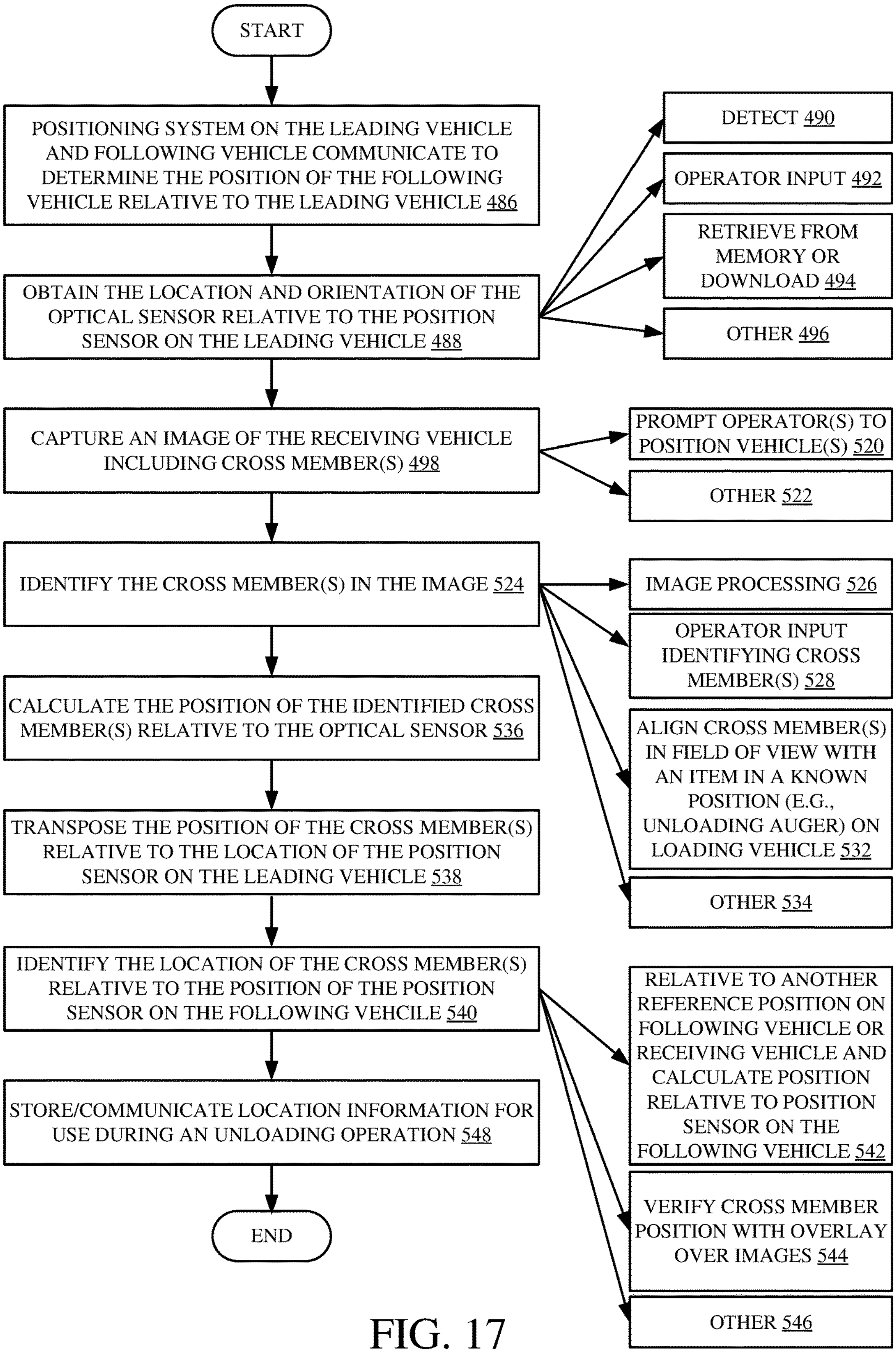

START
POSITIONING SYSTEM ON THE LEADING VEHICLE AND FOLLOWING VEHICLE COMMUNICATE TO DETERMINE THE POSITION OF THE FOLLOWING VEHICLE RELATIVE TO THE LEADING VEHICLE 486
OBTAIN THE LOCATION AND ORIENTATION OF THE OPTICAL SENSOR RELATIVE TO THE POSITION SENSOR ON THE LEADING VEHICLE 488
DETECT 490
OPERATOR INPUT 492
RETRIEVE FROM MEMORY OR DOWNLOAD 494
OTHER 496
CAPTURE AN IMAGE OF THE RECEIVING VEHICLE INCLUDING CROSS MEMBER(S) 498
PROMPT OPERATOR(S) TO POSITION VEHICLE(S) 520
OTHER 522
IDENTIFY THE CROSS MEMBER(S) IN THE IMAGE 524
IMAGE PROCESSING 526
OPERATOR INPUT IDENTIFYING CROSS MEMBER(S) 528
ALIGN CROSS MEMBER(S) IN FIELD OF VIEW WITH AN ITEM IN A KNOWN POSITION (E.G., UNLOADING AUGER) ON LOADING VEHICLE 532
OTHER 534
CALCULATE THE POSITION OF THE IDENTIFIED CROSS MEMBER(S) RELATIVE TO THE OPTICAL SENSOR 536
TRANSPOSE THE POSITION OF THE CROSS MEMBER(S) RELATIVE TO THE LOCATION OF THE POSITION SENSOR ON THE LEADING VEHICLE 538
IDENTIFY THE LOCATION OF THE CROSS MEMBER(S) RELATIVE TO THE POSITION OF THE POSITION SENSOR ON THE FOLLOWING VEHCILE 540
RELATIVE TO ANOTHER REFERENCE POSITION ON FOLLOWING VEHICLE OR RECEIVING VEHICLE AND CALCULATE POSITION RELATIVE TO POSITION SENSOR ON THE FOLLOWING VEHICLE 542
VERIFY CROSS MEMBER POSITION WITH OVERLAY OVER IMAGES 544
OTHER 546
STORE/COMMUNICATE LOCATION INFORMATION FOR USE DURING AN UNLOADING OPERATION 548
END

FIG. 17

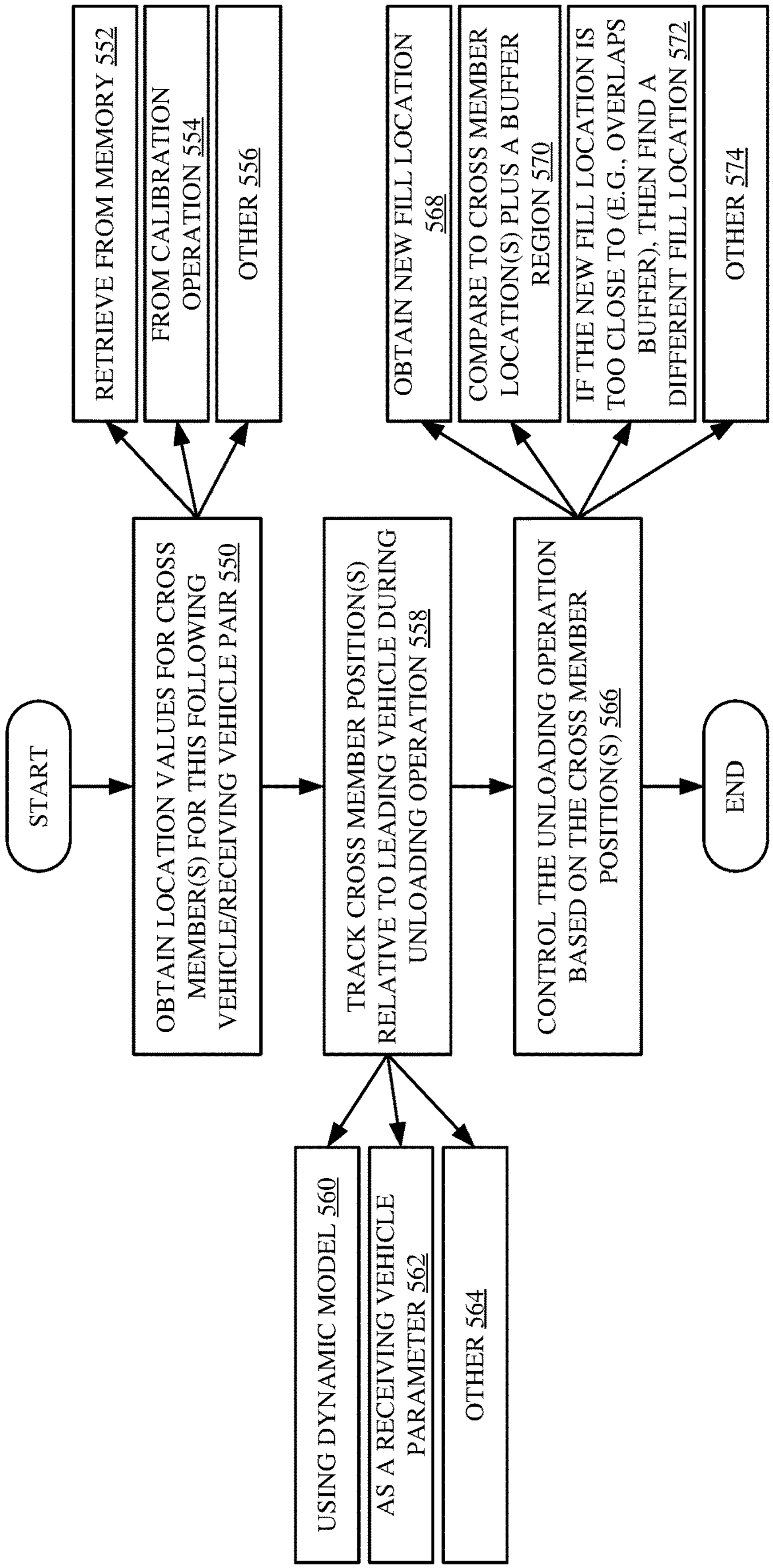

START
OBTAIN LOCATION VALUES FOR CROSS MEMBER(S) FOR THIS FOLLOWING VEHICLE/RECEIVING VEHICLE PAIR 550
RETRIEVE FROM MEMORY 552
FROM CALIBRATION OPERATION 554
OTHER 556
TRACK CROSS MEMBER POSITION(S) RELATIVE TO LEADING VEHICLE DURING UNLOADING OPERATION 558
USING DYNAMIC MODEL 560
AS A RECEIVING VEHICLE PARAMETER 562
OTHER 564
CONTROL THE UNLOADING OPERATION BASED ON THE CROSS MEMBER POSITION(S) 566
OBTAIN NEW FILL LOCATION 568
COMPARE TO CROSS MEMBER LOCATION(S) PLUS A BUFFER REGION 570
IF THE NEW FILL LOCATION IS TOO CLOSE TO (E.G., OVERLAPS BUFFER), THEN FIND A DIFFERENT FILL LOCATION 572
OTHER 574
END

FIG. 18

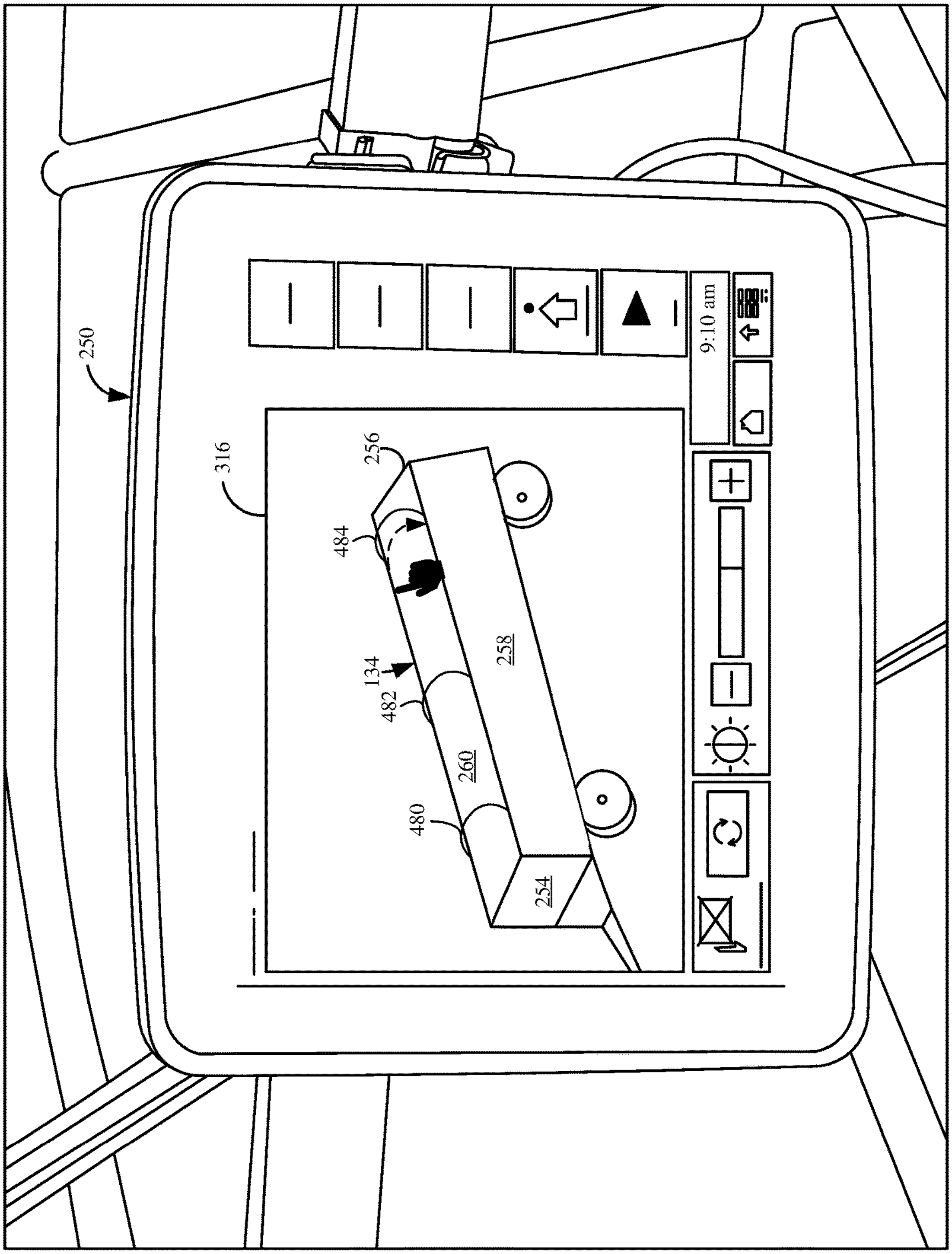


FIG. 19

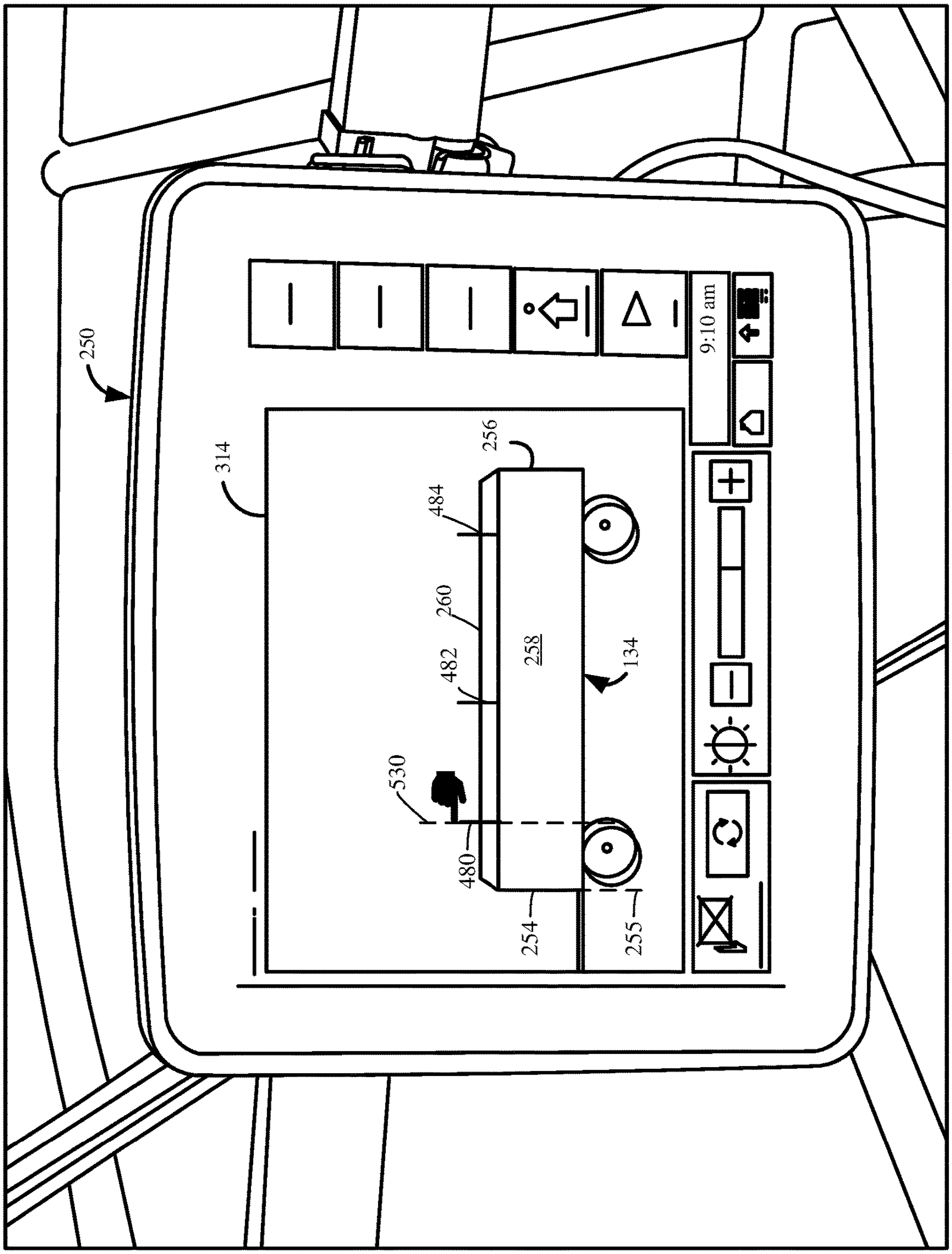


FIG. 20

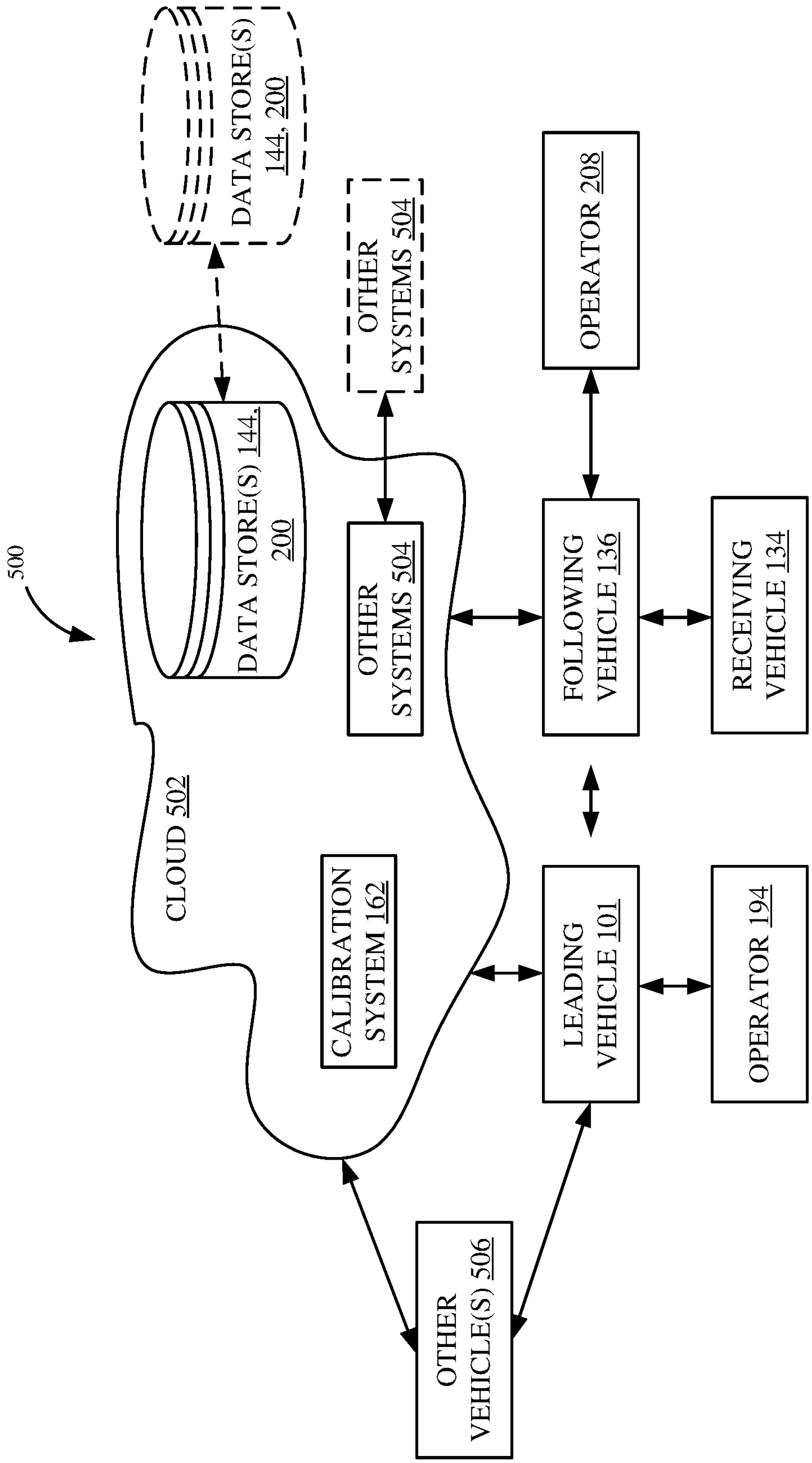

500
DATA STORE(S) 144, 200
OTHER SYSTEMS 504
OPERATOR 208
DATA STORE(S) 144, 200
OTHER SYSTEMS 504
FOLLOWING VEHICLE 136
RECEIVING VEHICLE 134
CLOUD 502
CALIBRATION SYSTEM 162
LEADING VEHICLE 101
OPERATOR 194
OTHER VEHICLE(S) 506

FIG. 21

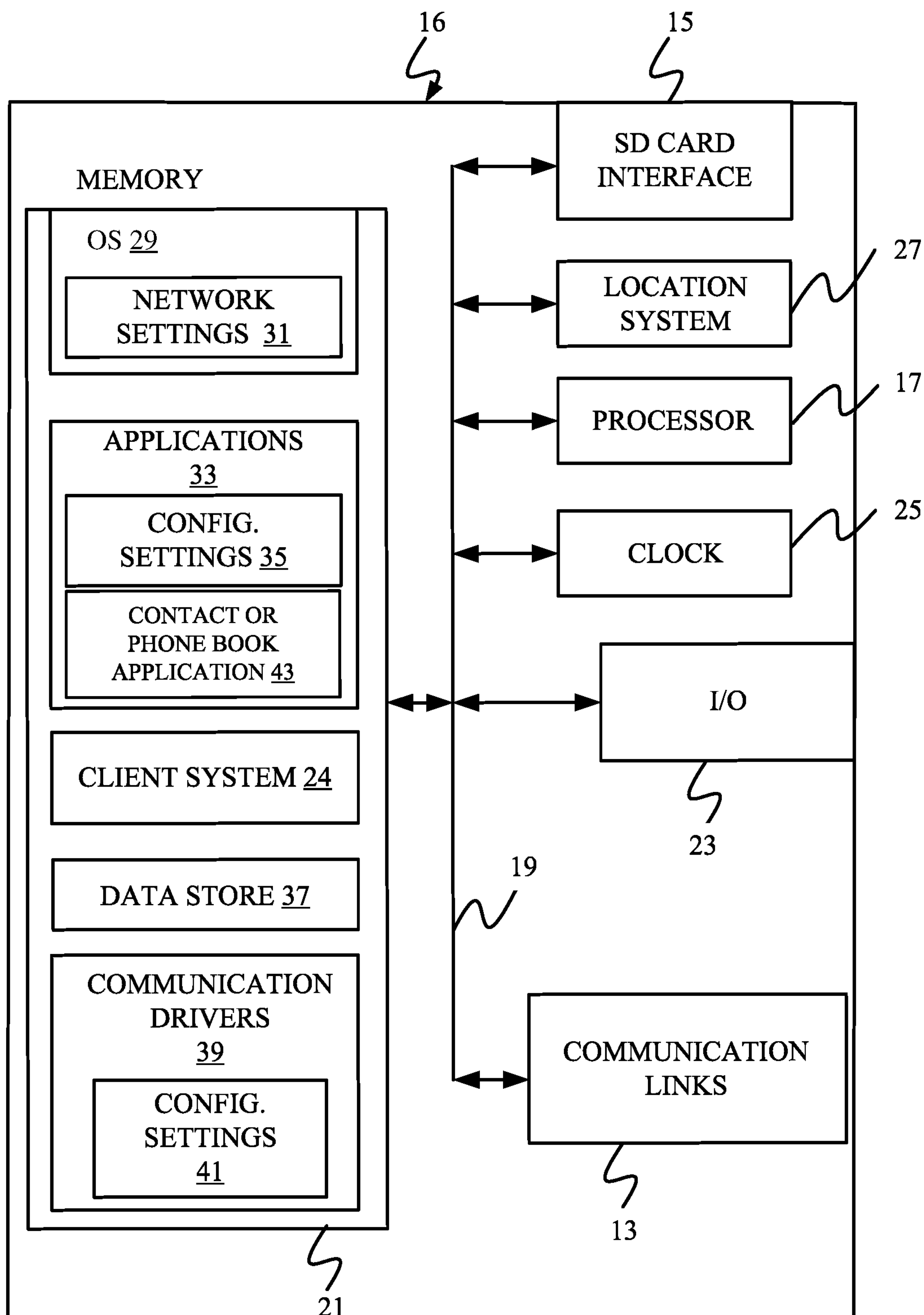

16
15
MEMORY
OS 29
NETWORK SETTINGS 31
APPLICATIONS 33
CONFIG. SETTINGS 35
CONTACT OR PHONE BOOK APPLICATION 43
CLIENT SYSTEM 24
DATA STORE 37
COMMUNICATION DRIVERS 39
CONFIG. SETTINGS 41
21
SD CARD INTERFACE
LOCATION SYSTEM
27
PROCESSOR
17
CLOCK
25
I/O
23
19
COMMUNICATION LINKS
13

FIG. 22

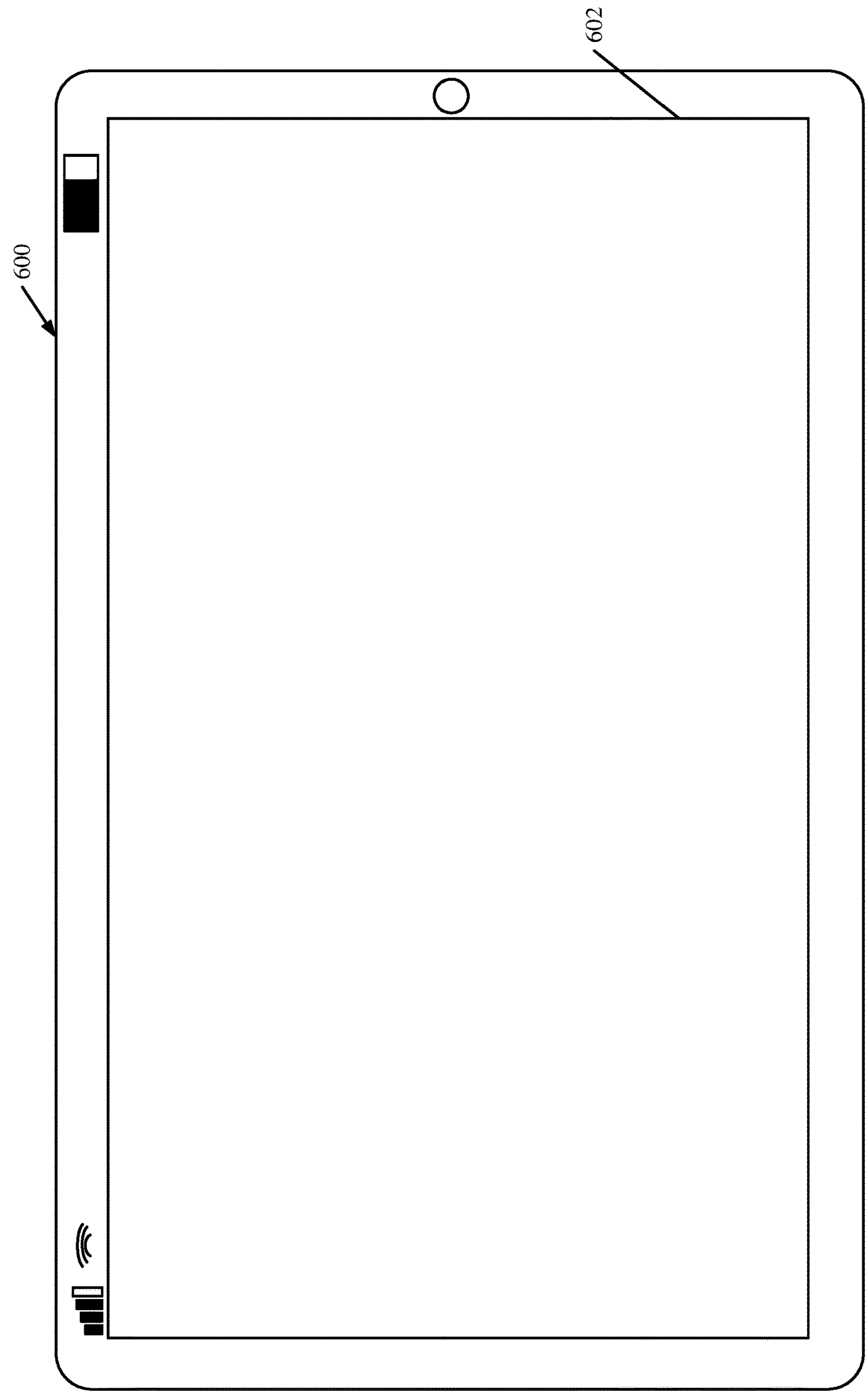


FIG. 23

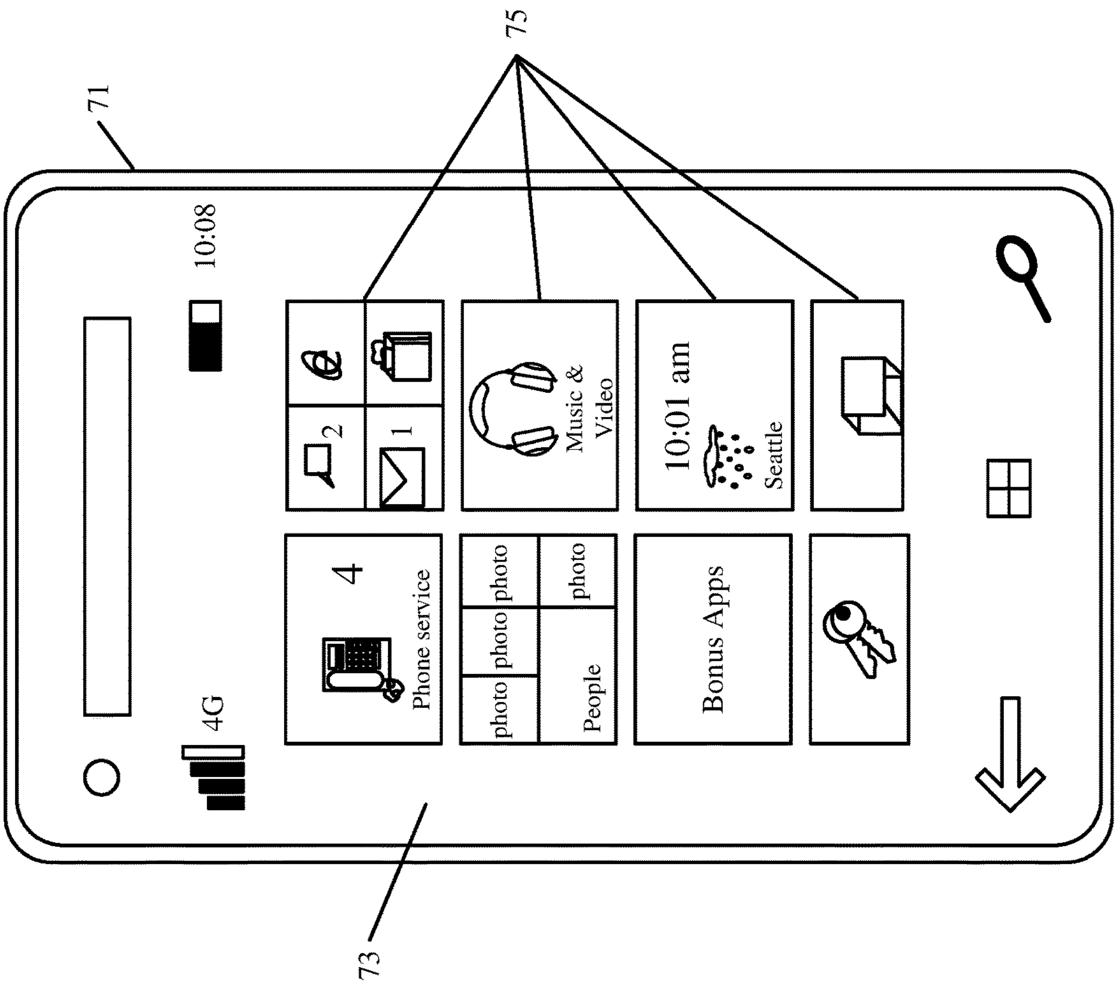
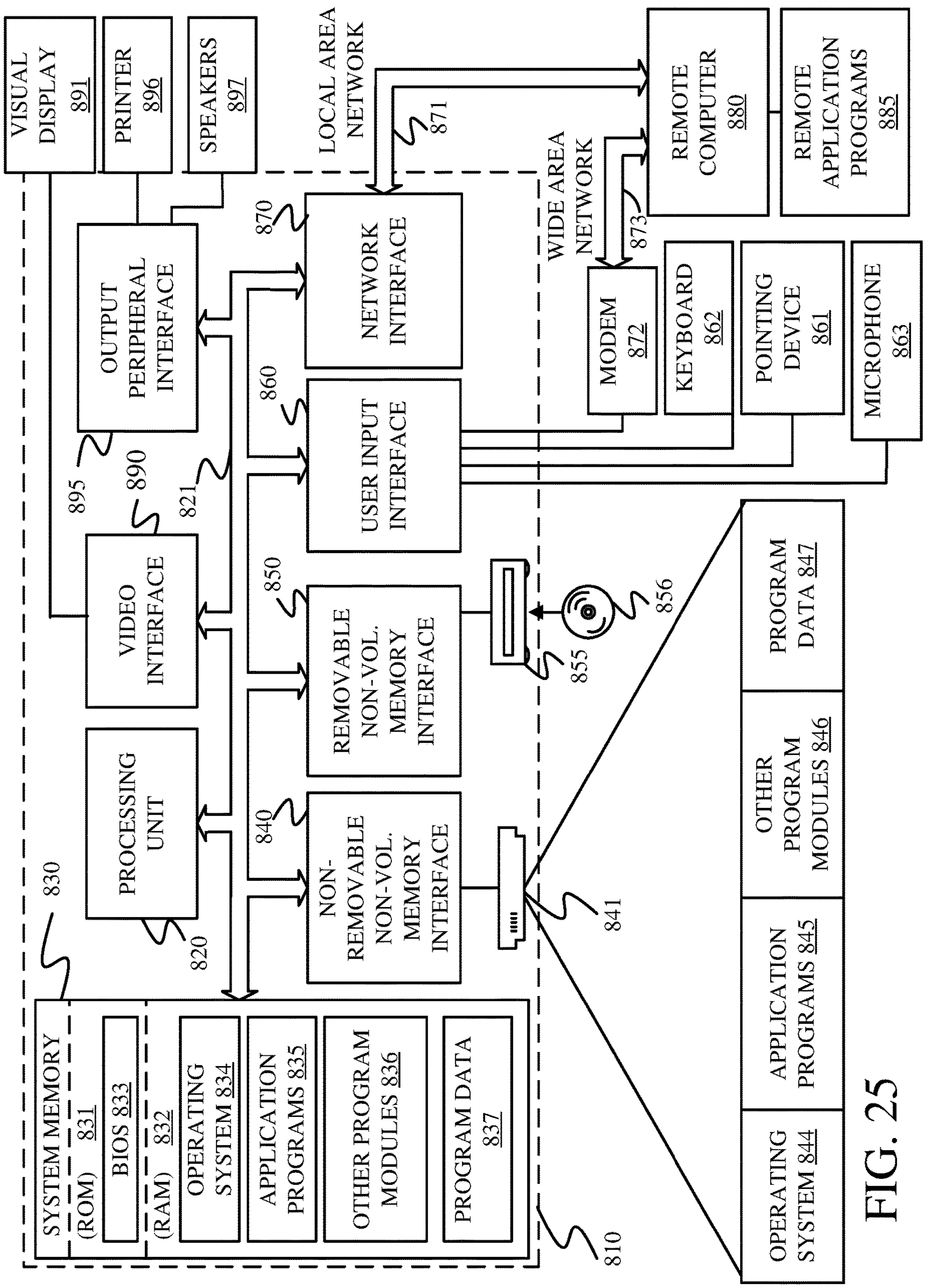

71
75
10:08
4G
2
1
Music & Video
10:01 am
Seattle
4
Phone service
photo
photo
photo
photo
People
Bonus Apps
73

FIG. 24



VISUAL DISPLAY 891
PRINTER 896
SPEAKERS 897
LOCAL AREA NETWORK
871
REMOTE COMPUTER 880
REMOTE APPLICATION PROGRAMS 885
WIDE AREA NETWORK
873
870
NETWORK INTERFACE
OUTPUT PERIPHERAL INTERFACE
895
890
821
860
USER INPUT INTERFACE
MODEM 872
KEYBOARD 862
POINTING DEVICE 861
MICROPHONE 863
VIDEO INTERFACE
850
REMOVABLE NON-VOL. MEMORY INTERFACE
855
856
PROGRAM DATA 847
OTHER PROGRAM MODULES 846
APPLICATION PROGRAMS 845
OPERATING SYSTEM 844
830
PROCESSING UNIT
820
840
NON-REMOVABLE NON-VOL. MEMORY INTERFACE
841
SYSTEM MEMORY
(ROM) 831
BIOS 833
(RAM) 832
OPERATING SYSTEM 834
APPLICATION PROGRAMS 835
OTHER PROGRAM MODULES 836
PROGRAM DATA 837
810

FIG. 25

CROSS MEMBER LOCATION AND AVOIDANCE DURING AN UNLOADING OPERATION

CROSS-REFERENCE TO RELATED APPLICATION

The present application is based on and claims the benefit of U.S. provisional patent application Ser. No. 63/381,178, filed Oct. 27, 2022, and U.S. provisional patent application Ser. No. 63/381,187, filed Oct. 27, 2022, the content of which is hereby incorporated by reference in its entirety.

FIELD OF THE DESCRIPTION

The present description generally relates to machines that load material into receiving vehicles, such as harvesting machines that fill carts, semitrailers, or other agricultural receiving vehicles. More specifically, but not by limitation, the present description relates to automated control of an unloading operation with automatic receiving vehicle cross member location.

BACKGROUND

There are a wide variety of different types of vehicles that load material into other vehicles. Some such vehicles include agricultural vehicles such as forage harvesters or other harvesters (such as combine harvesters, sugarcane harvesters, silage harvesters, etc.), that harvest grain or other crop. Such harvesters often unload material into carts, which may be pulled by tractors, or semitrailers, as the harvesters are moving. Other vehicles that unload into receiving vehicles include construction vehicles, such as cold planers that unload into a dump truck, and other vehicles.

Taking an agricultural harvester as an example, while harvesting in a field using a forage harvester or combine harvester, an operator attempts to control the harvester to maintain harvesting efficiency, during many different types of conditions. The soil conditions, crop conditions, etc. can all change. This may result in the operator changing control settings. This means the operator needs to devote a relatively large amount of attention to controlling the forage harvester or combine harvester.

At the same time, a semitruck or tractor-pulled cart (a receiving vehicle), is often in position relative to the harvester (e.g., alongside the harvester or behind the harvester) so that the harvester can fill the truck or cart, while moving through the field. In some current systems, this requires the operator of the harvester to control the position of the unloading spout and flap so that the truck or cart is filled evenly, but not over filled. Even a momentary misalignment between the spout and the truck or cart may result in hundreds of pounds of harvested material being dumped on the ground, rather than in the truck or cart.

The discussion above is merely provided for general background information and is not intended to be used as an aid in determining the scope of the claimed subject matter.

SUMMARY

A cross member on a receiving vehicle which is coupled to a following vehicle, is located relative to a leading vehicle. The location of the cross member is tracked during an unloading operation so the leading vehicle avoids the cross member when unloading material into the receiving vehicle.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter. The claimed subject matter is not limited to implementations that solve any or all disadvantages noted in the background.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a pictorial illustration of one example of a forage harvester filling a tractor-pulled receiving vehicle, with the receiving vehicle following the forage harvester.

FIG. **2** is a pictorial illustration of one example of a forage harvester filling a receiving vehicle that is alongside the forage harvester.

FIG. **3** shows a combine harvester filling a receiving vehicle.

FIG. **4** is a block diagram of one example of an agricultural system.

FIG. **5** is a block diagram of one example of a calibration system.

FIG. **6** is one example of an operator interface display.

FIGS. **7**A and **7**B (collectively referred to as FIG. **7**) show a flow diagram illustrating an example operation of a harvesting machine.

FIGS. **8** and **9** show examples of operator interface displays.

FIG. **10** is a block diagram of one example of an offset value management system in more detail.

FIGS. **11**A and **11**B (collectively referred to herein as FIG. **11**) show a flow diagram illustrating one example of the operation of an offset value management system.

FIG. **12** is a flow diagram illustrating on example of the operation of a mismatch processing system.

FIGS. **13**A, **13**B, **13**C, **13**D, and **13**E show examples of how the position of a receiving vehicle changes.

FIG. **14** is a block diagram showing one example of a vehicle position detection system.

FIGS. **15**A and **15**B (collectively referred to as FIG. **15**) show a flow diagram illustrating one example of the operation of the vehicle position detection system.

FIG. **16** shows one example of a combine harvester filling a receiving vehicle, where the receiving vehicle has a cross member.

FIG. **17** is a flow diagram showing one example of the operation of a calibration system in locating cross members on a receiving vehicle.

FIG. **18** is a flow diagram showing one example of the operation of an unloading control system in controlling an unloading operation based on the location of the cross member on the receiving vehicle.

FIGS. **19** and **20** show examples of operator interfaces.

FIG. **21** is a block diagram showing one example of a harvesting machine deployed in a remote server architecture.

FIGS. **22-24** show examples of mobile devices that can be used in the machines and systems described in other figures.

FIG. **25** is a block diagram showing one example of a computing environment that can be used in the machines and systems described with respect to previous figures.

DETAILED DESCRIPTION

The present discussion proceeds with respect to an agricultural harvester, but it will be appreciated that the present discussion is also applicable to construction machines or other material loading vehicles as well. As discussed above, it can be very difficult for an operator to maintain high efficiency in controlling a harvester, and also to optimally monitor the position of the receiving vehicle during an unloading (or filling) operation. This difficulty can even be exacerbated when the receiving vehicle is located behind the harvester (such as a forage harvester), so that the forage harvester is executing a rear unloading operation, but the difficulty also exists in side-by-side unloading scenarios.

The difficulty can be further exacerbated when the receiving vehicle has cross members or hoops. Such cross members or hoops may be rigid or flexible (e.g., belts or straps) and are often used to hold a cover for the receiving vehicle. If grain or other material contacts the cross members, the grain or other material often bounces out of the receiving vehicle onto the ground and/or may damage the cross member.

In order to address these issues, some automatic cart filling control systems have been developed to automate portions of the filling process. One such automatic fill control system uses a stereo camera on the spout of the harvester to capture an image of the receiving vehicle. An image processing system determines dimensions of the receiving vehicle and the distribution of crop deposited inside the receiving vehicle. The system also detects crop height within the receiving vehicle, in order to automatically aim the spout toward empty spots and control the flap position (and thus material trajectory) to achieve a more even fill, while reducing spillage. Such systems can fill the receiving vehicle according to a fill strategy (such as front-to-back, back-to-front, etc.) that is set by the operator or that is set in other ways, but often do not avoid cross members.

In addition, some current harvesters are provided with a machine synchronization **17** control system. The harvester may be a combine harvester so that the spout is not movable relative to the frame during normal unloading operations. Instead, the relative position of the receiving vehicle and the combine harvester is changed in order to fill the receiving vehicle as desired. Thus, in a front-to-back fill strategy, for instance, the relative position of the receiving vehicle, relative to the combine harvester, is changed so that the spout is first filling the receiving vehicle at the front end, and then gradually fills the receiving vehicle moving rearward. In such an example, the combine harvester and receiving vehicle may have machine synchronization systems which communicate with one another. When the relative position of the two vehicles is to change, the machine synchronization system on the combine harvester can send a message to the machine synchronization system on the towing vehicle to nudge the towing vehicle slightly forward or rearward relative to the combine harvester, as desired. By way of example, the machine synchronization system on the combine harvester may receive a signal from the fill control system on the combine harvester indicating that the position in the receiving vehicle that is currently being filled is approaching its desired fill level. In that case, the machine synchronization system on the combine harvester can send a "nudge" signal to the machine synchronization system on the towing vehicle. The "nudge", once received by the machine synchronization system on the towing vehicle, causes the towing vehicle to momentarily speed up or slow down, thus nudging the position of the receiving vehicle forward or rearward, respectively, relative to the combine harvester.

In all of the systems that attempt to automate part or all of the unloading process from a harvester into a receiving vehicle, the automated system attempts to understand where the receiving vehicle is located over time relative to the towing vehicle (e.g., the tractor pulling the receiving vehicle—also referred to has the following vehicle), and relative to the leading vehicle (e.g., the harvester or the vehicle that is controlling the following vehicle). For purposes of the present discussion, the term leading vehicle will be the vehicle that is unloading material into the receiving vehicle. The term following vehicle will refer to the propulsion vehicle, or towing vehicle (such as a tractor), that is providing propulsion to the receiving vehicle (such as a cart).

Determining the location of the receiving vehicle over time can be accomplished using different types of systems. In some current systems, a camera and image processor are used to capture an image (static or video) of parts of the receiving vehicle (the edges of the receiving area of the cart, the walls of the cart, the front end and rear end of the cart, the hoops or cross members on the receiving vehicle, etc., collectively referred to herein as receiving vehicle parameters) and an image processor processes that image in attempt to identify the receiving vehicle parameters, in real-time, during the harvesting operation. The image processor identifies the receiving vehicle parameters in the image and a controller then attempts to identify the location of the receiving vehicle parameters relative to the leading vehicle (e.g., relative to the harvester), in real-time, during harvesting and unloading.

However, this can be prone to errors. For instance, during the harvesting and unloading operation, the environment can be relatively dusty or have other obscurants so that it can be difficult to continuously identify the receiving vehicle parameters and then calculate their location relative to the leading vehicle. The dust or other obscurants in the environment can lead to an image that is difficult to process, and therefore, the accuracy in identifying the receiving vehicle parameters (and thus locating them relative to the leading vehicle) can take additional time, and can be error prone.

The present description thus proceeds, in one example, with respect to a system that conducts a calibration operation that identifies one or more receiving vehicle parameters and the position of the parameter(s) relative to a reference point on the following vehicle. A detector on the leading vehicle detects the receiving vehicle parameters. Positioning systems (e.g., global navigation satellite systems-GNSS receivers) on the leading vehicle and the following vehicle communicate with one another so that the relative position of the leading vehicle, relative to the following vehicle, is known. An offset on the following vehicle between the positioning system and a reference point (such as a hitch, wheelbase, etc.) is also known. A calibration system thus determines the relative position of the receiving vehicle parameters, relative to the location of the leading vehicle (as identified by the positioning system on the leading vehicle) and transposes that information into a location of the receiving vehicle parameters relative to the reference point on the following vehicle. This is referred to as the calibrated offset value corresponding to the receiving vehicle parameter. Then, during an unloading operation, the leading vehicle (e.g., the harvester) need only receive the position of the following vehicle (e.g., the GPS coordinates of the tractor). The leading vehicle can then calculate where the receiving vehicle parameter (e.g., the front wall, the side walls, the rear wall, hoops or cross members, etc. of the receiving vehicle) is located relative to the reference location on the trailing vehicle (e.g., the trailer hitch of the tractor) based upon the calibrated offset value for the particular receiving vehicle parameter under consideration.

In another example, the location of the hitch point (or pivot point) where the receiving vehicle is coupled to the following vehicle is identified relative to the leading vehicle. The position and heading or route of the following vehicle is detected relative to the leading vehicle. A dynamic model identifies the location of the receiving vehicle relative to the leading vehicle based on the location of the hitch relative to the leading vehicle and based on the position and heading or route of the following vehicle relative to the loading vehicle.

In this way, the leading vehicle need not rely on real-time images captured in a noisy (e.g., dusty) environment to attempt to identify the location of the receiving vehicle during the unloading process. Instead, during harvesting and unloading, once the GNSS location of the hitch point on the following vehicle is known (or the relative position of the hitch point on the following vehicle is known relative to the leading vehicle), along with the route or heading of the following vehicle relative to the leading vehicle, then the location of the receiving vehicle can be identified using the dynamic model, without performing image processing. The unloading operation can thus be controlled based on the output from the dynamic model.

FIG. **1** is a pictorial illustration showing one example of a self-propelled forage harvester **100** (a material loading vehicle also referred to as a leading vehicle) filling a tractor-pulled grain cart (or receiving vehicle) **102**. Cart **102** thus defines an interior that forms a receiving vessel **103** for receiving harvested material through a receiving area **112**. In the example shown in FIG. **1**, a tractor **104** (a towing vehicle also referred to as a following vehicle), that is pulling grain cart **102**, is positioned directly behind forage harvester **100** Also, in the example illustrated in FIG. **1**, forage harvester **100** has a detector such as camera **106** mounted on the spout **108** through which the harvested material **110** is traveling. The spout **108** can be pivotally or rotatably mounted to a frame **107** of harvester **100**. In the example shown in FIG. **1**, the detector **106** is a stereo-camera or a mono-camera that captures an image (e.g., a still image or video) of the receiving area **112** of cart **102**. Also, in the example shown in FIG. **1**, the receiving area **112** is defined by an upper edge of the walls of cart **102**.

When harvester **100** has an automatic fill control system that includes image processing, as discussed above, the automatic fill control system attempts to identify the location of the receiving area **112** by identifying the edges or walls of the receiving area and can then gauge the height of harvested material in cart **102**, and the location of that material in the receiving vehicle. The system thus automatically controls the position of spout **108** and flap **109** to direct the trajectory of material **110** into the receiving area **112** of cart **102** to obtain an even fill throughout the entire length and width of cart **102**, while not overfilling cart **102**. By automatically, it is meant, for example, that the operation is performed without further human involvement except, perhaps, to initiate or authorize the operation.

For example, when executing a back-to-front automatic fill strategy the automatic fill control system may attempt to move the spout and flap so the material begins landing at a first landing point in the back of vessel **103** of receiving vehicle **102**. Then, once a desired fill level is reached in the back of vessel **103**, the automatic fill control system moves the spout and flap so the material begins landing just forward of the first landing point in vessel **103**.

There can be problems with this approach. The environment of receiving area **112** can have dust or other obscurants making it difficult to visually identify the location and bounds of receiving area **112**. Thus, it can be difficult to accurately control the trajectory of material **110** to achieve the desired fill strategy.

FIG. **2** is a pictorial illustration showing another example of a self-propelled forage harvester **100**, this time loading a semi-trailer (or receiving vessel on a receiving vehicle) **122** in a configuration in which a semi-tractor (also referred to as a following vehicle) is pulling semi-trailer **122** alongside forage harvester **100**. Therefore, the spout **108** and flap **109** are positioned to unload the harvested material **110** to fill trailer **122** according to a pre-defined side-by-side fill strategy. Again, FIG. **2** shows that camera **106** can capture an image (which can include a still image or **6** video) of semi-trailer **122**. In the example illustrated in FIG. **2**, the field of view of camera **106** is directed toward the receiving area **120** of trailer **122** so that image processing can be performed to identify a landing point for the harvested material in trailer **122**.

FIG. **3** shows an example in which leading vehicle **101** is a combine harvester, with an operators compartment **121** and with a header **130** that engages crop. The crop is processed and placed in a clean grain tank **132**, where it is unloaded (such as using an auger) through spout **108** into a receiving vehicle **134** (e.g., a grain cart) that is pulled by a following vehicle **136** (e.g., a tractor). FIG. **3** shows that receiving vehicle **134** is coupled to following vehicle **136** at a hitch point, or pivot point, **137**. When harvester **101** is a combine harvester, it may be that the spout **108** is not moved relative to the frame of harvester **101** during normal unloading operations. Instead, the relative position of the receiving vehicle **134** and the combine harvester **101** is changed in order to fill the receiving vessel as desired. Thus, if a front-to-back fill strategy is to be employed, then the relative position of the receiving vessel in receiving vehicle **134**, relative to the combine harvester **101**, is changed so that the spout **108** is first filling the receiving vehicle **134** at the front end, and then gradually fills the receiving vessel moving rearward.

In the configuration shown in FIGS. **2** and **3**, spout **108** illustratively has a detector, such as a stereo camera, that again attempts to identify parameters of the receiving vehicle so that the receiving vehicle can be located and so that the unloading operation can be controlled to unload material at a desired location in the receiving vehicle to accomplish a desired fill strategy. Again, as with the configuration illustrated in FIG. **1**, the environment of the receiving vehicles **122** and **134** in FIGS. **2** and **3**, respectively, may have dust or other obscurants making it difficult to identify the parameters of the receiving vehicle (e.g., the edges or walls that define the receiving vessel) during runtime.

Thus, the present description proceeds with respect to a system that conducts a calibration operation for the following vehicle and receiving vehicle to identify an offset between one of the receiving vehicle parameters (e.g., the front wall, either or both sidewalls, the rear wall, hoop(s), cross members, etc.) and a known reference location on the following vehicle on the following vehicle (such as the tractor hitch, point **137** the wheelbase, etc.). The offset is referred to as the calibrated offset value. The calibrated offset value can then be used during the harvesting operation to locate the receiving vehicle relative to the following vehicle without the need to identify the receiving vehicle parameters in an image that may be captured in a noisy environment (such as a dusty environment or an environment that has other obscurants) during the harvesting and unloading operation. Instead, the control system simply needs to obtain the location of the following vehicle (such as through a GNSS receiver or another location detection system) and then use that location to calculate the location of the receiving vehicle using the calibrated offset value.

For instance, by tracking the position of hitch point **137** and the route of following vehicle **136** relative to leading vehicle **101**, a dynamic model can be used to identify the location **12** of receiving vehicle **134** relative to leading vehicle **101**. This location can be used to control the unloading operation (e.g., control the trajectory of material being unloaded, control the position of the vehicles, controlling the unloading subsystem, etc.).

FIG. **4** is a block diagram showing one example of an agricultural system **140** which includes leading vehicle (in the present example, a combine harvester) **101** which is followed by following vehicle (in the present example, a tractor or another propulsion vehicle) **136**. Following vehicle **136** is pulling a receiving vehicle **134**. It will be appreciated that while agricultural system shown in FIG. **4** includes leading vehicle **101**, following vehicle **136**, and receiving vehicle **134** (e.g., the vehicles shown in the example illustrated in FIG. **3**) other leading vehicles, following vehicles, and receiving vehicles can be used as well. The example shown in FIG. **4** is shown for the sake of example only.

Leading vehicle **101** includes one or more processors or servers **142**, data store **144** (which can include machine dimension information **145**, vehicle parameter offset values—e.g., calibrated offset value(s) for vehicle parameter(s), **147** which may be indexed by vehicle identification number-VIN or vehicle model number or other vehicle identifier, and other information **149**), position sensor **146**, communication system **148**, unloading control system **150**, receiving vehicle sensors **152**, operator interface system **154**, controllable subsystems **156**, and other vehicle functionality **158**. Unloading control system **150** can include following/receiving vehicle pair detector **160**, offset value management system **161**, calibration system **162**, vehicle position detection system **164**, control signal generator **166**, and other control system functionality **168**. Receiving vehicle sensors **152** can include optical sensor **169**, RADAR sensor **170**, LIDAR sensor **172**, and/or other sensors **174**. Optical sensor **169** can include camera **106**, image processor **171**, and/or other items **173**. Operator interface system **154** can include interface generation system **176**, output generator **178**, operator interaction detector **180**, and other interface devices and/or functionality **182**. Controllable subsystems **156** can include header subsystem **184**, material **6** conveyance subsystem (e.g., blower, spout, flap, etc.) **186**, propulsion subsystem **188**, steering subsystem **190**, optical sensor positioning system **191**, and other items **192**. FIG. **4** also shows that leading vehicle **101** can be operated by an operator **194**.

Following vehicle **136** can include position sensor **196**, communication system **198**, one or more processors or servers **195**, data store **200**, control system **202**, operator interface system **204**, and any of a wide variety other functionality **206**. FIG. **4** also shows that following vehicle **136** is operated by operator **208**. Receiving vehicle **134** can include an identifier **210** and/or other items **212**. Before describing the overall operation of agricultural system **140** in more detail, a description of some of the items in system **140**, and their operation, will first be provided.

Position sensor **146** can be a global navigation satellite system (GNSS) receiver, a dead reckoning system, a cellular triangulation system, or any of a wide variety of other systems that identify the coordinates or location of leading vehicle **101** in a global or local coordinate system. Data store **144** can store dimension information and orientation information, such as information that identifies the location and orientation of optical sensor **169** relative to the material conveyance system (e.g., blower, spout, flap, etc.) **186**. Data store **144** can store calibrated offset values described in greater detail elsewhere here, as well as other information.

Communication system **148** enables the communication of items on vehicle **101** with other items on vehicle **101**, as well as communication with following vehicle **136** and other communication. Therefore, communication system **148** can be a controller area network (CAN) bus and bus controller, a cellular communication device, a Wi-Fi communication device, a local or wide area network communication device, a Bluetooth communication device, and/or any of a wide variety of devices or systems that enable communication over different types of networks or combinations of networks.

Receiving vehicle sensors **152** sense the receiving vehicle **134** and/or parameters of receiving vehicle **134**. In the example discussed herein, the parameters of receiving vehicle **134** are structural portions of receiving vehicle **134** that allow the location of the receiving area of receiving vehicle **134** to be determined. The receiving vehicle parameters, for example, may be the front wall or top front edge of the receiving vehicle **134**, the side walls or top side edges of receiving vehicle **134**, the rear wall or the top rear edge of receiving vehicle **134**, etc. Therefore, optical sensor **169** can include camera **106** and image processor **171**. During the calibration process, camera **106** can capture an image (static or video) of receiving vehicle **134** and image processor **171** can identify the location of the receiving vehicle parameters within that image. Thus, image processor **171** can identify the location of the front wall or front edge of receiving vehicle **134** within the captured image, and/or the other receiving vehicle parameters. In other examples, RADAR sensor **170** and/or LIDAR sensor **172** can be used to identify the receiving vehicle parameters in different ways. Sensors **170** and **172** can have signal processing systems that process the signals generated by RADAR and LIDAR sensors to identify the receiving vehicle parameters.

Unloading control system **150** controls the unloading process by which material conveyance subsystem **186** conveys material from leading vehicle **101** to receiving vehicle **134**. Following vehicle/receiving vehicle pair detector **160** detects the identity of following vehicle **136** and receiving vehicle **134** (e.g., the identity of this tractor/cart pair). Offset value management system **161** can then determine whether calibration data (e.g., calibration offset value(s)) have already been generated for this particular pair of vehicles. If so, offset value management system **161** can retrieve the calibration data such as calibrated offset values **147** from data store **144** and provide those values to vehicle position detection system **164** so the values can be used to locate receiving vehicle **134** and to control the unloading process. Offset value management system **161** can receive vehicle parameter offset values from calibration system **162** (e.g., as calibrated offset values) or from operator inputs from operator **194** and/or operator **208** or from other places. System **161** then stores the vehicle parameter offset values **147**. Values **147** can be indexed in data store by an identifier for following vehicle **136** (e.g., the VIN), by the model number or other identifier, by a receiving vehicle identifier of one or both vehicles **134**, **136**, by a vehicle pair identifier, or otherwise. System **161** can also identify a mismatch between the vehicle parameter offset values **147** that have been returned and the actual vehicle parameter offset values for the following vehicle/receiving vehicle pair. For example, if the operator of following vehicle **136** hooks up a different receiving vehicle, this can be identified and processed by system **161** to notify the operator(s) and/or to obtain the correct vehicle parameter offset values, as is described in greater detail below with respect to FIGS. **10-12**. If calibrated offset values have not yet been generated for this following vehicle **136**/receiving vehicle **134** pair, then calibration system **162** can be used to perform a calibration operation for this particular following vehicle **136**/receiving vehicle **134** pair.

The calibration operation identifies the location of the receiving vehicle parameters (e.g., the front wall, rear wall, side walls, etc., of the receiving vehicle) relative to a reference location on the following vehicle **136** (e.g., relative to the hitch point **137**, wheelbase, etc. of following vehicle **136**). The calibration operation can also identify the location of hitch point **137** relative to the position sensor **196** or other known location on following vehicle **136**. These locations are referred to as the calibrated offset values for this particular following vehicle/receiving vehicle pair. The calibrated offset values can then be stored in data store **144** for use in identifying the location of receiving vehicle **134** and controlling the unloading operation.

Vehicle position detection system **164** detects the position of leading vehicle **101** and following vehicle **136** either in terms of absolute coordinates within a global or local coordinate system, or in terms of a relative position in which the positions of vehicles **101** and **136** are determined relative to one another. For instance, vehicle position detection system **164** can receive an input from position sensor **146** on vehicle **101** and from position sensor **196** (which May also be a GNSS receiver, etc.) on following vehicle **136** to determine where the two vehicles are located relative to one another. Vehicle position detection system **164** can then detect the location **19** of receiving vehicle **134** relative to the material conveyance subsystem **186** using the calibration offset value for this particular following vehicle/receiving vehicle pair and/or using a dynamic model as described elsewhere herein, such as below with respect to FIGS. **13**A-**15**B.

For instance, by knowing the location of following vehicle **136**, and by knowing the calibrated offset values, which locate the walls (or other receiving vehicle parameter(s)), of receiving vehicle **134** relative to a reference position on following vehicle **136**, vehicle position detection system **164** can identify the location of the walls of receiving vehicle **134** relative to the material conveyance subsystem **186** on leading vehicle **101**. In another example, by knowing the route and location of following vehicle **136**, and by knowing the location of hitch point **137**, position detection system **164** can use a dynamic model that models the kinematics of receiving vehicle **134** to identify the location of receiving vehicle **134** relative to the material conveyance subsystem **186** on leading vehicle **101**. This location can then be used to determine how to control vehicles **101** and **136** to perform an unloading operation so that material conveyance system **186** loads material into receiving vehicle **134** according to a desired fill pattern.

Control signal generator **166** generates control signals that can be used to control vehicle **101** and following vehicle **136** to accomplish the desired fill pattern. For instance, control signal generator **166** can generate control signals to control the material conveyance subsystem **186** to start or stop material conveyance, to control the spout position or flat position in order to control the trajectory of material that is being conveyed to receiving vehicle **134**, or to control the propulsion system **188** or steering subsystem **190**. Control signal generator **166** can also generate control signals that are sent by communication system **148** to the following vehicle **136** to "nudge" the following vehicle forward or rearward relative to leading vehicle **101**, to instruct the operator **208** of following vehicle **136** to perform a desired operation, or to generate other control signals.

Header subsystem **184** controls the header of the harvester. Material conveyance subsystem **186** may include a blower, spout, flap, auger, etc., which control conveyance of harvested material from leading vehicle **101** to receiving vehicle **134**, as well as the trajectory of such material. Propulsion subsystem **188** can be an engine that powers one or more different motors, electric motors, or other systems that provide propulsion to leading vehicle **101**. Steering subsystem **190** can be used to control the heading and forward/backward directions of travel of leading vehicle **101**. Optical sensor positioning subsystem **191** can be a controllable actuator that points or aims or otherwise controls the orientation of optical sensor **169** to change the location of the field of view of sensor **169** (or other sensors).

Operator interface system **154** can generate interfaces for operator **194** and receive inputs from operator **194**. Therefore, operator interface system **154** can include interface mechanisms such as a steering wheel, joysticks, pedals, buttons, displays, levers, linkages, etc. Interface generation system **176** can generate interfaces for interaction by operator **194**, such as on a display screen, a touch sensitive displays screen, or in other ways. Output generator **178** outputs that interface on a display screen or in other ways and operator interaction detector **180** can detect operator interactions with the displayed interface, such as the operator actuating icons, links, buttons, etc. Operator **194** can interact with the interface using a point and click device, touch gestures, speech commands (where speech recognition and/or speech synthesis are provided), or in other ways.

As mentioned above, position sensor **196** on following vehicle **136** may be a global navigation satellite system (GNSS) receiver, a dead reckoning system, a cellular triangulation system, or any of a wide variety of other systems that provide coordinates of following vehicle **136** in a global or local coordinate system, or that provide an output indicating the position of following vehicle **136** relative to a reference point (such as relative to leading vehicle **101**), etc. Communication system **198** allows the communication of items on vehicle **136** with one another, and also provides for communication with leading vehicle **101**, and/or other systems. Therefore, communication system **198** can be similar to communication system **148** discussed above, or different. It will be assumed for the purpose of the present discussion that communication systems and **198** are similar, although this is for the sake of example only. Data store **200** can store dimension data which identify different dimensions of following vehicle **136**, the location and/or orientation of different sensors on vehicle **136**, kinematic information describing vehicle **134** and/or vehicle **136**, and other information. Control system **202** can be used to receive inputs and generate control signals. The control signals can be used to control communication system **198**, operator interface system **204**, data store **200**, the propulsion and/or steering subsystem on following vehicle **136**, and/or other items. Operator interface system **204** can also include operator interface mechanisms, such as a steering wheel, joysticks, buttons, levers, pedals, linkages, etc. Operator interface system **204** can also include a display screen that can be used to display operator interfaces for interaction by operator **208**.

Operator **208** can interact with the operator interfaces using a point and click device, touch gestures, voice commands, etc.

Identifier **210** on receiving vehicle **134** may be visual indicia, or electronic indicia, or another item that specifically identifies receiving vehicle **134**. Identifier **210** may also simply be the make or model of receiving vehicle **134**, or another marker that identifies receiving vehicle **134**.

FIG. **5** is a block diagram showing one example of calibration system **162** in more detail. In the example shown in FIG. **5**, calibration system **162** includes trigger detector **220**, data store interaction system **222**, operator prompt generator **224**, receiving vehicle parameter locator system **226**, parameter location output generator **228**, and other calibration system functionality **230**. Receiving vehicle parameter locator system **226** includes receiving vehicle parameter selector **232**, leading vehicle reference locator system **234**, vehicle-to-vehicle location system **236**, cross member locator system **237**, following vehicle reference locator system **238**, overlay generator **239**, and other locator functionality **240**. Before describing the operation of calibration system **162** in more detail, a description of some of the items in calibration system **162**, and their operation, will first be described.

Trigger detector **220** detects a trigger indicating that calibration system **162** is to perform a calibration operation to identify the calibrated offset value that locates one or more receiving vehicle parameters (front wall, rear wall, side walls, etc.) relative to a reference point on a following vehicle (e.g., a towing vehicle or tractor that is providing propulsion to the receiving vehicle). In one example, trigger detector **220** detects an operator input indicating that the operator **8** wishes to perform a calibration operation. In another example, the receiving vehicle sensors **152** (shown in FIG. **4**) may detect a new following vehicle/receiving vehicle pair for which no calibrated offset value has been generated. This may trigger the calibration system **162** to perform a calibration operation. The calibration system may be triggered by leading vehicle **101** beginning to perform a harvesting operation (e.g., where the harvesting functionality is engaged) or for other reasons.

Operator prompt generator **224** then prompts the operators of one or more of leading vehicle **101** and following vehicle **136** to position receiving vehicle **134** so that the receiving vehicle parameter may be detected by one or more of the receiving vehicle sensors **152**. For instance, where the receiving vehicle sensors **152** include an optical sensor (such as camera **106**) then the prompt may direct the operators of the vehicles to move the vehicles in place relative to one another so that the camera **106** can capture an image of the receiving vehicle parameters and so that those parameters can be identified by image processor **171** within the image.

FIG. **6**, for instance, shows one example of a user interface display device **250** displaying a display **252** that can be generated for the operator of either leading vehicle **101** or following vehicle **136** or both. Operator interface display **252** includes displays an image (static or video) taken by camera **106**. In the example shown in FIG. **6**, the operators have moved the vehicles into position relative to one another so that an image of receiving vehicle **134** can be captured by camera **106**. In the example shown in FIG. **6**, receiving vehicle **134** includes a front wall **254**, a rear wall **256**, a near wall **258** (which is near camera **106**), and a far wall **260** which is further from camera **106** than near wall **258**). The top edges of each of the walls **254**-**260** are also visible in the image illustrated in display **252**.

Returning to the description of FIG. **5**, data store interaction system **222** can interact with data store **144** to obtain dimension information indicating the location and orientation of camera **106** (or other receiving vehicle sensors **152**) that is used to detect the receiving vehicle parameters. Image processor **171** then processes the image to identify the receiving vehicle parameters (such as the walls **254**-**260**) within the captured image. Receiving vehicle parameter locator system **226** can then process the location of the receiving vehicle parameters in the captured image to identify the location of the receiving vehicle parameters relative to a reference point on the following vehicle (e.g., tractor) **134**. In doing so, receiving vehicle parameter selector **232** selects which receiving vehicle parameter is to be processed first (such as front wall **254**, rear wall **256**, or side walls **258** and/or **260**). Leading vehicle reference locator system **234** then identifies the location of the selected parameter (for purposes of the present description it will be assumed that the selected receiving vehicle parameter is front wall **254**) relative to a reference point on the leading vehicle **101**. For instance, system **234** can identify the location of the front wall **254** of receiving vehicle **134** relative to the location of the GPS receiver (or other position sensor) **146** on leading vehicle **101**. Vehicle-to-vehicle location system **236** then communicates with the following vehicle **136** to identify the location of leading vehicle **101** relative to the location of following vehicle **136**. In particular, system **236** may identify the location of the position sensor **146** on leading vehicle **101** relative to the location of the position sensor **196** on following vehicle **136**.

Cross member locator system **237** can be used to identify the locations of cross members so the unloading operation can be controlled to avoid dumping material onto the cross members. System **237** is described in greater detail elsewhere herein. Following vehicle reference locator system **238** then identifies the location of the selected parameter (the front wall **254**) of receiving vehicle **134** relative to the reference point on following vehicle **136**. For instance, where the reference point on following vehicle **136** is the hitch point **137**, then following vehicle reference locator system **238** first identifies the location of front wall **254** relative to the position sensor **196** on following vehicle **136** and then, using dimension information or other information about following vehicle **136**, identifies the offset between the reference position (e.g., the hitch point **137**) on following vehicle **136** and the position sensor **196** on following vehicle **136**. Once this offset is known, then the location of the front wall **254** of receiving vehicle **134** to the hitch can be calculated by following vehicle reference locator system **238**. The result is that system **238** generates an output indicating the location of the selected receiving vehicle parameter (in this case the front wall **254** of receiving vehicle **134**) relative to the reference point on the following vehicle **136** (in this case the hitch of following vehicle **136**). This is referred to herein as the calibrated offset value.

Overlay generator **239** can be used to overlay the calculated locations of receiving vehicle parameters (including the walls, cross members, etc.) over an optical image of the receiving vehicle **134** so the operator can easily see if the calculated locations coincide with those on the image. If not, the operator can illustratively interact with the image (e.g., by moving items on the overlay so they match the image), and the locations of the moved overlay items can then be connected.

Parameter location output generator **228** generates an output from calibration system **162** to store the calibration offset value in data store **144** for his particular following vehicle **136**/receiving vehicle **134** pair. Thus, when vehicle position detection system **164** on leading vehicle **101** encounters this following vehicle **136**/receiving vehicle **134** pair during the harvesting operation, the calibrated offset value can be retrieved and used in controlling the unloading operation during which harvested material is unloaded from leading vehicle **101** into receiving vehicle **134**.

FIGS. **7**A and **7**B (collectively referred to herein as FIG. **7**) show a flow diagram illustrating one example of the operation of agricultural system **140** in performing a calibration operation to identify the calibrated offset value corresponding to one or more receiving vehicle parameters of receiving vehicle **134**. It is first assumed that the work machines are configured so that a calibration operation can be performed, as indicated by block **270** in the flow diagram of FIG. **7**. In one example, the leading vehicle **101** is a harvester as indicated by block **272** and the following vehicle is a tractor or other towing vehicle as indicated by block **274**. Also in the example, the receiving vehicle is a grain cart as indicated by block **276**. Also, in the present example, it is assumed that both the leading vehicle and the following vehicle have a position sensing system **146**, **196**, respectively, as indicated by block **278** in the flow diagram of FIG. **7**. Further, it is assumed that leading vehicle **101** has a receiving vehicle sensor **152** that is at a known location and orientation on leading vehicle **101**, as indicated by block **280** in the flow diagram of FIG. **7**. The receiving vehicle sensor **152** may be an image capture device, such as a stereo camera **106**, a RADAR sensor **170**, a LIDAR sensor **172**, etc. The work machines may be configured in other ways to perform the calibration operation as well, as indicated by block **282** in the flow diagram of FIG. **7**.

Detecting a calibration trigger is indicated by block **284** in the flow diagram of FIG. In one example, operator **194** may provide an operator input through operator interface system to trigger a calibration operation for the following vehicle **136**/receiving vehicle **134** pair. Detecting a trigger based on an operator input is indicated by block **286** in the flow diagram of FIG. **7**. Calibration trigger detector **220** may detect a trigger based upon leading vehicle **101** beginning to perform the harvesting operation, which may be detected by detecting engagement of the header or other harvesting functionality, or in another way. Detecting a trigger based upon the beginning of the machine operation is indicated by block **288** in the flow diagram of FIG. **7**. The calibration trigger may be detected based on following vehicle/receiving vehicle pair detector **160** detecting that the current following vehicle/receiving vehicle pair is a new pair for which no calibration offset data has been generated. Detecting a trigger based on the detection of a new vehicle (one for which no calibration offset data is stored) is indicated by block **290** in the flow diagram of FIG. **7**. The calibration trigger can be detected in a variety of other ways, based upon other trigger criteria as well, as indicated by block **292** in the flow diagram of FIG. **7**.

Once the calibration operation has been triggered, operator prompt generator **224** generates a prompt that can be displayed or otherwise output to operator **194** and/or operator **208** by operator interface systems **154**, **204**, respectively. The prompt prompts the operator, to move the vehicles so the material receiving vehicle **134** is in a position where at least one of the receiving vehicle parameters is detectable by the receiving vehicle sensor(s) **152** on leading vehicle **101**. Outputting such a prompt is indicated by block **294** in the flow diagram of FIG. **7**, and outputting the prompt to one or both operators is indicated by block **296**. Again, the receiving vehicle parameters to be detected may include the front wall **254**, rear wall **256**, near wall **258**, far all **260**, etc., as indicated by block **298** in the flow diagram of FIG. **7**.

Therefore, for instance, the operators **194**, **208** of the vehicles **101**, **136** may position receiving vehicle **134** so that the receiving vehicle parameter to be located is in the field of view of the image sensor or camera **106**, as indicated by block **300** in the flow diagram of FIG. **7**. In another example, the receiving vehicle parameter to be located is detectable by one of the other sensors **170**-**174** on leading vehicle **101**, as indicated by block **302** in the flow diagram of FIG. **7**. In yet another example, the receiving vehicle parameter to be located (e.g., the front wall **254** of vehicle **134**) is aligned with a known point of reference on leading vehicle **101**. For instance, it may be that the spout **108** on the combine harvester **101** is at a known location relative to the position sensor **146** on combine harvester **101**. In that case, the front wall **254** of receiving vehicle **134** may be aligned with the spout **108** so that the location of front wall **254**, relative to the reference point (e.g., spout **108**) on leading vehicle **101** is known. Aligning the receiving vehicle parameter to be located with a known reference point on the leading vehicle **101** is indicated by block **304** in the flow diagram of FIG. **7**. The prompt can be output to the operators in other ways as well, as indicated by block **306** in the flow diagram of FIG. **7**.

Leading vehicle reference locator system **234** then detects a location of the receiving vehicle parameter (e.g., front wall **254**) relative to the sensor **152** on the leading vehicle as indicated by block **308** in the flow diagram of FIG. **7**. In one example, camera **106** captures an image of receiving vehicle **134** and image processor **171** processes the image captured by camera **106** using a machine learned processor, or disparity image processor, etc., in order to identify the location of the receiving vehicle parameter (e.g., front wall **254**) in the captured image, as indicated by block **310**. System **234** can then obtain data identifying the known sensor location and/or orientation of the receiving vehicle sensor **152** (e.g., camera **106**) on leading vehicle **101**, as indicated by block **312**. System **234** uses the location of the front wall **254** in the captured image and the location and orientation of the camera **106** to calculate the location of front wall **254** relative to camera **106**.

It will be noted that, instead of using image processing to identify the location of front wall **254** (or another receiving vehicle parameter) in the captured image, an operator input can be used to identify the receiving vehicle parameter in the captured image. FIG. **8**, for instance, shows one example of an operator interface display **314** on a display device **250** displaying a side view of the receiving vehicle **134**. The side view is slightly elevated so that the front wall **254**, rear wall **256**, near wall **258**, and far wall **260**, of receiving vehicle **134** are all visible. In the example shown in FIG. **8**, operator **194** can use a touch gesture (or point and click device) to trace along the front wall **254** (as indicated in FIG. **8**) to identify the location of front wall **254** in the captured image. Also, of course, where the receiving vehicle parameter to be located is the rear wall, or the side walls, operator **194** can trace along those walls.

In another example, system **226** can project a line on the video displayed to the operator and the operator can then align the receiving vehicle parameter (e.g., front wall **254**) with the line. For example, in FIG. **8**, system **226** can project line **255** on the display **314** (which May be a live video feed from camera **106**) so the operator(s) can align front wall **254** with line **255**. System **226** knows the pixels used to display line **255**. Therefore, once front wall **254** is aligned with line

255 (which can be indicated by an operator input), system 226 will know the location of front wall 254 in the image 314.

FIG. 9, for example, is similar to FIG. 8 and similar items are similarly numbered. However, the view of receiving vehicle 134 is slightly more elevated and from a slightly different perspective than that shown in FIG. 8. In FIG. 9, it can be seen that the operator has traced along the top edge of the far wall 260 (or a line has been projected at that location and wall 260 has been aligned with the line) to identify the location of the far wall 260 in the displayed image 316. Identifying the receiving vehicle parameter (e.g., one of walls of the receiving vehicle or top edge of the wall, etc.) based on operator interaction with the displayed image is indicated by block 318 in the flow diagram of FIG. 7. Projecting a line and aligning the receiving vehicle parameter with the line is indicated by block 319. Detecting a location of the receiving vehicle parameter in the image and relative to the camera 106 can be done in other ways as well, as indicated by block 320.

Again, once the location of the receiving vehicle parameter is identified in the image, then using the known location and orientation of the camera 106, the location of the receiving vehicle parameter can be identified relative to one or more other reference points on receiving vehicle 101.

Calculating or otherwise obtaining the location of the receiving vehicle parameter relative to the location of a reference point on the leading vehicle 101 is indicated by block 322 in the flow diagram of FIG. 7. In one example, the reference point on leading vehicle 101 is the location of the position sensor 146. Thus, leading vehicle reference locator system 234 identifies the location of the receiving vehicle parameter (e.g., front wall 254) on receiving vehicle 134 relative to the location of the position sensor 146 on leading vehicle 101, as indicated by block 324 in the flow diagram of FIG. 7. Of course, where the reference point is a different reference point on leading vehicle 101, then the location of the receiving vehicle parameter (e.g., front wall 254) relative to that reference point can be calculated as well. Identifying the location of the receiving vehicle parameter relative to the location of another reference point on leading vehicle 101 is indicated by block 236 in the flow diagram of FIG. 7.

Vehicle-to-vehicle location system 236 uses communication system 148 and communication system 198 to communicate with one another so that the position of following vehicle 136 can be identified relative to the position of the leading vehicle 101 as indicated by block 328. In one example, the position of one vehicle relative to the other can be calculated using the absolute positions of both vehicles sensed by the corresponding position sensors 146 and 196. In another example, other sensors can be used (such as RADAR, LIDAR, etc.) to detect the relative position of the two vehicles.

Once vehicle-to-vehicle location system 236 identifies the relative locations of the two vehicles relative to one another, then following vehicle reference locator 238 can identify the location of the receiving vehicle parameter (e.g., front wall 254) relative to the coordinates of a reference point on the following vehicle 136, as indicated by block 330 in the flow diagram of FIG. 7. The reference point on following vehicle 136 can be any of a wide variety of different reference points, such as the location of the position sensor 196, the location of a hitch or wheelbase, etc. Determining the location of the receiving vehicle parameter (e.g., front wall 254) relative to the location of position sensor 196 on following vehicle 136 is indicated by block 332 in the flow diagram of FIG. 7. Determining the location of the receiving vehicle parameter (e.g., front wall 254) relative to the hitch or wheelbase of the following vehicle 136 can be done by retrieving vehicle dimension information from data store 200 or data store 144 or elsewhere, where the dimension information identifies the location of the reference point relative to the position sensor 196. Identifying the location of the receiving vehicle parameter relative to another reference point on following vehicle 136 in this way is indicated by block 334 in the flow diagram of FIG. 7. The location of the receiving vehicle parameter (e.g., front wall 254) relative to the location of a reference point on the following vehicle 136 can be done in a wide variety of other ways as well, as indicated by block 336 in the flow diagram of FIG. 7.

When more receiving vehicle parameters (e.g., rear wall, side walls, etc.) are to be located relative to the reference point on following vehicle 136, as indicated by block 338 in the flow diagram of FIG. 7, processing reverts to block 294 where the operators are prompted to position the two vehicles relative to one another so that the next receiving vehicle parameter can be detected by the receiving vehicle sensors 152 (e.g., camera 106) on leading vehicle 101. It will be noted that, in one example, the location of the receiving vehicle parameters relative to a reference position on the following vehicle 136 can be determined or calculated in a priority order. The priority may be to first locate the front wall 254, then the rear wall 256, then the near wall 258 and finally the far wall 260. In another example, only the front wall 254 is located and then known dimensional information (e.g., the length of the receiving vehicle 134) is used to identify the location of the rear wall. Similarly, a center point on the front wall 254 can be located and then width information that defines the width dimension of receiving vehicle 134 can be used to locate the side walls 258 and 260. In yet another example, the top edges of the walls 254-260 are identified to define the material-receiving opening in material receiving vehicle 134. These are just examples of the different receiving vehicle parameters that can be located relative to a reference point on the following vehicle 134. Other receiving vehicle parameters can be located as well, and they can be located in different orders.

Parameter location output generator 228 can generate an output indicative of the locations of the receiving vehicle parameters relative to the reference point on the following vehicle 134, as calibrated offset values, to data store interaction system 222 which can store the calibrated offset values in data store 144, data store 200, or elsewhere, where the values can be retrieved by leading vehicle 101 when performing the harvesting operation, and when locating the receiving vehicle 134 during an unloading operation. Storing the receiving vehicle parameter locations relative to the reference point on the following vehicle 136 is indicated by block 340 in the flow diagram of FIG. 7.

In one example, the calibrated offset values are stored and indexed by the particular following vehicle 136/receiving vehicle 134 pair for which the calibrated offset values are calculated, as indicated by block 342 so that the values can be looked up during later operation, when a harvester is unloading to this particular following vehicle 136/receiving vehicle 134 pair (or a similar pair). In one example, the calibration offset values are stored locally in data store 144 on vehicle 101, or locally in data store 200 on following vehicle 136, as indicated by block 344.

In another example, the calibrated offset values can be stored remotely in a cloud-based system, in another remote server architecture, on a different machine, or in a different system which can then be accessed by leading vehicle 101 at an appropriate time, as indicated by block 346. In another example, the calibrated offset values can be transmitted to other vehicles (such as other harvesters, etc.) so that the calibration need not be performed by all of the other leading vehicles which may encounter this particular following vehicle **136**/receiving vehicle **134** pair. Sending the calibrated offset values to other vehicles is indicated by block **348** in the flow diagram of FIG. **7**. The calibrated offset values can be stored in other ways, and used in other ways (such as in controlling the unloading operation during a subsequent process), as indicated by block **350** in the flow diagram of FIG. **7**. Retrieving and using the calibrated offset values to control an unloading operation is indicated by block **352** in the flow diagram of FIG. **7**.

FIG. **10** is a block diagram showing one example of offset value management system **161** in more detail. In the example shown in FIG. **10**, offset value management system **161** includes vehicle parameter offset value receiving system **270**, data store interaction system **272**, retrieval trigger detector **274**, offset value retrieval system **276**, mismatch processing system **278**, and other offset value management functionality **280**. Offset value retrieval system **276** can include vehicle identification system **282**, current vehicle comparison system **284**, data retrieval system **286**, and other items **288**. Mismatch processing system **278** can include mismatch identification system **290**, operator interface interaction system **292**, retrieval control system **294**, and other items **296**. Before describing the operation of offset value management system **161** in more detail, a description of some of the items in system **161**, and their operation, will first be provided.

Vehicle parameter offset value receiving system **270** receives vehicle parameter offset values so that they can be stored and managed. In one example, system **270** controls operator interface system **154** to generate an operator interface that allows operator **194** to enter the vehicle parameter offset values, manually. In another example, system **270** can use communication system **148** to communicate with operator interface system **204** on vehicle **136** to generate an operator interface so that operator **208** can enter the vehicle parameter offset values manually. Those values can then be communicated back to vehicle parameter offset value receiving system **270**. In another example, system **270** can receive, as the vehicle parameter offset values, the calibrated offset values from calibration system **162**.

System **270** can then use data store interaction system **272** to interact with data store to store the vehicle parameter offset values as calibrated offset values **147**. In one example, values **147** are indexed by a vehicle identifier, such as an identifier that identifies the following vehicle **136**/receiving vehicle **134** pair, by the VIN of the following vehicle, by a vehicle identifier that identifies the receiving vehicle **134**, or in other ways. The values **147** can be stored in a look-up table or in another data structure.

Thus, when following/receiving vehicle pair detector **160** encounters following vehicle **136**/receiving vehicle **134** pair (such as when a pair approaches leading vehicle **101** for unloading), the corresponding vehicle identifier can be obtained and the vehicle parameter offset values can be easily obtained from data store **144** using the corresponding vehicle identifier. In another example, system **270** uses data store interaction system **272** to store the vehicle parameter offset values based upon the model number of the receiving vehicle **134** or a combination of the model numbers of the following vehicle **136** and receiving vehicle **134**. In this way, when leading vehicle **101** encounters a following vehicle/receiving vehicle pair that is not identically the same as a pair for which vehicle parameter offset values have been generated, but which has the same vehicle model numbers, then the vehicle parameter offset value(s) for that combination of model numbers (the same model number for the following vehicle **136**, and the same model number for the receiving vehicle **134**) can be used to control the unloading operation.

Retrieval trigger detector **274** detects a trigger indicating that a set of vehicle parameter offset values **147** should be retrieved. For example, operator **194** may provide an input indicating that it is time to unload into a new following vehicle **136**/receiving vehicle **134** pair. In that case, trigger detector **274** detects this as a trigger indicating that it is time to obtain the vehicle parameter offset values for this particular following vehicle **136**/receiving vehicle **134** pair. In another example, the control system **202**, or operator **208**, of following vehicle **136** may send a signal to leading vehicle **101** indicating that following vehicle **136** is approaching leading vehicle **101** for an unloading operation. In that case, trigger detector **274** can detect that a new following vehicle **136**/receiving vehicle **134** pair is approaching and so the vehicle parameter offset values corresponding to that vehicle pair should be obtained. In yet another example, trigger detector **274** may detect that the clean grain tank of the leading vehicle **101** is nearly full so that an unloading operation needs to be commenced in the near future, in which case the vehicle parameter offset values can be obtained to prepare for the unloading operation.

Once triggered, offset value retrieval system **276** retrieves the vehicle parameter offset values for the following vehicle **136**/receiving vehicle **134** pair to which the harvested material is to be unloaded, based on a vehicle identifier. Vehicle identification system **282** identifies a vehicle identifier corresponding to the following vehicle **136**/receiving vehicle **134** pair and current vehicle comparison system **284** determines whether that pair is the same as the pair for which the vehicle parameter offset values are currently loaded into the control system. If so, then no additional vehicle parameter offset values need to be retrieved. If not, however, then data retrieval system **268** uses data store interaction system **272** to retrieve the vehicle parameter offset values **147** for this particular following vehicle **136**/receiving vehicle **134** pair. Those offset values are then loaded into vehicle position detection system **164** so that the unloading operation can be controlled using the correct vehicle parameter offset values.

Mismatch processing system **278** can detect whether there is a mismatch between the vehicle parameter offset values that are being used for the current following vehicle **136**/receiving vehicle **134** pair and the actual offset values for that vehicle pair. For instance, it **6** may be that operator **208** of following vehicle **136** has switched receiving vehicles (e.g., switched grain carts). In that case, the vehicle parameter offset values that are stored and correspond to the VIN number of the following vehicle **136** may not be accurate because they correspond to a different receiving vehicle. Therefore, mismatch identification system **290** identifies that the vehicle parameter offset values are inaccurate.

Identifying a mismatch can be done in a number of different ways. For instance, system **290** may identify the receiving vehicle **134**, itself, and determine that the identified receiving vehicle **134** is not the same as the receiving vehicle that is assumed to be connected to following vehicle **136**. As one example, where the receiving vehicle **134** may include a unique identifier (such as visual indicia, a visual signature or visual characteristic extracted from visual features of the receiving vehicle **134**, a transmission device that transmits a unique identifier, or that stores a unique identifier that can be read by mismatch identification system **290**), then the identity of the particular receiving vehicle **134** can be compared against the identity of the receiving vehicle that is assumed to be connected to following vehicle **136** to determine whether they are the same.

If a mismatch is identified (e.g., the receiving vehicle **134** that is actually being used is not the same as that to which the vehicle parameter offset values correspond and that are currently loaded into vehicle position detection system **164**) then operator interface interaction system **292** can generate a notification to operator **194** and/or operator **208**. The notification May identify the mismatch, and the notification may also include other information. For instance, the notification may indicate that a calibration operation is automatically being performed, or the notification may provide an operator actuatable button (such as an icon) that can be actuated by one of the operators **194**, **208** in order to initiate a calibration operation. In another example, the operator interface can include data input fields which allow one of the operators to input the vehicle parameter offset values for the receiving vehicle **134** that is being used.

It may also be that there are already vehicle parameter offset values **147** stored for this particular following vehicle **136**/receiving vehicle **134** pair (even though it was initially assumed that a different receiving vehicle **134** was connected to following vehicle **136**). In that case, retrieval control system **294** can retrieve the already-existing vehicle parameter offset values for this particular following vehicle **136**/receiving vehicle **134** pair.

In another example, retrieval control system **294** can obtain the vehicle parameter offset values from calibration system **162**, once the calibration operation has been performed. The retrieval control system **294** can retrieve the vehicle parameter offset values from the operator interface, once they have been manually entered. The retrieval control system **294** can retrieve the vehicle parameter offset values for the current following vehicle **136**/receiving vehicle **134** pair in other ways as well. Mismatch processing system **278** can then output the correct calibrated offset values to vehicle position detection system **164** for use in loading material into the receiving vehicle **134**.

FIGS. **11**A and **11**B (collectively referred to as FIG. **11**) show a flow diagram illustrating one example of the operation of offset value management system **161**. It is first assumed that vehicle identification system **282** obtains a vehicle identifier corresponding to the following vehicle **136**/receiving vehicle **134** pair, as indicated by block **291**. The vehicle identifier can be for the vehicle pair or for one or both vehicles **136**, **134** individually. The vehicle identifier can be input by an operator as indicated by block **293**, or automatically detected, as indicated by block **295**. The vehicle identifier can be received from the following vehicle **136**, as indicated by block **297** or in other ways, as indicated by block **299**.

The vehicle parameter offset values (which identify an offset between receiving vehicle parameter and a reference point on following vehicle **136**) for the receiving vehicle and following vehicle are detected, for different vehicle parameters. Detecting the vehicle parameter offset values for the different vehicle parameters is indicated by block **298** in the flow diagram of FIG. **11**. In one example, the vehicle parameter offset values can be obtained by accessing dimension information **145** in data store **144**, and calculating dimensional information indicative of the vehicle parameter offset values, as indicated by block **300**. In another example, a display screen can be generated on a user interface on the following vehicle **136** or leading vehicle **101**, or both, for manual entry of the vehicle parameter offset values, as indicated by block **302**. In yet another example, the vehicle parameter offset values are obtained by calibration system **162** performing a calibration operation, as indicated by block **304**. In one example, the vehicle parameter offset values include one or more positions of reference locations on following vehicle **136**, as indicated by block **306** in the flow vehicle of FIG. **11**, as well as the positions defining the receiving vessel on the receiving vehicle **134** (e.g., the positions of receiving vehicle parameters) relative to the reference locations on vehicle **136**, as indicated by block **308**.

The machine or vehicle parameters for which offset values are obtained can be parameters for both the following vehicle **136** and/or the receiving vehicle **134**. For instance, vehicle parameter offset values can be obtained for the location of the front wall, side walls, and rear wall of the receiving vehicle **134** relative to one or more different reference points on the following vehicle **136**. The receiving vehicle parameter offset values and the following vehicle parameter offset values can be detected in other ways as well, as indicated by block **310** in the flow diagram of FIG. **11**.

The vehicle parameter offset values are received by vehicle parameter offset value receiving system **270**. System **270** then interacts with data store interaction system **272** to store the vehicle parameter offset values, values **147** in data store **144**. In one example, the vehicle parameter offset values **147** are stored in a lookup table that is indexed by the vehicle identifier corresponding to this particular following vehicle **136**/receiving vehicle **134** pair. One example of the vehicle identifier may include a unique identifier corresponding to the following vehicle **136** and/or receiving vehicle **134** (such as a VIN number, etc.). In another example, the lookup table may be indexed by vehicle model number (including the model numbers of one or both of the following vehicle **136** and receiving vehicle **134**), by another vehicle identifier for the following vehicle **136** and/or receiving vehicle **134**, or in other ways. Storing the receiving vehicle parameter offset values and following vehicle parameter offset values is indicated by block **312** in the flow diagram of FIG. **11**. Storing those values in a lookup table or in another data structure indexed by the vehicle identifier (e.g., VIN) in indicated by block **314**. Storing the vehicle parameter offset values by vehicle model identifiers is indicated by block **316** in the flow diagram of FIG. **11**. Storing the vehicle parameter offset values in a data structure that is accessible by using other following vehicle/receiving vehicle identifiers is indicated by block **318** in the flow diagram of FIG. **11**.

It should also be noted that the vehicle parameter offset values can be stored either on local data store **144**, or on a remote data store. The remote data store may be a data store in a remote server environment (e.g., the cloud), a data store on a different vehicle, a data store at a farm manager computing system, a data store in a mobile device, or another data stores.

At some point, retrieval trigger detector **274** detects a trigger indicating that the vehicle parameter offset values are to obtained for this particular following vehicle **136**/receiving vehicle **134** pair. Detecting a trigger is indicated by block **320** in the flow diagram of FIG. **11**. The trigger may be based on an operator input **322** provided by operator **194** or operator **208**. For example, operator **208** may send a message through communication system **148** to receiving vehicle **101** identifying the VIN number or other vehicle identifier corresponding to vehicle **136** to indicate that an unloading operation is about to commence with respect to vehicle **136** and receiving vehicle **134**. An operator input may also be provided by operator **194** indicating that the vehicle offset parameter values are to be obtained.

In another example, following vehicle **136** may automatically send a signal identifying itself to receiving vehicle **101** (and specifically to unloading control system **150**). Having the following vehicle **136** automatically send a signal with a vehicle identifier is indicated by block **324**. Such a signal may be detected or interpreted by retrieval trigger detector **274** as a trigger to obtain the vehicle parameter offset values for the following vehicle **136** and receiving vehicle **134**.

In another example, unloading control system **150** may generate a signal to retrieval trigger detector **274** indicating that an unloading operation is about to commence. This signal may also be a trigger to obtain the vehicle parameter offset values for the following vehicle **136**/receiving vehicle **134** pair that is about to be loaded with material. Detecting a signal indicating that leading vehicle **101** is ready to unload signal is indicated by block **326** in the flow diagram of FIG. **11**. The trigger detector **274** can detect a retrieval trigger based on other trigger criteria, or in other ways, as indicated by block **328**.

Vehicle identification system **282** then obtains a vehicle identifier which identifies the following vehicle **136**/receiving vehicle **134** pair, as indicated by block **330** in the flow diagram of FIG. **11**. Such a vehicle identifier may be sent from receiving vehicle **134** or from following vehicle **136** to leading vehicle **101**, as indicated by block **332**. The vehicle identifier may be automatically recognized on the leading vehicle **101**, as indicated by block **334**. For instance, following vehicle/receiving vehicle pair detector **160** may detect the identifiers of following vehicle **136** and receiving vehicle **134** using optical recognition that identifies optical features or other optical indicia that identify those vehicles. In another example, the vehicles **136** and **134**, themselves, send vehicle identifiers (such as the VIN, model number, or other identifiers that identify those vehicles). The vehicle identifiers of the following vehicle **136**/receiving vehicle **134** pair (or one or both of those vehicles individually) can be obtained in a wide variety of other ways as well, as indicated by block **336** in the flow diagram of FIG. **11**.

Current vehicle comparison system **284** compares the vehicle identified by the vehicle identifier obtained at block **330** to that of the following vehicle **136** and/or receiving vehicle **134** whose offset values are currently loaded into vehicle position detection system **164**. If the two are the same, as determined at block **338**, then processing continues at block **340**. However, if, at block **338**, current vehicle comparison system **284** determines that the vehicles identified by the vehicle identifier obtained at block **330** are different than the vehicles for which the vehicle parameter offset values are currently loaded into vehicle position detection system **164**, then data retrieval system **268** uses the vehicle identifier from block **330** to access the correct vehicle parameter offset values as indicated by block **342**.

The vehicle parameter offset values **147** can be accessed from a local data store **144**, as indicated by block **344** in the flow diagram of FIG. **11**. The vehicle parameter offset values can be accessed from a remote store, as indicated by block **346**, or from another vehicle, as indicated by block **348**. For instance, if a different leading vehicle (different from leading vehicle **101**) performed a calibration operation to identify the calibrated offset values for this particular following vehicle **136**/receiving vehicle **134** pair, then those calibrated offset values can be accessed from that other leading vehicle and used on leading vehicle **101**. The vehicle parameter offset values can be accessed from other locations, or in other ways, as well, as indicated by block **350** in the flow diagram of FIG. **11**.

Once the vehicle parameter offset values have been obtained, then data retrieval system **268** loads those values into vehicle position detection system **164** for use in controlling the unloading operation. Loading the vehicle parameter offset values to control the unloading operation is indicated by block **352** in the flow diagram of FIG. **11**.

Mismatch processing system **278** can also process any mismatches between the vehicle parameter offset values that are loaded into the vehicle position detection system **164** and those for the actual vehicles being loaded. Processing any mismatches is indicated by block **354** in the flow diagram of FIG. **11**. Processing the mismatches is described in greater detail below with respect to FIG. **12** as well.

FIG. **12** is a flow diagram illustrating one example of the operation of mismatch processing system **278** in performing processing when the vehicle parameter offset values that have been loaded into vehicle position detection system **164** do not appear to correspond to the receiving vehicle **134** that is actually being loaded (or is about to be loaded). Mismatch identification system **290** first detects that the vehicle parameter offset values for the following vehicle and/or the receiving vehicle are not for this particular following vehicle **136**/receiving vehicle **134** pair. Detecting this mismatch is indicated by block **360** in the flow diagram of FIG. **12**.

For instance, it may be that system **290** has detected that the receiving vehicle **134** is not the one in the following vehicle **136**/receiving vehicle **134** pair that was used to access the vehicle parameter offset values, as indicated by block **362**. This may be the case, for instance, where the operator **208** has changed grain carts, etc. It may also be that one of the operators **194**, has provided an input indicating that there is a mismatch, or the mismatch may be detected in any of a variety of other ways, as indicated by block **364** in the flow diagram of FIG. **12**.

Operator interface interaction system **292** then generates a notice to one or more of the operators **194** and/or **208**, indicating that a mismatch has been detected. Generating the notice is indicated by block **366** in the flow diagram of FIG. **12**. The notice may prompt the operator to trigger calibration system **162** to perform a calibration operation for this particular following vehicle **136**/receiving vehicle **134** pair, as indicated by block **368**. The notice may prompt the manual or automated input of the actual vehicle parameter offset values, as indicated by block **370**. For instance, the notice may provide a text entry field where one of the operators can input the vehicle parameter offset values, or the system **292** can expose an interface that can be automatically called by control system **202** on vehicle **136** to automatically input the vehicle parameter offset values. The notice can be generated for the operators in a wide variety of other ways as well, as indicated by block **372**.

Retrieval control system **294** then obtains the correct vehicle parameter offset values, as indicated by block **376** in the flow diagram of FIG. **12**. For instance, the vehicle parameter offset values can be received (as calibrated offset values) from calibration system **162** or through an operator input, as indicated by block **378**. The vehicle parameter offset values can be obtained by identifying the correct following vehicle **136**/receiving vehicle **134** pair and automatically retrieving the vehicle parameter offset values from data store **144** or elsewhere, as indicated by block **380** in the flow diagram of FIG. **12**. Retrieval control system **294** can obtain the correct vehicle parameter offset values in any of a wide variety of other ways as well, as indicated by block **382**.

Mismatch processing system **278** generates an output indicative of the correct vehicle parameter offset values so that those values can be loaded into vehicle position detection system **164** for use in controlling the unloading operation, as indicated by block **384** in the flow diagram of FIG. **12**.

It can thus be seen that the present description proceeds with respect to a system that automatically stores the vehicle parameter offset values corresponding to a following vehicle **136**/receiving vehicle **134** pair so that the values can be automatically retrieved and used in controlling an unloading operation when the leading vehicle **101** encounters that following vehicle **136**/receiving vehicle **134** pair in the future. The vehicle parameter offset values can be stored in a lookup table or in another data structure that is indexed by a vehicle identifier that identifies the following vehicle **136**/receiving vehicle **134** pair or one or both of those vehicle individually. The vehicle identifier may be, for instance, the VIN of the following vehicle **136** and/or of the receiving vehicle **134**. The vehicle identifier may be a model number of one or both of the vehicles, an optical identifier, or another type of identifier that identifies the following vehicle **136**/receiving vehicle **134** pair or one or both of the vehicles individually, corresponding to the set of vehicle parameter offset values. The vehicle parameter offset values can be stored locally on one of the machines, remotely in a remote server environment, or on a different system. The vehicle parameter offset values can be stored on a mobile device, or on a different vehicle where they can be accessed at a later time.

When a mismatch is identified, in which the following vehicle **136**/receiving vehicle **134** pair does not match that corresponding to the vehicle parameter offset values that are currently being used to control the unloading operation, then a notification can be generated for one or more of the operators, and operations can be performed to obtain the correct set of vehicle parameter offset values. The correct set of vehicle parameter offset values is then loaded into the vehicle position detection system for use in controlling the unloading operation.

It can thus be seen that the present description has also described a system which performs a calibration operation that can be used to locate different receiving vehicle parameters relative to a reference point on a following vehicle. This calibrated offset values can then be stored and used in locating the receiving vehicle during subsequent unloading operations so that the receiving vehicle need not be located using visual image capture and image processing, which can be error prone. This increases the accuracy of the unloading operation.

As mentioned above, in one example, the location of the hitch point **137** (hitch point location) on following vehicle **136** can be identified relative to the location of the position sensor (e.g., the GPS receiver) **196** on following vehicle **136**. Also, the hitch point location and the route of following vehicle **136** and position of vehicle **136** relative to leading vehicle **101** can be applied to a dynamic model that models the kinematics of receiving vehicle **134**. For instance, the dynamic model may model how receiving vehicle **134** moves as the pivot point or hitch point **137** to which it is coupled moves along the route of following vehicle **136**. As an example, if following vehicle **136** turns, vehicle **136** pivots about hitch point **137**. However, receiving vehicle **134** will eventually also turn to follow the same heading as following vehicle **136**. The dynamic model models the motion of various points (e.g., the receiving vehicle parameters or other points) on receiving vehicle **134** through space, based upon the location of pivot point **137** and the route of the following vehicle **136**. In such an example, once the calibrated offset value is known which indicates the location of hitch point **137** relative to the position sensor **196** on following vehicle **136**, and once the route of following vehicle **136** is known, then the dynamic model can be used to compute the location of receiving vehicle **134** relative to the position sensor **196** or hitch point **137** or other known reference value on following vehicle **136**. The location of receiving vehicle **134**, relative leading vehicle **101**, can thus be determined as well.

FIGS. **13A-13**E show examples of how the movement of receiving vehicle **134** changes based upon the position and heading or route of following vehicle **136**. In FIG. **13**A, for instance, it can see that vehicle **136** is moving in a forward direction along a route generally indicated by arrow **400**, and it can be seen that following vehicle **136** and receiving vehicle **134** have just recently traveled along a route indicated by arrow **402**. FIG. **13**B, however, shows that following vehicle **136** has now turned to follow a route indicated by arrow **404**, in order to avoid an obstacle **406**, but receiving vehicle **134** is still traveling as shown by arrow **402**.

Eventually, because pivot point **137** will move with following vehicle **136**, the receiving vehicle **134** will also turn to follow following vehicle **136**, as illustrated in FIG. **13**C. FIG. **13**D shows that following vehicle **136** is now moving in a direction indicated by arrow **408**. Receiving vehicle **134** is traveling along a different route indicated by arrow **410**, but will soon turn to follow following vehicle **136**, as shown in FIG. **13**E. This type of movement of receiving vehicle **134** can be modeled by a dynamic model which models the kinematics of receiving vehicle **134** as the hitch point **137** moves through space (e.g., as hitch point **137** moves to follow the route taken by following vehicle **136**). The dynamic model may be a machine learned kinematic model that automatically learns the kinematics of receiving vehicle **134**, or a model that receives data indicative of the dimensions and movement characteristics of receiving vehicle **134**. It can be seen in FIGS. **13A-13**E that, by knowing the location of hitch point **137** relative to leading vehicle **101** (e.g., by knowing the offset of hitch point **137** from the position sensor **196** on following vehicle **136** and by knowing the relative position of the position sensor **196** relative to the position sensor on leading vehicle **101**), and by knowing the route that following vehicle **136** is following, then the dynamic model can estimate the position of receiving vehicle **134** relative to hitch point **137**, or relative to the position sensor **196** on receiving vehicle **136**, and/or relative to the position of leading vehicle **101** and/or relative to a subsystem on leading vehicle **101**.

Therefore, vehicle position detection system **164** on leading vehicle **101** simply needs to obtain the location of position sensor **196** on following vehicle **136** as well as the route of following vehicle **136** and the offset of hitch point **137** relative to position sensor **196**. From that information, and using a dynamic model, vehicle position detection system **164** can detect the position of receiving vehicle **134** relative to spout **108** on leading vehicle **101**. Such information can be used to control the unloading operation (e.g., to position spout **108**, to position vehicles **136** and **101** relative to one another, to turn on and off the material conveyance subsystem **186** on leading vehicle **101**, etc.).

FIG. **14** is a block diagram showing one example of vehicle position detection system **164**, in more detail. In the example shown in FIG. **14**, vehicle position detection system **164** includes following vehicle parameter loading system **414**, receiving vehicle parameter loading system **416**, position detection trigger detector **418**, cross member location comparison system **419**, following vehicle position and heading detection system **420**, hitch point location system **422**, dynamic receiving vehicle locator model **424**, and other items **426**.

Following vehicle parameter loading system **414** can obtain parameters, such as dimensions, calibrated offset values, etc., with respect to following vehicle **136**. The parameter information may include such things as the offset between hitch point **137** and the position sensor **196** on following vehicle **136**, and/or other dimension information or kinematic information. Receiving vehicle parameter loading system **416** receives or otherwise obtains access to parameters of receiving vehicle **134**, such as calibrated offset values, kinematic information, dimensional information, and/or other parameter values. Systems **414** and **416** can obtain the information from following vehicle **136** and/or receiving vehicle **134**, from data store **144**, from another remote system or another vehicle, etc. Also, the information can be retrieved based upon a vehicle identifier or another identifier that identifies receiving vehicle **134** and/or following vehicle **136**, or the pair comprising following vehicle **136** and receiving vehicle **134**.

Position detection trigger detector **418** detects trigger criteria indicating when the position or location of receiving vehicle **134** is to be detected. For instance, when trigger detector **418** detects that the following vehicle **136**/receiving vehicle **134** pair is in position for leading vehicle **101** to begin an unloading operation, this may trigger detector **418** to detect the position of receiving vehicle **136**. The detection may be continuous or intermittent, so long as the unloading operation is being performed, or the trigger criteria can be other criteria or the criteria can be detected in other ways.

Cross member location comparison system **419** is described in more detail elsewhere herein. Briefly, system **419** compares a fill location where material is to be unloaded into receiving vehicle **134** to a location of the cross members that span at least a portion of a material-receiving area of receiving vehicle **134**. If the two locations overlap, system **419** generates a signal so control signal generator **166** can generate appropriate control signals (such as to change the fill location so material is not inadvertently dumped on the hoops or cross members).

Following vehicle position and heading detection system **420** detects the position and heading of following vehicle **136**. The position and heading can be in absolute terms, or relative to the position and/or heading of leading vehicle **101**. For instance, it may be that system **420** controls communication system **148** to obtain the current position and heading of following vehicle **136** from position sensor **196**. In another example, the position and heading of following vehicle **136** can be detected using sensors **152** or other sensors on leading vehicle **101**. In another example, system **420** can obtain multiple position indicators from position sensor **196** and calculate the heading of following vehicle **136** based upon the multiple position indicators. System **420** can obtain the position and heading or route of following vehicle **136** in other ways as well.

Hitch point location system **422** can be used to locate the hitch point **137** relative to leading vehicle **101**. For instance, once the position of position sensor **196** on following vehicle is known, and once the offset between sensor **196** and hitch point **137** is known, hitch point location system **422** can identify the location of hitch point **137** relative to position sensor **196** on following vehicle **136**. Then, obtaining the position of leading vehicle **101** from position sensor **146**, hitch point location system **422** can locate the hitch point **137** relative to the position sensor **146** on leading vehicle **101**, and relative to any other items on leading vehicle **101** (given that the offset between those items and position sensor **146** is known).

Dynamic receiving vehicle locator model **424** is illustratively a machine learned dynamic model that receives, as an input, the location of hitch point **137** and the route of following vehicle **136** and/or receiving vehicle **134** and calculates the position of receiving vehicle **134** relative to the hitch point **137**, relative to following vehicle **136**, and/or relative to vehicle **101** (or any subsystem or reference point on vehicle **101**). In this way, system **164** can calculate the location of receiving vehicle **134** relative to spout **108** (or other known items on leading vehicle **101**) and can use that information to control the unloading operation (such as to turn on or off the material conveyance subsystem **186**, to control the position or orientation of material conveyance subsystem **186**, to control the relative position of vehicles **101** and **136** or **134** relative to one another, etc.).

FIGS. **15**A and **15**B (collectively referred to herein as FIG. **15**) show a flow diagram illustrating one example of the operation of vehicle position detection system **164** in using a dynamic model **424** along with the location of hitch point **137** and route of vehicle **136** and/or vehicle **134** to identify the position of receiving vehicle **134** relative to leading vehicle **101**. It is first assumed that leading vehicle **101** is configured to perform an unloading operation to unload material into receiving vehicle **134**, as indicated by block **428** in the flow diagram of FIG. **15**. Following vehicle parameter loading system **414** then obtains the following vehicle parameter information, as indicated by block **430**. The following vehicle parameter information can include the location of following vehicle **136** relative to leading vehicle **101**, or relative to a ground plane, as indicated by block **432**. The following vehicle parameter information can include the location of hitch point **137** relative to the wheelbase or position sensor **196** on following vehicle **136**, as indicated by block **434**. The following vehicle parameter information can include the location of the wheelbase of following vehicle **136** relative to the position sensor **196** on following vehicle **136**, as indicated by block **436**. The following vehicle parameter information can account for the articulation angle when the following vehicle is an articulated vehicle, such as an articulated tractor. The articulation angle can be sensed and a fixed transform, a lookup table, or another technique can be used to identify or calculate the pose of following vehicle **136** and receiving vehicle **134**, as indicated by block **435**. In addition, the following vehicle parameter information can include the height of the hitch above the ground, as indicated by block **437**. The following vehicle parameter information can be manually entered into the leading vehicle **101**, through a user interface, as indicated by block **438**, or it can be accessed in a lookup table or other data structure (e.g., in data store **144**) that may be indexed based on vehicle type, vehicle identifier, etc., as indicated by block **440** in the flow diagram of FIG. **15**. The following vehicle parameter information can be obtained by running the calibration operation discussed above, as indicated by block **442**, or the following vehicle parameter information can be sent from following vehicle **136** (where it may be sent manually or automatically) as indicated by block **444**. The following vehicle parameter information can include other types of information and it can be obtained in other ways as well, as indicated by block **446**.

Receiving vehicle parameter loading system **416** then obtains the receiving vehicle parameter information, as indicated by block **448** in the flow diagram of FIG. **15**. The receiving vehicle parameter information can include such things as the location of the front wall (or origin) and/or axle of the receiving vehicle **134** relative to hitch point **137** (or relative to the position sensor **196**) on following vehicle **136**, as indicated by block **450**. The receiving vehicle parameter information can include dimension information such as the length of the receiving vehicle (or the receiving vessel in the receiving vehicle) **134**, or the location of the rear wall relative to the front wall or origin, or relative to the positioning sensor **196** on following vehicle **146**, or other information, as indicated by block **452** in the flow diagram of FIG. **15**. The receiving vehicle parameter information can include the width of the receiving vehicle (e.g., the distance from the near wall to the far wall—or lateral offset of the two walls) or the distance to the near and/or far walls from the position sensor **196** on the following vehicle **136**. Obtaining information indicative of the width of the receiving vehicle **134** is indicated by block **454** in the flow diagram of FIG. **15**. The receiving vehicle parameter information can be manually input, automatically input, looked up in a data store, or obtained by performing the calibration operation, discussed above, as indicated by block **456** in the flow diagram of FIG. **15**. The receiving vehicle parameter information can be obtained in other ways as well, as indicated by block **458**.

Position detection trigger detector **418** then detects whether it is time to detect the position of receiving vehicle **134**, as indicated by block **460** in the flow diagram of FIG. **15**. As discussed above, the trigger criteria may be based on the fact that following vehicle **136** and **6** receiving vehicle **134** are in position for leading vehicle **101** to commence an unloading operation. The trigger criteria may be that the unloading operation has commenced, or the trigger criteria May be an operator input or automated input from following vehicle **136** (that is detected on leading vehicle **101**), an operator input or automated input on leading vehicle **101**, or other trigger criteria. If the trigger criteria are not yet met, then processing reverts to block **430** where the following vehicle parameter information and receiving vehicle parameter information can be obtained (in case a different following vehicle **136** and receiving vehicle **134** pair have been encountered by leading vehicle **101**).

However, once the trigger criteria are met, then following vehicle position and heading detection system **420** detects or calculates the following vehicle position and the heading or route of the following vehicle **136** relative to leading vehicle **101**, as indicated by block **462** in the flow diagram of FIG. **15**. The following vehicle heading or route and position can be communicated from following vehicle **136** to detection system **420** automatically or manually or the following vehicle position and route can be detected using one or more of the sensors **152** or in other ways.

Hitch point location system **422** then identifies the location of hitch point **137** relative to the leading vehicle **101**, as indicated by block **464** in the flow diagram of FIG. **15**. Again, the location of the hitch point **137** can be identified based upon the location of position sensor **196** on following vehicle **136** relative to leading vehicle **101**, and based on the offset between the position sensor **196** and hitch point **137**. The location of the hitch point **137** relative to leading vehicle **101** can be identified or detected in other ways as well.

Vehicle position detection system **164** then uses dynamic receiving vehicle locator model **424** to identify the location of the receiving vehicle boundaries (the walls of receiving vehicle **134** or the center point of receiving vehicle **134**) or to otherwise locate receiving vehicle **134**. Receiving vehicle **134** is located based upon the location of the hitch point **137**, the route of following vehicle **136**, and/or any other desired vehicle parameters (either following vehicle parameters or receiving vehicle parameters) as indicated by block **466** in the flow diagram of FIG. **15**. Dynamic model **424** can model the kinematics of receiving vehicle **134** given the location of hitch point **137** and the route of the vehicles. Model **424** can be a machine learned model, or an artificial intelligence (AI) model, or another model.

Vehicle position detection system **164** outputs the location of receiving vehicle **136** relative to leading vehicle **101** to control signal generator **166**. Control signal generator **166** then generates control signals based upon the location of the receiving vehicle **134**, in order to control the unloading operation, as indicated by block **468** in the flow diagram of FIG. **15**. In one example, control signal generator **166** generates control signals to control optical sensor positioning subsystem **191**. System **191** can be actuated to point or aim camera **106** based on the location of receiving vehicle **134** relative to leading vehicle **101**. For example, when optical sensor **169** is being used to observe the fill profile in receiving vehicle **136**, or to observe other items, then optical sensor positioning system **191** can be controlled so that the field of view of sensor **169** obtains a best or satisfactory view of the fill level in receiving vehicle **134**. Controlling the position of optical sensor **169** based upon the location of the receiving vehicle **134** relative to the leading vehicle **101** is indicated by block **470** in the flow diagram of FIG. **15**.

Control signal generator **166** can generate control signals to control the propulsion subsystem **188** and steering subsystem **190** on leading vehicle **101** and/or the propulsion subsystem and steering subsystem on following vehicle **136**. Control signal generator **166** can generate other control signals to control the relative vehicle positions (such as to nudge one forward or backward relative to the other) as indicated by block **472** in the flow diagram of FIG. **15**. In another example, control signal generator **166** can generate control signals to control material conveyance subsystem **186** (such as to turn on and off the blower, auger, etc.), to position the spout and/or flap, etc. as indicated by block **474** in the flow diagram of FIG. **15**.

Also, some control algorithms calculate buffer zones proximate the edges (or walls) of the receiving vehicle **134** so that the filling operation does not fill material in those zones in order to avoid inadvertent spillage of material over the side of receiving vehicle **134**. Thus, the location of the receiving vehicle **134** relative to leading vehicle **101** can be used to define the buffer zones in performing the unloading operation. Calculating and using buffer zones or buffer areas on receiving vehicle **134** is indicated by block **476** in the flow diagram of FIG. **15**. Control signal generator **166** can generate any of a wide variety of other control signals to control other operations or subsystems in order to control the unloading operation, as indicated by block **478** in the flow diagram of FIG. **15**.

It can thus be seen that the present description describes a system that models the kinematics of a receiving vehicle based on the location of the hitch point and the heading of the following vehicle relative to the leading vehicle. The heading and position of the following vehicle can be communicated to the leading vehicle so that the dynamic model can locate the receiving vehicle relative to the leading vehicle without needing to rely on images captured by an optical sensor. This improves the accuracy and robustness of the location and control systems.

FIG. **16** is similar to FIG. **3**, and some more items are similarly numbered. However, FIG. **16** shows that receiving vehicle **134** has a plurality of cross members, or hoops, **480**, **482**, and **484** (hereinafter referred to as cross members). Cross members **480-484** may be rigid or flexible (e.g., straps) that span part or all of the material-receiving area of receiving vehicle **134**. Cross members **480-484** can be used to hold a cover on receiving vehicle **134**. When grain exits spout **108** and hits one of the cross members **480**, **482**, or **484**, the grain may bounce out of receiving vehicle **134** and/or damage the cross member that it strikes. Therefore, in accordance with one example, the location of cross members **480-484** are determined relative to leading vehicle **101** (and/or the unloading end of spout **108**) so that spout **108** can be controlled to avoid unloading grain onto cross members **480**, **482**, and **484**.

FIG. **17** is a flow diagram illustrating one example of the operation of calibration system **162** in identifying the location of the cross members **480-484** relative to position sensor **196** on following vehicle **136**. Once the location of the cross members **480-484** relative to position sensor **196** (or relative another reference point on following vehicle **136**) is known, then the unloading control system **150** on leading vehicle **101**can control the unloading operation to unload grain from leading vehicle **101** into receiving vehicle **104**, while avoiding cross members **480-484**.

It is first assumed that the positioning systems **146** and **196** communicate their positions to one another so that vehicle-to-vehicle location system **236** can calculate or otherwise obtain the position of the following vehicle **136** relative to the leading vehicle **101**. Determining this position is indicated by block **486** in the flow diagram of FIG. **17**. Receiving vehicle parameter locator system **226** then obtains the location and orientation of the optical sensor **106** relative to the position sensor **146** on the leading vehicle, as indicated by block **488** in the flow diagram of FIG. **17**. This location and orientation can be detected as indicted by block **490** or entered through an operator interface generated by operator interface system **154**, as indicated by block **492** in the flow diagram of FIG. **17**. In another example, the location and orientation of the camera **106** relative to the position sensor **146** on leading vehicle **101** can be pre-calculated and stored so that data store interaction system **222** can retrieve that information from a data store or memory, as indicated by block **494**. The location and orientation of camera **106** relative to position sensor **146** can be obtained in other ways as well, as indicated by block **496** in the flow diagram of FIG. **17**.

Camera **106** then captures an image of the receiving vehicle **134**, including one or more of the cross members **480-484**. For instance, operator prompt generator **224** can prompt the operator of one or both vehicles **101**, **136** to position the vehicles so that the image can be captured. Capturing the image of the receiving vehicle **134** including at least one of the cross members is indicated by block **498** in the flow diagram of FIG. **17**. Prompting the operator to position the vehicles appropriately is indicated by block **520** in the flow diagram of FIG. **17**. The image can be captured in other ways as well, as indicated by block **522**.

Optical sensor **169** then identifies the location of the hoops in the captured image, as indicated by block **524**. For instance, image processor **174** can automatically identify the locations of the cross members in the image by processing the image (e.g., identify which pixels correspond to the cross members), as indicated by block **526**. In another example, an operator input can be used to identify the cross members in the image, as indicated by block **528**. For instance, FIG. **19** shows an operator interface that is similar to that shown in FIG. **9**, and similar items are similarly numbered. However, FIG. **19** also shows cross members **480-484** on receiving vehicle **134**. The operator can trace over the cross members **480-484** using a point and click device, or using a touch gesture, as illustrated in FIG. **19**, in order to identify the locations of the cross members **480-484** in the captured image. In another example, such as that shown in FIG. **20**, the operator can trace or otherwise indicate a marker or line **530** on the image, and then cause vehicles **136** and/or **101** to move to align a cross member (e.g., cross member **480** shown in FIG. **20**) with that line **530** to identify the location of the cross member in the captured image. The examples shown in FIGS. **19**, and **20** are examples only, and the operator can identify the cross members in the captured image in other ways as well.

In yet another example, the cross members can be identified on the image by aligning the cross members in the field of view of camera **106** with an item that is in a known position relative to the camera **106** (e.g., the unloading auger or spout **108**) on leading vehicle **101**, so the position of the cross member in the image coincides with the position of the known item with which the cross member is aligned. This way of identifying the cross members in the image is indicated by block **532** in the flow diagram of FIG. **17**. The cross members can be identified in the image in other ways as well, as indicated by block **534**.

Once the position of a cross member in the image is identified, then cross member locator system **237** can calculate the position of the identified cross member relative to camera **106** (or optical sensor **169**) as indicated by block **536** in the flow diagram of FIG. **17**. Given that the location and orientation of camera **106** is known relative to position sensor **146**, then the position of the cross members relative to camera **106** can be transposed by cross member locator system **237** to obtain the position of the cross members relative to the location of the position sensor **146** on leading vehicle **101**, as indicated by block **538**. Also, because vehicle-to-vehicle location system **236** has identified the relative positions of the position sensors **146** and **196**, then the location of the cross members relative to the location or position of position sensor **196** on the following vehicle **136** can also be generated, as indicated by block **540**. The location of the cross members **480-484** can also be identified relative to another reference position on following vehicle **136** (such as the hitch point or other reference position) or a reference position on receiving vehicle **134** and that reference position can then be used to calculate the position of the cross members **480-484** relative to position sensor **196** on the following vehicle **136**. This involves an additional step in calculation, but may be performed as well, as indicated by block **542** in the flow diagram of FIG. **17**.

Also, in accordance with one example, overlay generator **239** can verify the position of the cross member so that the user can make corrections to that position. For instance, in an operator interface display such as that shown in FIG. **20**, overlay generator **239** (shown in FIG. **5**) can generate an optical overlay, overlaying the calculated position of the cross members **480-484** onto the image shown in FIG. **20**, so that the operator can determine whether the overlayed image matches the actual location of the hoops or cross members **480-484**. If not, then the operator may illustratively use a touch gesture or another input to correct the location of the overlayed cross members to match the actual cross members shown in the image. Verifying the cross member position with an overlay on the captured image is indicated by block **544** in the flow diagram of FIG. **17**. The location of the cross members relative to the position of the position sensor **196** or another reference point on following vehicle **136** can be determined in other ways as well, as indicated by block **546** in the flow diagram of FIG. **17**.

Once the location of the cross members **480-482** has been calculated relative to the location of position sensor **196** (e.g., the calibrated cross member offset values), that information can be stored (on vehicle **136**, vehicle **101**, at a remote server location, etc.) and/or the information can be communicated to another vehicle, or to remote storge locations, for use during an unloading operation, the next time a leading vehicle **101** encounters the pair of following vehicle **136** and receiving vehicle **134**. Storing or communicating the location information in this way is indicated by block **548** in the flow diagram of FIG. **17**.

FIG. **18** is a flow diagram illustrating one example of the operation of unloading control system **150** in controlling the unloading operation based upon the location of the cross members **480-484** relative to the position of position sensor **196** (or another reference position) on following vehicle **136**.

Following vehicle/receiving vehicle pair detector **160** can identify the vehicle pair to obtain the location values (e.g., the calibrated cross member offset values) for the cross members for this particular vehicle pair. The calibrated cross member offset values will indicate the offset (e.g., distance and direction or just distance) of the cross members relative to a reference point on following vehicle **136** (such as relative to the position sensor **196**). Obtaining the calibrated cross member offset values for the cross members for this following vehicle/receiving vehicle pair is indicated by block **550** in the flow diagram of FIG. **18**. The values can be retrieved from a memory **552** on leading vehicle **101**, or from a remote location (such as from another vehicle or remote server location). The values can be obtained by running a calibration operation, as described above, and as indicated by block **554** in the flow diagram of FIG. **18**. The calibrated cross member offset values can be obtained in other ways as well, as indicated by block **556**.

Vehicle position detection system **164** then tracks the positions of the cross members relative to the leading vehicle during the unloading operation, as indicated by block **558**. For instance, the dynamic receiving vehicle locator model **424** can be used to track the movement of those cross members relative to the leading vehicle **101**, as indicated by block **560**. The location of the cross members can be tracked by updating the relative position of the following vehicle **136** relative to receiving vehicle **101** using communication between position sensors **146** and **196**. The position of the cross members can be tracked in a similar way as other receiving vehicle parameters, as indicated by block **562**, or in other ways, as indicated by block **564**.

Control signal generator **166** then generates control signals to control the unloading operation based upon the cross member positions, as indicated by block **566** in the flow diagram of FIG. **18**. For instance, assume that spout **108** is positioned relative to receiving vehicle **134** to fill at a particular location in receiving vehicle **134**. Assume also that the fill level has reached an appropriate level so that the vehicles should be nudged relative to one another to move spout **108** relative to receiving vehicle **134** to a new fill location. Obtaining the new fill location is indicated by block **568** in the flow diagram of FIG. **18**. The cross member location comparison system **419** can then compare the new fill location to the location of the cross members to determine whether the new fill location is appropriate for filling. Comparing the locations is indicated by block **570** in the flow diagram of FIG. **18**.

In one example, the location of the cross members is increased to include a buffer region on either side of the cross members to ensure that no material inadvertently hits the cross members. Therefore, cross member location comparison system **419** can compare the location of the cross members, including the buffer regions, to the new fill location to determine whether the new fill location is too close to (e.g., overlaps with) the cross member location (plus the buffer). If so, then a different fill location is obtained (e.g., a fill location that may be closely proximate the previous fill location but adjusted by a distance so the fill location is outside the location of the cross members plus the buffer region). Finding a different fill location in this way is indicated by block **572** in the flow diagram of FIG. **18**. Control signal generator **166** then generates control signals to move the vehicles so spout **108** is unloading material at the adjusted fill location. Unloading control system **150** can generate control signals to control the unloading operation based upon the location of the cross members in other ways as well, as indicated by block **574** in the flow diagram of FIG. **18**.

It can thus be seen that the present description describes a system which automatically identifies the locations of cross members on a receiving vehicle, and controls the material unloading operation to avoid those locations. This reduces the likelihood that grain or other material that is being unloaded will come into contact with the cross members. Therefore, inadvertent spillage or damage to the cross members can be avoided.

FIG. **21** is a block diagram illustrating agricultural machine **140**, shown in FIG. **4**, except that system **140** is disposed in a remote server architecture **500**. In an example, remote server architecture **500** can provide computation, software, data access, and storage services that do not require end-user knowledge of the physical location or configuration of the system that delivers the services. In various examples, remote servers can deliver the services over a wide area network, such as the internet, using appropriate protocols. For instance, remote servers can deliver applications over a wide area network and they can be accessed through a web browser or any other computing component. Software or components shown in previous FIGS. as well as the corresponding data, can be stored on servers at a remote location. The computing resources in a remote server environment can be consolidated at a remote data center location or they can be dispersed. Remote server infrastructures can deliver services through shared data centers, even though they appear as a single point of access for the user. Thus, the components and functions described herein can be provided from a remote server at a remote location using a remote server architecture. Alternatively, they can be provided from a conventional server, or they can be installed on client devices directly, or in other ways.

In the example shown in FIG. **21**, some items are similar to those shown in FIG. **4** and they are similarly numbered. FIG. **21** specifically shows that data stores **144**, **200**, calibration system **162**, and other systems **504**, can be located at a remote server location **502**. Therefore, vehicles **101**, **136** can access those systems through remote server location **502**.

FIG. **21** also depicts another example of a remote server architecture. FIG. **21** shows that it is also contemplated that some elements of FIG. **4** can be disposed at remote server location **502** while others are not. By way of example, one or more of data stores **144**, **200** and other systems **504**, or other items can be disposed at a location separate from location **502**, and accessed through the remote server at location **502**. Regardless of where the items are located, the items can be accessed either directly by machine **101** and/or machine **136**, through a network (either a wide area network or a local area network), the items can be hosted at a remote site by a service, or the items can be provided as a service, or accessed by a connection service that resides in a remote location. Also, the data can be stored in substantially any location and intermittently accessed by, or forwarded to, interested parties. All of these architectures are contemplated herein.

FIG. **21** shows that other vehicles **506** can communicate with one or more vehicles **101**, **136**, or with remote server environment **502** to obtain the calibrated offset values and/or other information. It will also be noted that the elements of FIG. **4**, or portions of them, can be disposed on a wide variety of different devices. Some of those devices include servers, desktop computers, laptop computers, tablet computers, or other mobile devices, such as palm top computers, cell phones, smart phones, multimedia players, personal digital assistants, etc.

FIG. **22** is a simplified block diagram of one illustrative example of a handheld or mobile computing device that can be used as a user's or client's hand held device **16**, in which the present system (or parts of it) can be deployed. For instance, a mobile device can be deployed in the operator compartment of one or both of vehicles **101**, **136** for use in generating, processing, or displaying the calibrated offset values. FIGS. **23-24** are examples of handheld or mobile devices.

FIG. **22** provides a general block diagram of the components of a client device **16** that can run some components shown in FIG. **4**, that interacts with them, or both. In the device **16**, a communications link **13** is provided that allows the handheld device to communicate with other computing devices and in some examples provides a channel for receiving information automatically, such as by scanning. Examples of communications link **13** include allowing communication though one or more communication protocols, such as wireless services used to provide cellular access to a network, as well as protocols that provide local wireless connections to networks.

In other examples, applications can be received on a removable Secure Digital (SD) card that is connected to an interface **15**. Interface **15** and communication links **13** communicate with a processor **17** (which can also embody processors from previous FIGS.) along a bus **19** that is also connected to memory **21** and input/output (I/O) components **23**, as well as clock **25** and location system **27**.

I/O components **23**, in one example, are provided to facilitate input and output operations. I/O components **23** for various examples of the device **16** can include input components such as buttons, touch sensors, optical sensors, microphones, touch screens, proximity sensors, accelerometers, orientation sensors and output components such as a display device, a speaker, and or a printer port. Other I/O components **23** can be used as well. **27**

Clock **25** illustratively comprises a real time clock component that outputs a time and date. It can also, illustratively, provide timing functions for processor **17**.

Location system **27** illustratively includes a component that outputs a current geographical location of device **16**. This can include, for instance, a global positioning system (GPS) receiver, a dead reckoning system, a cellular triangulation system, or other positioning system. Location system **27** can also include, for example, mapping software or navigation software that generates desired maps, navigation routes and other geographic functions.

Memory **21** stores operating system **29**, network settings **31**, applications **33**, application configuration settings **35**, contact or phone book application **43**, client system **24**, data store **37**, communication drivers **39**, and communication configuration settings **41**. Memory **21** can include all types of tangible volatile and non-volatile computer-readable memory devices. Memory **21** can also include computer storage media (described below). Memory **21** stores computer readable instructions that, when executed by processor **17**, cause the processor to perform computer-implemented steps or functions according to the instructions. Processor **17** can be activated by other components to facilitate their functionality as well.

FIG. **23** shows one example in which device **16** is a tablet computer **600**. In FIG. **23**, computer **600** is shown with user interface display screen **602**. Screen **602** can be a touch screen or a pen-enabled interface that receives inputs from a pen or stylus. Tablet computer **600** can also **14** use an on-screen virtual keyboard. Of course, computer **600** might also be attached to a keyboard or other user input device through a suitable attachment mechanism, such as a wireless link or USB port, for instance. Computer **600** can also illustratively receive voice inputs as well.

FIG. **24** shows that the device can be a smart phone **71**. Smart phone **71** has a touch sensitive display **73** that displays icons or tiles or other user input mechanisms **75**. Mechanisms **75** can be used by a user to run applications, make calls, perform data transfer operations, etc. In general, smart phone **71** is built on a mobile operating system and offers more advanced computing capability and connectivity than a feature phone.

Note that other forms of the devices **16** are possible.

FIG. **25** is one example of a computing environment in which elements of FIG. **4**, or parts of it, (for example) can be deployed. With reference to FIG. **25**, an example system for implementing some embodiments includes a computing device in the form of a computer **810** programmed to operate as discussed above. Components of computer **810** may include, but are not limited to, a processing unit **820** (which can comprise processors from previous FIGS.), a system memory **830**, and a system bus **821** that couples various system components including the system memory to the processing unit **820**. The system bus **821** may be any of several types of bus structures including a memory bus or memory controller, a peripheral bus, and a local bus using any of a variety of bus architectures. Memory and programs described with respect to FIG. **4** can be deployed in corresponding portions of FIG. **25**.

Computer **810** typically includes a variety of computer readable media. Computer readable media can be any available media that can be accessed by computer **810** and includes both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer readable media may comprise computer storage media and communication media. Computer storage media is different from, and does not include, a modulated data signal or carrier wave. Computer storage media includes hardware storage media including both volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. Computer storage media includes, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-

ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store the desired information and which can be accessed by computer **810**. Communication media may embody computer readable instructions, data structures, program modules or other data in a transport mechanism and includes any information delivery media. The term "modulated data signal" means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal.

The system memory **830** includes computer storage media in the form of volatile and/or nonvolatile memory such as read only memory (ROM) **831** and random access memory (RAM) **832**. A basic input/output system **833** (BIOS), containing the basic routines that help to transfer information between elements within computer **810**, such as during start-up, is typically stored in ROM **831**. RAM **832** typically contains data and/or program modules that are immediately accessible to and/or presently being operated on by processing unit **820**. By way of example, and not limitation, FIG. **25** illustrates operating system **834**, application programs **835**, other program modules **836**, and program data **837**.

The computer **810** may also include other removable/non-removable volatile/nonvolatile computer storage media. By way of example only, FIG. **25** illustrates a hard disk drive **841** that reads from or writes to non-removable, nonvolatile magnetic media, an optical disk drive **855**, and nonvolatile optical disk **856**. The hard disk drive **841** is typically connected to the system bus **821** through a non-removable memory interface such as interface **840**, and optical disk drive **855** are typically connected to the system bus **821** by a removable memory interface, such as interface **850**.

Alternatively, or in addition, the functionality described herein can be performed, at least in part, by one or more hardware logic components. For example, and without limitation, illustrative types of hardware logic components that can be used include Field-programmable Gate Arrays (FPGAs), Application-specific Integrated Circuits (e.g., ASICs), Application-specific Standard Products (e.g., ASSPs), System-on-a-chip systems (SOCs), Complex Programmable Logic Devices (CPLDs), etc.

The drives and their associated computer storage media discussed above and illustrated in FIG. **25**, provide storage of computer readable instructions, data structures, program modules and other data for the computer **810**. In FIG. **25**, for example, hard disk drive **841** is illustrated as storing operating system **844**, application programs **845**, other program modules **846**, and program data **847**. Note that these components can either be the same as or different from operating system **834**, application programs **835**, other program modules **836**, and program data **837**.

A user may enter commands and information into the computer **810** through input devices such as a keyboard **862**, a microphone **863**, and a pointing device **861**, such as a mouse, trackball or touch pad. Other input devices (not shown) may include a joystick, game pad, satellite dish, scanner, or the like. These and other input devices are often connected to the processing unit **820** through a user input interface **860** that is coupled to the system bus, but may be connected by other interface and bus structures. A visual display **891** or other type of display device is also connected to the system bus **821** via an interface, such as a video interface **890**. In addition to the monitor, computers may also include other peripheral output devices such as speakers **897** and printer **896**, which may be connected through an output peripheral interface **895**.

The computer **810** is operated in a networked environment using logical connections (such as a controller area network-CAN, local area network-LAN, or wide area network WAN) to one or more remote computers, such as a remote computer **880**.

When used in a LAN networking environment, the computer **810** is connected to the LAN **871** through a network interface or adapter **870**. When used in a WAN networking environment, the computer **810** typically includes a modem **872** or other means for establishing communications over the WAN **873**, such as the Internet. In a networked environment, program modules may be stored in a remote memory storage device. FIG. **25** illustrates, for example, that remote application programs **885** can reside on remote computer **880**.

It should also be noted that the different examples described herein can be combined in different ways. That is, parts of one or more examples can be combined with parts of one or more other examples. All of this is contemplated herein. **6**

Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing the claims.

What is claimed is:

1. A work machine system including a leading vehicle configured to unload material into a receiving vehicle during an unloading operation, the receiving vehicle being configured to be propelled by a following vehicle, the work machine system comprising:

a receiving vehicle sensor mounted to the leading vehicle, the receiving vehicle sensor being configured to detect a cross member disposed across a material-receiving opening of the receiving vehicle and generate a sensor signal responsive to the detected cross member;

one or more processors; and

memory storing instructions executable by the one or more processors that, when executed by the one or more processors, configure the work machine system to:

identify, responsive to the sensor signal, a first offset value that is indicative of a location of the cross member relative to a first reference point on the leading vehicle;

identify a second offset value that is indicative of a location of a second reference point on the following vehicle relative to the first reference point on the leading vehicle;

identify a calibrated cross member offset value indicative of a location of the cross member relative to the second reference point on the following vehicle based on the first offset value and the second offset value; and

control the unloading operation based on the calibrated cross member offset value.

2. The work machine system of claim **1**, wherein the instructions, when executed by the one or more processors, further configure the work machine system to:

begin unloading material at a fill location in the receiving vehicle;

detect a location of the following vehicle and a location of the leading vehicle and to identify a location of the cross member based on the location of the following vehicle, the location of the leading vehicle, and the calibrated cross member offset value; and

compare the fill location with the location of the cross member and generate a comparison result signal indicative of a result of the comparison.

3. The work machine system of claim **2**, wherein the instructions, when executed by the one or more processors, further configure the work machine system to:

generate a position control signal to adjust the fill location based on the comparison result signal.

4. The work machine system of claim **3**, wherein the instructions, when executed by the one or more processors, further configure the work machine system to generate the comparison result signal to indicate whether the fill location overlaps with the location of the cross member and to generate the control signal to change the fill location to a new fill location that does not overlap with the location of the cross member.

5. The work machine system of claim **4**, wherein the instructions, when executed by the one or more processors, further configure the work machine system to identify, as part of the location of the cross member, a buffer area.

6. The work machine system of claim **1**, wherein the instructions, when executed by the one or more processors, further configure the work machine system to:

generate an operator prompt on an operator interface on the leading vehicle, the operator prompt prompting an operator to position the receiving vehicle relative to the leading vehicle so the receiving vehicle sensor can detect the cross member.

7. The work machine system of claim **1**, wherein the receiving vehicle sensor comprises:

an optical sensor configured to capture an image of a portion of the receiving vehicle; and

an image processor configured to identify a location of the cross member in the captured image.

8. The work machine system of claim **7**, wherein the instructions, when executed by the one or more processors, further configure the work machine system to obtain a location and orientation of the optical sensor on the leading vehicle and to identify the first offset value based on the location of the cross member in the captured image and based on the location and orientation of the optical sensor on the leading vehicle.

9. The work machine system of claim **8**, wherein the instructions, when executed by the one or more processors, further configure the work machine system to:

obtain a first vehicle location based on a location of a position sensor on the following vehicle and a second vehicle location based on a location of a position sensor on the leading vehicle and identify, as the calibrated cross member offset value, a location of the cross member relative to the location of the position sensor on the following vehicle based on the first offset value, the first vehicle location, and the second vehicle location.

10. The work machine system of claim **1**, wherein the receiving vehicle sensor comprises:

an image sensor configured to capture an image of the receiving vehicle;

an operator display device configured to display the image of the receiving vehicle; and

an operator interaction detector configured to detect operator interaction with the image of the receiving vehicle, the operator interaction identifying the cross member in the image of the receiving vehicle.

11. The work machine system of claim **1**, wherein the instructions, when executed by the one or more processors, further configure the work machine system to:

store the calibrated cross member offset value in a data store, identified in the data store by a vehicle pair identifier identifying, as a following vehicle and receiving vehicle pair, the following vehicle and receiving vehicle; and

identify the following vehicle/receiving vehicle pair and obtain, from the data store, the calibrated cross member offset value corresponding to the identified following vehicle/receiving vehicle pair.

12. The work machine system of claim **11**, wherein the instructions, when executed by the one or more processors, further configure the work machine system to:

obtain a following vehicle position signal indicative of a position of the following vehicle relative to the leading vehicle and to determine a position of the cross member based on the calibrated cross member offset value obtained from the data store and based on the following vehicle position signal.

13. A computer implemented method of controlling an unloading operation for unloading material from a leading vehicle into a receiving vehicle that is propelled by a following vehicle, the method comprising:

detecting a cross member on the receiving vehicle with a receiving vehicle sensor on the leading vehicle;

identifying a first offset value that is indicative of a location of the cross member relative to a first reference point on the leading vehicle;

identifying a second offset value that is indicative of a location of a second reference point on the following vehicle relative to the first reference point on the leading vehicle;

identifying a calibrated cross member offset value indicative of a location of the cross member relative to the second reference point on the following vehicle based on the first offset value and the second offset value; and

controlling the unloading operation based on the calibrated cross member offset value.

14. The computer implemented method of claim **13** wherein detecting the cross member comprises:

displaying an image of the receiving vehicle on an operator display device;

detecting an operator interaction with the image of the receiving vehicle; and

identifying the cross member in the image based on the detected operator interaction.

15. The computer implemented method of claim **13** wherein detecting a cross member comprises:

capturing an image of a portion of the receiving vehicle with an optical sensor;

generating an operator prompt on an operator interface on the leading vehicle, the operator prompt prompting an operator to position the receiving vehicle relative to the leading vehicle so a reference item on the leading vehicle is aligned with the cross member in the image; and

identifying the cross member in the image based on the location of the reference item in the image.

16. The computer implemented method of claim **13** wherein detecting the cross member comprises:

capturing an image of a portion of the receiving vehicle with an optical sensor; and

automatically identifying a location of the cross member in the captured image.

17. The computer implemented method of claim 16 wherein identifying the first offset value comprises:

obtaining a location and orientation of the optical sensor on the leading vehicle; and

identifying the first offset value based on the location of the cross member in the captured image and based on the location and orientation of the optical sensor on the leading vehicle.

18. The computer implemented method of claim 17 wherein identifying the calibrated cross member offset value comprises:

obtaining a first vehicle location based on a location of a position sensor on the following vehicle;

obtaining a second vehicle location based on a location of a position sensor on the leading vehicle; and

identifying, as the calibrated cross member offset value, a location of the cross member relative to the location of the position sensor on the following vehicle based on the first offset value, the first vehicle location, and the second vehicle location.

19. A control system comprising:

memory storing instructions executable by the one or more processors that, when executed by the one or more processors, configure the control system to

receive a sensor signal indicative a cross member spanning at least a portion of a receiving area of the receiving vehicle and to identify a first offset value that is indicative of a location of the cross member relative to a first reference point on a leading vehicle that is configured to perform an unloading operation to unload material into a receiving vehicle propelled by a following vehicle;

identify a second offset value that is indicative of a location of a second reference point on the following vehicle relative to the first reference point on the leading vehicle; and

identify a calibrated cross member offset value indicative of a location of the cross member relative to the second reference point on the following vehicle based on the first offset value and the second offset value; and

control the unloading operation based on the calibrated cross member offset value.

20. The control system of claim 19, wherein the instructions, when executed by the one or more processors, further configure the control system to:

an operator prompt generator configured to generate an operator prompt on an operator interface on the leading vehicle, the operator prompt prompting an operator to position the receiving vehicle relative to the leading vehicle so the receiving vehicle sensor can detect the cross member.

* * * * *